

발 간 등 록 번 호

11-B090039-000877-01

Policy Research 2022-22-04



2022 K-Innovation Partnership Program with Indonesia

Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: S&T Think Tank, R&D Program Evaluation, and S&T Laws and Regulations

2022 K-Innovation Partnership Program with Indonesia

Title	Policy Consultation on the Arrangement of Science, Technology and Innovation (STI) Governance: S&T Think Tank, R&D Program Evaluation, and S&T Laws and Regulations
Institute	Science and Technology Policy Institute (STEP I)
Partner country	Republic of Indonesia
In cooperation with	National Research and Innovation Agency (BRIN)
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Published on 30 December 2022

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Printed by Mirae Media

Tel: +82-2-815-0407 Fax: +82-2-822-1269

ISBN 978-89-6112-876-6 93300



Executive Summary

Since 2014, the Science and Technology Policy Institute (STPEI) has maintained long-term cooperation with Indonesia by providing policy consultation related to science and technology and capacity-building programs for the science and technology institutes (Agency for the Assessment and Application of Technology (BPPT) and the Indonesian Institute of Sciences (LIPI)) in Indonesia.

Recently, STI governance issues have become more critical in Indonesia with the establishment of its National Research and Innovation Agency (BRIN) in 2021. BRIN requested from STPEI a policy consultation on effective R&D innovation governance, and STPEI duly gave an STI policy consultation based on a three-year framework (2020-2022).

The arrangement of STI Governance (Coordination Mechanism, Institutional Framework, Financial and Investment System) in Indonesia for the period 2020-2022 comprised various contents. In 2020, STPEI analyzed Indonesia's STI governance with regard to its coordination mechanisms, investment systems, and R&D program evaluations. In 2021, STPEI conducted various analyses of Indonesia's coordination mechanisms, policy think tanks, and the law's regulatory framework (No. 2019-11). The research topics for 2021 included (i) the adjustment mechanism (or coordination mechanism) for the establishment of mission-based research institutes, (ii) the national evaluation system for R&D programs, and (iii) the contents of law no. 209-11 in comparison with Korea's S&T framework Act. In 2022, STPEI conducted research focused on (i) BRIN's role in green growth policy, (ii) development of STI human capital in Indonesia, and (iii) improvement of BRIN's management regime for mission-oriented research and innovation.

In 2022, the research was conducted through joint research and performance dissemination workshops with Indonesian partners. Researchers from both Korea and Indonesia (BRIN)

collaborated on the research, while Korean experts passed on their experience and provided policy advice.

The 2022 project achieved the following outcomes:

First, by benchmarking Korea's green growth success cases, the research team presents directions and strategies for BRIN's role, mission, and operational systems in implementing Indonesia's green growth policy.

Second, the research team provides experience and implications for Korea's S&T human resources policy in light of the current situation and the problems associated with Indonesia's STI human capital. It also proposes a promotion strategy for developing Indonesia's human capital and sets directions for improving the relevant laws and systems.

Third, the research team suggests ways to operate BRIN more effectively, and proposes a new operation and governance system for BRIN to facilitate cooperation between the three deputies (Research & Innovation Facilitation, Research & Innovation Infrastructure, and Research & Innovation Utilization) and cooperation with external industries and innovation actors.

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CHAPTER 1

Project Overview





Chapter 1. Project Overview

1. Project Background

Since 2014, the Science and Technology Policy Institute (STEPI) has been providing policy consultation and a capacity-building program for the experts of Indonesia's Institute of Science (LIPI) and its Agency for the Assessment and Application of Technology (BPPT) as part of a long-term partnership. In particular, it has engaged in practical collaboration with LIPI under the K-Innovation Partnership Program since 2018. Accordingly, the 2020-2022 K-Innovation Partnership Program with Indonesia is also a cooperative project initiated at LIPI's request. Over the past two years, the STEPI and LIPI project teams have conducted joint research and policy consultation on reforms of Indonesia's STI governance, its plan for the establishment of a think tank, and its national R&D program management and evaluation system.

Recently, there have been significant changes in science and technology governance in Indonesia. For instance, on September 1, 2021 various national R&D institutions were integrated into the National Research and Innovation Agency (BRIN), a presidential agency of Indonesia, while LIPI was merged into BRIN.

In response to these changes and the altered demands of the partner, in 2022 the overall theme of the project was switched from 'Financial System, Evaluation, and Investment Strategy for STI Governance in Indonesia' to 'Effective Implementation of the National STI Policy System with BRIN' under an agreement between the two institutions.

2. Project Objectives

As per BRIN's request, the third-year project topic (2023) was 'Effective Implementation of the National STI Policy System with BRIN'. The detailed topics for each year of this project are as follows.

[Table 1-1] Objectives of the project, 2020-2022

Project Title	Arrangement of STI Governance: Coordination Mechanism, Institutional Framework and Financial System, and Investment in Indonesia	
Year	Topic	Sub-topics
2020	Analysis and Establishment of the Coordination Mechanism for Indonesia's STI Governance	Building a New Think Tank for Science and Technology Innovation in Indonesia
		R&D Funding and Governance of the Research Endowment Fund in Indonesia
		Management and Evaluation System of the National Research and Development Program in Indonesia
2021	Analysis and Establishment of the Institutional Framework for Indonesia's STI Governance	Building a New Think Tank for Science and Technology Innovation
		Building Monitoring and Evaluation Systems for the National Research & Development Program in Indonesia
		Laws& Regulations on S&T
2022	Effective Implementation of the National STI Policy System with BRIN	BRIN's Role in the Green Growth Policy
		Developing STI Human Capital in Indonesia
		Improving the Management Regime of BRIN for Mission-oriented Research and Innovation

The K-Innovation Partnership Program with Indonesia fundamentally aims to improve the science and technology system and the overall institutional conditions. Moreover, it also aims to purposefully promote cooperation between the two partner countries through policy consultation and capacity-building activities based on their respective requests and needs.

3. Project Framework

In essence, the 2022 project consists of two axes: i) international joint research and ii) capacity-building activities. The purpose of this project is not to provide one-way consulting from Korean experts, but rather to enable local policy researchers to diagnose themselves and devise improvement strategies together by directly participating in the research. In addition, the training workshops are designed to promote capacity building through knowledge sharing and case studies from Korea's experiences. Strengthening the research capabilities of individual researchers will eventually act as an engine for exploring the direction of national development.

First, the international joint research team was formed with researchers from BRIN in Indonesia and Korea. As regards the detailed subjects presented for policy consultation in 2022, the Indonesian research team studied Indonesia's current circumstances and situation, and identified limitations and challenges based on an analysis of their findings.

At the same time, the Korean research team diagnosed the situation in Indonesian and researched Korean cases. (Ultimately, policy recommendations were proposed based on the results of the analysis of the current situation and Korean case studies. As for the training workshop, it was designed to provide more practical methodologies on policy research by providing diverse cases. Indonesian researchers strengthened their policy research capabilities, which influenced the derivation of the final strategies in the international joint research.

Finally, an outcome dissemination forum was held to share the results of the international joint research and policy advice. In particular, the policy suggestions presented on the basis of comparisons with and analyses of the related systems and policies in Korea were promoted to local partner organizations and various stakeholders in the STI field. In the process, we expected not only to spread the results of this project but also requests for additional cooperation.

Therefore, the inputs and outputs of diverse activities and results of the project were systematically and convergently connected.

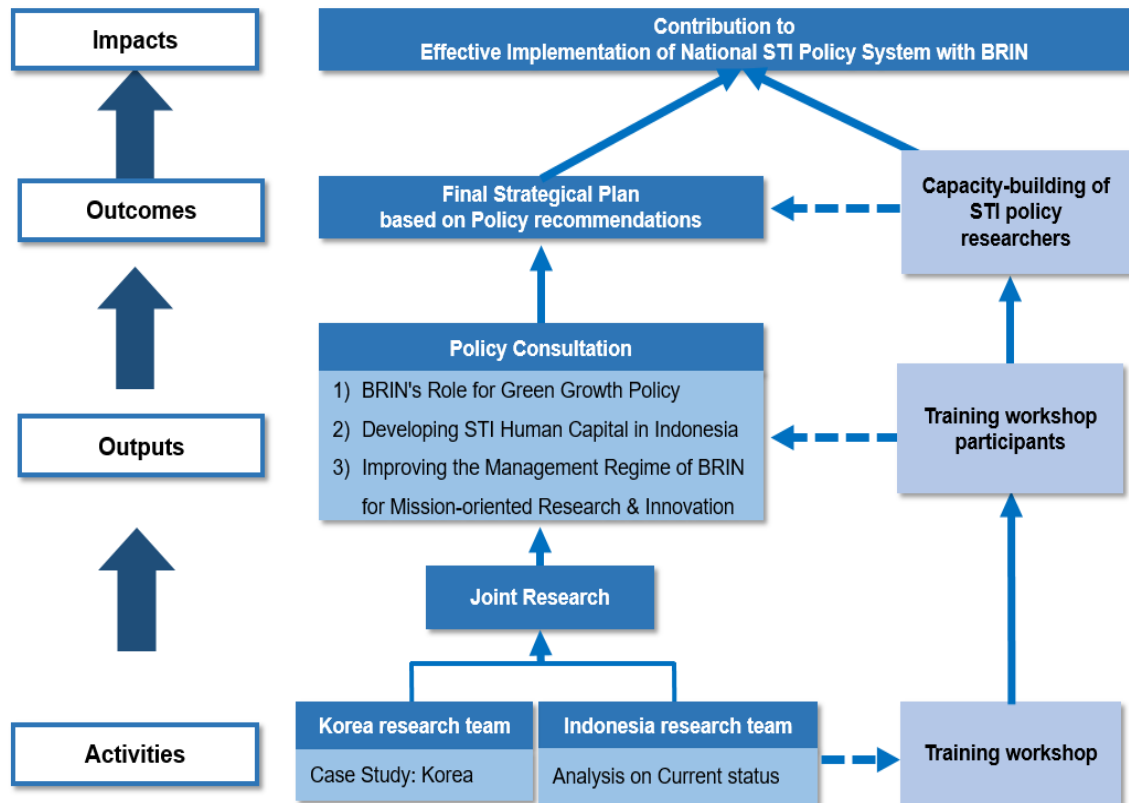
The inputs and outputs of the 2022 project are as follows:

- BRIN's Role in Green Growth Policy
 - Analysis of the policy promotion system and the role of the national think tank related to Korea's green growth policy.
 - Drafting detailed roadmaps and action plans for institutional readiness and securing resources such as funding, personnel, etc.

- Developing STI Human Capital in Indonesia
 - Understanding the history of STI human capital development in Indonesia.
 - Analysis of Indonesia STI human capital development status and problems.
 - Analysis of STI human capital development in Korea.
 - Presentation of Indonesia's STI human capital development strategy and means based on the implications derived from Korea's experience.

- Improving the Management Regime of BRIN for Mission-oriented Research and Innovation
 - Analysis and evaluation of the current status of BRIN governance.
 - Design of new operating systems for mission-oriented research and innovation.
 - Proposal of key tasks to realize a new operating system model.
 - Execution of joint research as part of this process in order to make BRIN's operating system feasible and appropriate to Indonesia's situation by researching Korean cases.

[Figure 1-1] Project Framework, 2022



4. Project Team

The project team consists of management and joint research teams. Notably, each joint research team is composed of BRIN researchers and policy analysts, and is led by Korean experts from the relevant fields. The roles and responsibilities of the STEPI project management team are as follows:

- Coordinating and managing the whole process of the 2022 project.
- Deriving the final policy recommendations based on the research results.

[Table 1-2] Project Team

Roles	Name	Position / Affiliation
Project Management	Dr. Jong-Seon Kim	Project Manager, Research Fellow, STEPI
	Dr. Deok Soon Yim	Senior Research Fellow, STEPI
	Dr. Eun Joo Kim	Principal Researcher, STEPI
	Ms. Shinae Kang	Project Coordinator, Researcher, STEPI
	Dr. Dudi Hydayat	Director, DRTIPF, BRIN
Joint Research Team1	Prof. Ki-Seok Kwon	Hanbat National University
	Mr. Anugerah Yuka Asmara	Researcher, BRIN
	Mr. Aditya Wisnu Pradana	Researcher, BRIN
	Ms. Siti Annisa Silvia Rosa	Policy Analyst, BRIN
	Ms. Sofia Yuniar Sani	Researcher, BRIN
	Ms. Ria Jayanthi	Policy Analyst, BRIN
Joint Research Team2	Prof. Seong-soo Kim	Hankuk University of Foreign Studies
	Dr. Changyul Lee	Hankuk University of Foreign Studies
	Mr. Maulana Akbar	Researcher, BRIN
	Ms. Dhian Kusumawardani	Policy Analyst, BRIN
	Ms. Indri Juwita Asmara	Researcher, BRIN
	Ms. Sri Supadmi	Policy Analyst, BRIN
	Ms. Anggini Dinaseviani	Researcher, BRIN
	Mr. Subkhi Abdul Aziz	Researcher, BRIN
Joint Research Team3	Dr. Youngjoo Ko	President, DISTEP
	Ms. Yustina Nita Sulistami	Policy Analyst, BRIN
	Ms. Nimas Maninggar	Researcher, BRIN
	Ms. Umi Salama	Policy Analyst, BRIN
	Ms. Mely Maulidiani	Policy Analyst, BRIN
	Ms. Stevani Anggina	Researcher, BRIN

5. Project Activities

5.1 Preliminary Meetings

5.1.1 Drafting the 2022 Project Objectives and Topics

- Date/Venue: 27 January 2022 (Online)
- Participants: Project Team of STEPI and BRIN
- Agenda: The changes of BRIN's functions and organizational structure, request for change of research topics

5.1.2 Drafting the 2022 Project Objectives and Topics

- Date/ Venue: 15 February 2022 (Online)
- Participants: Project Team of STEPI and BRIN
- Agenda: Details of the 2022 project plan, schedule of the kick-off seminar and training workshop

[Figure 1-2] Preliminary meetings



5.2 Kick-off Seminar

- Date/ Venue: 15:00-18:00, 7 April 2022 (On-off hybrid)
- Participants: Project Management Team of STEPI and BRIN, BRIN researchers, other stakeholders in Indonesia (in total 125 persons)
- Purpose: To reach a mutual agreement of the project plans and the objectives
- Detailed contents:
 - Introduction of the K-Innovation partnership program
 - Project objectives and detailed plans
 - Roles and responsibilities of the project team
 - (Session 1) BRIN's Role for Green Growth Policy" Benchmarking the Korean Case
 - a. Elaborating practice of South Korean's STI Policy Think Tank(s) in Supporting Green Growth Policy.
 - b. Describing role of DRTIPF - BRIN as STI Policy Think Tank in Supporting Green Growth Policy.
 - c. Elaborating DRTIPF - BRIN activities in collaboration with other task forces units inside and outside BRIN in Supporting Green Growth Policy.
 - (Session 2) Developing STI Human Capital in Korea
 - a. Collection and analysis of detailed data on the status of S&T human resources (Scientists, Engineers, Technicians, Women scientists, Science-Gifted Students etc.)
 - b. Analysis of the Special Act on Support of Scientists and Engineers
 - c. Analysis of Master Plans: how the Special Act has been implemented
 - d. Developing Technicians for the industry: Manufacturing is important
 - e. Discussing lessons and implications from the Korean experience
 - (Session 3) The Management Regime of Government-supported Research Institutes & National Research Council of S&T in Korea
 - a. Analysing on Management Regime in 5 perspectives of objectives, strategies, governance, funding, and organisation management.

- Program:

[Table 1-3] Kick-off seminar program

Time (ID)	Contents	Presenters
13:00-13:15	Opening Remarks	Dr. Hwanil Park Chief Director, Division of Global Innovation Strategy Research, STEPI
	Welcome Remarks	Mrs. Rr Nur Tri Aries Suestiningtyas Executive Secretary, BRIN
13:15-13:25	2020-2022 K-Innovation program with Indonesia	Dr. Jong-Seon Kim Project Manager, Research Fellow, STEPI
13:25-13:35	2022 Project overview <i>“Effective Implementation of the National STI Policy System with BRIN”</i>	Dr. Dudi Hidayat Director, Directorate for Research, Technology & Innovation Policy Formulation (DRTIPF), BRIN
13:35-13:50	Session 1. BRIN's Role in the Green Growth Policy	Prof. Ki-Seok Kwon Hanbat National University
13:50-14:05		Mr. Anugerah Yuka Asmar Coordinator, Directorate for Research, Technology & Innovation Policy Formulation (DRTIPF), BRIN
14:05-14:20	Session 2. Developing STI Human Capital in Indonesia	Prof. Seong-soo Kim Hankuk University of Foreign Studies
14:20-14:35		Mr. Maulana Akbar Policy Researcher, Directorate for Research, Technology & Innovation Policy Formulation (DRTIPF), BRIN
14:35-14:50	Session 3. Improving the Management Regime of BRIN for Mission-oriented Research and Innovation	Dr. Young Joo Ko President, Daejeon Institute of Science & Technology for Enterprise & People
14:50-15:05		Ms. Yustina Nita Sulistami Policy Analyst, BRIN
15:05-16:40	Q&A and Discussions	
15:40-16:00	Wrap-up	Dr. Deok Soon Yim Senior Research Fellow, STEPI

2022 K-Innovation Partnership Program with Indonesia

[Figure 1-3] Photos from the Kick-off Seminar

 2022 K-Innovation Partnership Program with Indonesia Dr. Hwanil Park Chief Director of Division of Global Innovation Strategy Research, STEPI	 Mrs. Rr Nur Tri Aries Suestiningtyas, MA Executive Secretary, BRIN
Dr. Hwanil Park (STEPI)	Mrs. Rr Nur Tri Aries Suestiningtyas (BRIN)
 2022 K-Innovation Partnership Program with Indonesia Dr. Jong-Seon Kim Director, DRTIPF, BRIN	 Dr. Dudi Hidayat Director, DRTIPF, BRIN
Dr. Jong-Seon Kim (STEPI)	Dr. Dudi Hidayat (BRIN)
 Kick-off Seminar 2022 K-Innovation Partnership Program with Indonesia Dr. Deok Soon Yim Senior Research Fellow, STEPI	 2022 K-Innovation Partnership Program with Indonesia Prof. Ki-Seok Kwon Hankuk Univ. of Foreign Studies
Dr. Deok Soon Yim (STEPI)	Prof. Ki-Seok Kwon (Hankuk Univ. of Foreign Studies)
 2022 K-Innovation Partnership Program with Indonesia	 Kick-off Seminar 2022 K-Innovation Partnership Program with Indonesia Effective Implementation of National STI Policy System with BRIN DATE: 1 - 7 April 2022
Kick-off Seminar Participants	Kick-off Seminar Speakers

5.3 International Joint Research

International joint research was conducted to derive final policy advice. Studies on the current status of STI in Indonesia and Korean case studies were conducted with respect to each of the three sub-topics, and three reports were produced. Based on these results, the following policy advice was implemented:

- After reflecting the current status of Indonesia's system and South Korea's experience, the strategy to promote Indonesia's green growth policy effectively was proposed.
- In Indonesia's STI human capital development, Korea's science and technology human resource development cases were introduced, and specific human resource development plans were presented based on the current situation.
- The policy implications and agendas were suggested for improving the operating system of BRIN according to the case analysis of public research institutes in Korea, especially the National Research Council for Science and Technology and government-funded research institutes.

The policy consultation contains the results of the above research and the implications raised by topic.

5.4 Training workshop

- Date/Venue: 20-21 July 2022 (On-offline hybrid)
- Participants: STI experts and researchers from Korea and Indonesia, government officials, enterprises (in total 150 persons)
- Purpose: To build the capacities and competences of BRIN's researchers and Indonesian experts by sharing expertise and know-how accumulated from Korea's experiences.
- Performance outcomes:
 - The results of the survey on the participants' satisfaction with the workshop raised the following suggestions for improvements: (1) inclusion of additional workshops, (2) separation of sessions to facilitate in-depth discussions.

- Some 150 people participated in the 2022 workshop, far exceeding the planned initial target of 40, i.e. an increase of 375%.

(*Formula for performance measurement: $\frac{\text{Number of participants}}{\text{Target number of participants}} \times 100$)

[Table 1-4] Results of the survey (Likert scale) on satisfaction with the training workshop

Questions	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I am satisfied with the workshop program overall.	32.7%	65.3%	1.3%	0.0%	0.7%
The topics covered in the workshop were relevant to our government's STI policy priorities.	27.3%	67.3%	5.3%	0.0%	0.0%
The topics covered in the workshop were relevant to our institute's primary mandate.	29.5%	61.1%	8.7%	0.7%	0.0%
The program was well organized and easy to follow.	34.7%	59.3%	6%	0.0%	0.0%
The presentations/lectures were well prepared and covered key information on the topics.	28%	63.3%	7%	0.0%	0.7%
The presenters/lecturers were fully prepared with expertise and professional experience on the topics they covered.	65.3%	32.7%	2%	0.0%	0.0%
The presentation materials facilitated our understanding of the presentation's contents.	28%	68%	4%	0.0%	0.0%
The Q&A and discussion session was helpful to enhancing my understanding of the topics.	25.3%	64.7%	9.3%	0.7%	0.0%
The information and knowledge I gained from the workshop are applicable to my work.	21.5%	63.8%	14.1	0.7%	0.0%

* Note: The sum of the response values is 100%, but some response rates are infinite decimals, so there is a margin of error of $\pm 0.1\%$ due to the processing of significant figures.

[Figure 1-4] Photos from the Training Workshop



5.5 Dissemination Forum

- Date/Venue: 11 October 2022, Jakarta, Indonesia
- Participants: 50 stakeholders from BRIN and STEPI
- Purpose: To present the performance achievements and results of the 2022 project outcomes

- Program:

[Table 1-5] Dissemination Forum Program

Time (ID)	Contents	Presenters
13:00-13:20	Opening Remarks	Dr. Boesdiastoeti Ontowirjo Deputy Chairman, Deputy for Research and Innovation Policy, BRIN
13:20-13:35	2020-2022 K-Innovation program with Indonesia	Dr. Jong-Seon Kim Project Manager, Research Fellow, STEPI
13:35-13:50	Overview of joint research	Mr. Yudho Baskoro Deputy secretary, Deputy for Research and Innovation Policy, BRIN
	Introduction of the 2022 Project Team	
14:00-15:00	BRIN's Role in the Green Growth Policy	Prof. Ki-Seok Kwon Hanbat National University
		Mr. Anugerah Yuka Asmara, Coordinator, Directorate for Research, Technology & Innovation Policy Formulation (DRTIPF), BRIN
15:00-16:00	Developing STI Human Capital in Indonesia	Prof. Seong-soo Kim Hankuk University of Foreign Studies
		Mr. Maulana Akbar Policy Researcher, BRIN
16:00-17:00	Improving the Management Regime of BRIN for Mission-oriented Research & Innovation	Dr. Young Joo Ko President, Daejeon Institute of Science & Technology for Enterprise & People (DISTEP)
		Ms. Yustina Nita Sulistami Policy Analyst, Directorate for Research, Technology & Innovation Policy Formulation (DRTIPF), BRIN
17:00-17:50	Q&A and Discussions	
17:50-18:00	Wrap-up	Dr. Dudi Hidayat Director, Directorate for Research, Technology & Innovation Policy Formulation(DRTIPF), BRIN

[Figure 1-5] Photos from the Dissemination Forum



6. Project Schedule

[Table 1-6] Project Schedule

Activity	2022												2023
	1	2	3	4	5	6	7	8	9	10	11	12	1
Project Planning													
Organizing the Project team													
Kick-off Seminar													
International Joint research													
Training Workshop													
Submission of Interim Report													
Dissemination Forum													
Submission of Final Report													
Presentation of Final Report													
Evaluation of Final Report													
Publication													



CHAPTER 2

Designing the STI Policy Think Tank for Green Growth Policy Formulation: Lessons Learned from South Korea's Practices and Experience

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Chapter 2. Designing the STI Policy Think Tank for Green Growth Policy Formulation: Lessons Learned from South Korea's Practices and Experience

1. Introduction

1.1 Background

The so-called “Green Economy” (GE) is a global and regional response to finding solutions to the economic, ecological, and social problems that is not separate from the Paris Agreement. The term “green economy” originated from developed countries as a policy trend to create a more efficient economic system and develop more environmentally-friendly technologies, and to save resources while reducing carbon emissions. Based on this, the GE concept has recently received worldwide attention as a solution to various global changes. These terms and approaches emerged in the field of economics relatively recently, i.e. about four decades ago (Ivlev & Ivleva, 2018).

GE leads to sustainable development, which is supported by technology advancement as an end goal in many countries (Ryszawska, 2017; Sulich, 2021). GE is a form of economic development aimed at achieving two goals: to reduce dangers and ecological scarcity and to enhance global well-being and social fairness (UNEP, 2014). All economy activities, ranging from the production and distribution of goods to their consumption, are pushed to human prosperity with minimal or no detrimental effect on the environment (Ivlev & Ivleva, 2018).

GE has been widely received in many countries around the world as a response to a global agreement, and translates to national green policies in many fields. In essence, GE implies a departure from the conventional 'business as usual' economic paradigm to that of 'business as usual' economics combined with strong regulatory measures and financial incentives aimed at promoting innovation, investment (e.g. in green technology), sustainable

consumption patterns and behaviors, and information sharing (UNEP, 2011). While green growth (GG) is a policy that aims to provide contingent practical prescriptions (Hamdouch & Depret, 2010), GE is a form of governmental intervention that attempts to achieve sustainable goals through GE development, and also aims to realize human well-being through green development (Samson & Capozza, 2019). Practically speaking, GE and GG are similar concepts that are used to define environmentally-friendly and sustainable economy policies. For the purposes of this study, we often use the two terms (GE and GG) interchangeably or sometimes combine them; hence, we refer to them as “green economy growth” (GEG).

GEG is aligned with science and technology innovation (STI) to promote sustainable socio-economic development (Ryszawska, 2017, Sulich, 2021). It focuses on the quality of economic growth and realizes economic prosperity by creating better social impacts and reducing pressure on the environment and natural capital in a given region. In practice, GEG is close to the concept of STI and sustainable development goals (SDGs). As such, governmental intervention (STI policy) is required to promote GEG to achieve SDGs (See Bakker & Konings, 2017; Stucki et al., 2018; Aguirre-Bastosa, & Weber, 2018). Therefore, GEG, STI Policy, and SDGs are inter-dependent governmental policies that should be carried out in both developed and developing countries.

The most challenging issue for many countries, especially for developing countries in the current decade, is not the need to foster economic development but rather the need to maintain sustainable environmental development. Though industrial-based economic growth and natural resource-based innovation are the main issues facing such countries (Aguirre-Bastosa, & Weber, 2018), they must face GEG issues in order to respond to global policies to save the earth (OECD, 2012). As a developing country, Indonesia faces major challenges to promoting and accelerating green-oriented economy growth. In addition, Indonesia needs to adopt GEG development at the same time. This means that Indonesia needs to adopt a pilot policy design from an advanced country that has succeeded in developing its industrial economy and green technology policy (GTP) concurrently.

As a part of the newly industrializing economies (NIEs), the government of South Korea has developed a comprehensive green growth strategy. The country has become one of the

pioneers in environmentally-friendly technologies in the world, and its STI policy think tank has greatly influenced the formulation of green and economic policies. Through the Green Technology Center (GTC), this country can bridge from STI to GEG development (GTC, 2022). It will be helpful for Indonesia to learn from South Korea about how the STI policy think tank engages with ministries and agencies in dealing with green and economics issues.

In the Indonesian context, the emergence of Indonesia's National Research and Innovation Agency (BRIN), formed in the middle of 2021, is expected to contribute to national GEG development by optimizing certain aspects of STI. As a new public institution, BRIN not only executes research and development (R&D) activities but also formulates and implements STI policies. In addition, BRIN played a role in supporting national GEG development through its active participation in the G20 Meeting at the end of 2022 (BRINa, 2022). This is in keeping with the function of BRIN, which is to contribute to GEG development based on STI in Indonesia. Therefore, BRIN could learn from GTC practices in South Korea in terms of formulating GTP policies in supporting GEG development at the national level.

1.2 Research Questions

- I. How does GTC contribute to GEG development through the STI policy in South Korea ?
- II. What are the key elements of South Korea's GTC practices that can be adopted by BRIN in Indonesia ?
- III. How does BRIN contribute to GEG development through the STI policy in Indonesia ?

1.3 Research Objectives

- I. Elaborate GTC's contribution to GEG development through the STI policy in South Korea.
- II. Identify and describe the key elements of South Korea's GTC practices that can be adopted by BRIN in Indonesia.
- III. Explore BRIN's contribution to GEG development through the STI policy in Indonesia.

1.4 Scope of Research

The scope of this research includes the STI policy think tank's contribution to supporting GEG development at the national level. In this regard, the role of the STI policy think tank, both in the GTC and BRIN, forms the main part of the discussion of this study. Ryszawska (2017) & Sulich (2021) revealed that a lack of financial resources and institutional capacity are obstacles both to implementing governmental policies on environmental issues and to developing green industries. Thus, the GTC's practices in South Korea were examined and analyzed to identify three key elements, namely resources, government strategy, and institutional governance, which are the focus of the present study and analysis.

In general terms, the STI think tank is an element of the overall public policy think tank. Its plays the role of a channel of knowledge from academicians to policy makers and society, incubating ideas concretely before delivering them to stakeholders (Mac Kenzie et al., 2015; Mc Gann, 2016). The principal functions of the STI policy think tank are as follows: i) to facilitate the establishment of hard and soft infrastructures in order to improve the innovation system, and to monitor the system, and ii) to organize and maintain multiple networks widely (funders, experts/academician, society, other partners) (Seo et al., 2021). These two main functions of the STI policy think tank are also included within the scope of this analysis.

2. Green Economy Growth and Policy Think Tanks

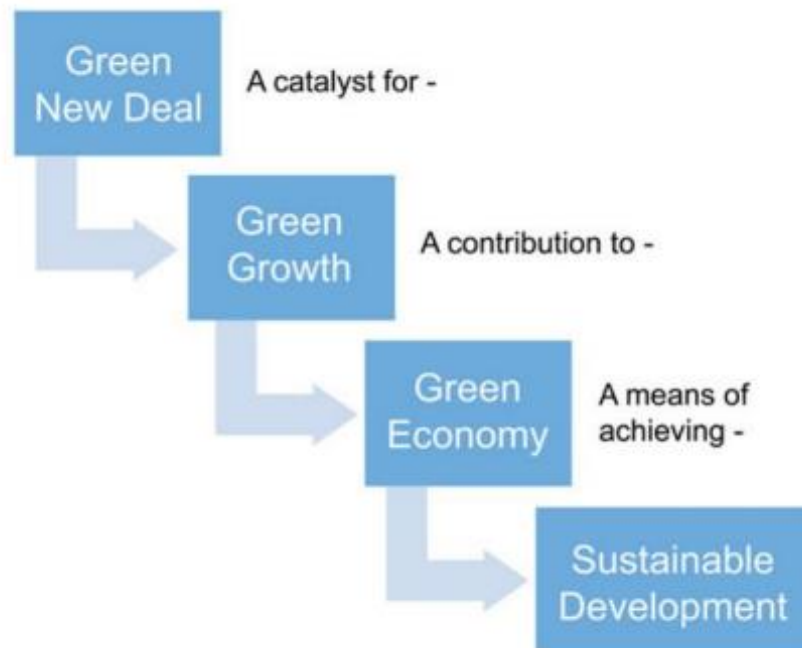
2.1 Green Economy Growth (GEG)

The Green Economy (GE) creates more economical systems and greener technologies, thereby reducing GHG emissions through better resource management. It has attracted worldwide attention due to changes in the global environment. During the late 1960s and early 1970s, certain environmental problems became prevalent in developed countries (Ivlev & Ivleva, 2018). The original idea of GE was proposed by David Pearce and his colleagues, who argued that the pricing system was the main cause of environmental variability in the allocation of natural resources. Social and environmental factors are rarely considered in pricing goods and services. In addition, interest in GE is driven by global and regional efforts to address economic, environmental, and social issues (Ryszawska, 2017; Sulich, 2021).

As such, the green economy should aim to minimize environmental damage and ecological scarcity while improving human well-being and social justice (UNEP, 2014). Therefore, the main goals of GE are to generate low-carbon, efficient and socially inclusive resources and to promote the transition to sustainability. Advocates of the GE believe its current business model is inefficient. GE is committed to improving the material well-being of a small portion of the world's population at the expense of the environment, as well as to mitigating widespread poverty, the effects of climate change, loss of biodiversity, and other serious problems. It helps the process of production, distribution, and consumption in the long run to promote human well-being without destroying the environment (Ivlev & Ivleva, 2018).

GE is not a singular term for green industry development. Green Growth (GG) is a general term used to promote economic growth and development. GG is defined as strong growth that takes into consideration the role of natural capital in natural disasters, environmental management, and the prevention of natural disasters, and which also minimizes pollution and other negative environmental impacts (World Bank, 2021).

[Figure 2-1] Hierarchy of Green Economy Concepts



Source: The Economics of Ecosystem and Biodiversity (TEEB), ten Brink et al. (2012).

GG's ultimate goal is to achieve a sustainable economy and improve people's lives. It focuses on the interactions and trade-offs between the economic and environmental pillars of sustainable development goals (SDGs). GG rejects the usual "grow first, clean later" paradigm. Instead, it aims to be more respectful of human well-being (social pillars) and to foster investment and innovation within an inclusive environment. Good governance, transparency, and equity are all required to achieve transformative growth (UNEP, 2014; OECD, 2011).

Figure 2.1 above shows how the Green New Deal (GND) is seen as an enabler and how GG is perceived as a series of governmental strategies or key tools to support the GE system. Although they are closely interrelated, they also have different scopes in terms of achieving SDGs. The transition to GE can be seen as an important means of achieving SDGs (ten Brink et al., 2012). Similar to the Rio+20 situation, GE partnerships have been established as a means of achieving SDGs and reducing broader poverty (UNDESA, 2012). The differences between the three concepts are as follows:

[Table 2-1] Definitions of the GE, GG, and GND Concepts

Concept	Source	Definition
Green Economy (GE)	UNEP (2010)	"...a system of economic activities related to the production, distribution, and consumption of goods and services that result in improved human well-being over the long term, while not exposing future generations to significant environmental risks or ecological scarcity."
Green Growth (GG)	OECD (2011)	"...aims to foster economic growth and development, while ensuring that natural assets are used sustainably, and continue to provide the resources and environmental services on which our well-being depends."
Green New Deal (GND)	UNEP (2009)	"...a set of globally coordinated, large-scale stimulus packages and policy measures that have the potential to bring about global economic recovery in the short term, while laying the foundations for sustained economic growth in the medium and long term."

Source: Author's elaboration

The use of the terms GE and GG differs slightly at the conceptual level. Both are related to the development of the green economy and human well-being in a country. The aim is to achieve sustainable economic development by conserving natural resources without damaging the environment. GE concerns a country's sustainable economic system, while GG focuses on how that country creates the GE system. GE and GG are governmental policies or strategies for leveraging the GE system in a country. Alternatively, GE is a system created with government policy support.

In practice, the use of the terms GE and GG is somewhat redundant. In this study the term GE is used in parallel with GG, and both are used synonymously with the term "green economic growth" (GEG). In fact, GEG is used in this study for three reasons. **First**, some studies use the three terms to imply a similar meaning, namely, ecological modernization, green industry development, sustainable economic growth, or economic sustainability (UNDESA, 2012; Global Green Growth Institute, 2014; Kasztelan, 2017).

Second, GE and GG are "dual loops." In the innovation policy literature, systems and policies are mutually supportive partners (Kuhlmann et al., 2010). GG can support the creation of a

GE system, and GG is also subject to GE system models. There is mutual interaction between GE and GG.

Third, GE and GG have similar goals in developing countries. Sustainability, climate resilience, and prosperity are the top three goals of GE and GG (UNEP, 2011, Lorek & Spangenberg, 2014). Kasztelan (2017) argues that GE and GG do not replace the concept of sustainable development, but both achieve SDGs. Similarly, Adamowicz (2022) and Lee & Woo (2020) suggest that GG became one of the pillars of sustainable development, and that GE became one of the three pillars of this concept, especially since the early 1970s. However, the two terms, GE and GG, share common elements. Their differences are fading, and they are often used interchangeably in many studies and literatures as well as in practice.

2.2 The Role of the Think Tank in the Policy Process

Although the origin of the term "think tank" is unknown, most reports (Rich, 2004) indicate that it was coined during World War II to refer to military research and development organizations. Think tanks are defined as "... organizations engaged in regular research and advocacy on any matter related to public policy." However, the use of think tanks, including in Asian countries, may reduce the importance of independent research. In modern democracies, they serve as a link between knowledge and power" (UNDP, 2003). Similarly, the Cambridge Dictionary (2022) defines a think tank as a group of experts brought together, usually by a government, to develop ideas and make recommendations for action on a specific subject.

The Anglo-American tradition views think tanks as relatively autonomous organizations with their own legal identity that conduct policy analysis independently of the government, political parties, and pressure groups. Think tanks are widely assumed to provide unbiased, well-researched advice and advocacy on complex social issues (Diane Stone, 2005). In general, think tanks generate accumulated ideas or thoughts that are inevitably disseminated to the targeted spheres, and more particularly to policy-relevant fields (Shaw, Russell, Parsons & Greenhalgh, 2015, McGann, 2018).

Think tanks are groups of professionals who research and analyze public policy challenges,

develop and evaluate potential solutions, including policy alternatives, and inform decision-makers, their staffs, and the public of the findings of their research, analyses, and assessments of potential solutions (Cheney, D.W., Heaton Jr., G.R., Hill, C.T. & Windhamet, P.H., 2016). In other words, think tanks are groups that conduct policy-oriented research, analysis, and interaction on domestic and global issues, all of which helps politicians and the general public to make well-informed decisions about public policy.

This study defines 'think tank' as a professional group within an independent and autonomous organization. Such groups are composed of academicians, experts, and practitioners whose function is to generate and incubate ideas before delivering them to policy makers. Those ideas are derived from academic and empirical discussion referring to research results, scientific discourse, experiences, and history. The goal is to reduce bias opinion among policy makers and strengthen evidence-based policy.

In the policy process, the role of think tanks is to serve as policy knowledge intermediaries, research centers, and idea incubators. Think tanks are popularly called "knowledge broker.". As knowledge brokers, think tanks connect academics, policy makers, and civil society. They also transform R&D results (theories, discourse) into policy recommendations, and demonstrate the power of entrepreneurial solutions in addressing public issues. Most importantly, think tanks should provide credible, simple, and relevant information to policy makers (CIPE, 2022).

This study focuses on the role of the think tank in the science-technology-innovation (STI) policy making process. However, the role of the think tank in STI policy-making is very limited. Seo et al (2021) discuss the role of the think tank in an open innovation system (OIS). According to Seo's study, a think tank can play a role in academic research, contractual research, concept promotion, and networking. Organizationally, a think tank is likely to affiliate with businesses, universities, government, and non-government organizations (NGOs). Lastly, it should be noted that NGOs are independent think tanks (no affiliation).

STI policy think tanks play a role in two areas: 1) facilitating the establishment of hard and soft infrastructures in order to improve the innovation system (IS), including monitoring overall performance of the system, and 2) organizing and maintaining various global-wide

networks, such as funders, experts/academicians, society, and other partners (Seo et al, 2021). Regarding the first role, think tanks are not only producers of ideas or policy advocacy groups, but also groups involved in formulating, implementing, and evaluating STI policy. Think tanks are prompted to facilitate innovation needs, guarantee the establishment of an OIS, and monitor the progress and impact of an OIS. As for the second one, think tanks are a hub both for policy makers who seek information and knowledge and for academicians who produce and provide information and knowledge. Think tanks are very important in that they can identify and provide the right information and knowledge, from among all the information and knowledge available in plenty of sources, to policy makers.

In the OIS, the role of think tanks could be expanded. Their role should not be limited to facilitating the establishment of innovation infrastructure and keeping with pluralistic networks; rather, a think tank could also assist in creating IS. As Seo et al. (2021:325) point out, “a new think tank serves as an intermediate in the ecosystem, assisting innovative players in establishing interactive linkages with partners in order to develop, communicate, and turn knowledge into economic values, all of which contribute to more mature innovation systems.” However, a think tank’s role in the STI policy making process is very broad, comprising completed phases ranging from agenda setting, formulation and implementation to monitoring and evaluation. A think tank moves from policy advocates to policy entrepreneurs.

3. Green Economy Growth Policies and the STI Policy Think Tank in Indonesia

3.1 Policies for Green Economy Growth (GEG) in Indonesia

As discussed previously, the terms GE, GG, and GEG are used inter-changeably. The Government of Indonesia's policies related to GE, GG, or GEG are classified as follows: 1) Indonesia's national long-term development plan (RPJPN) 2005-2025; 2) Indonesia's mid-to-long-term development plan (RPJMN) 2020-2024; 3) green growth programs; 4) the G20 development working group; 5) circular economy programs; and 6) the Green Economy Index. Each of these policies is elaborated in more detail below.

3.1.1 Indonesia's national long-term development plan (RPJPN) 2005-2025

Indonesia's national long-term development plan (RPJPN) 2005-2025 is a national policy that covers all aspects of national development including CEG over a period of twenty years. Under this policy, the Government of Indonesia has set up programmes for food resilience with sustainable agriculture, sustainable forestry management, energy efficiency, renewable energy usage, clean technology support, waste management, low carbon transportation management, and green infrastructure development.

3.1.2 Indonesia's national mid-term development plan (RPJMN) 2020-2024

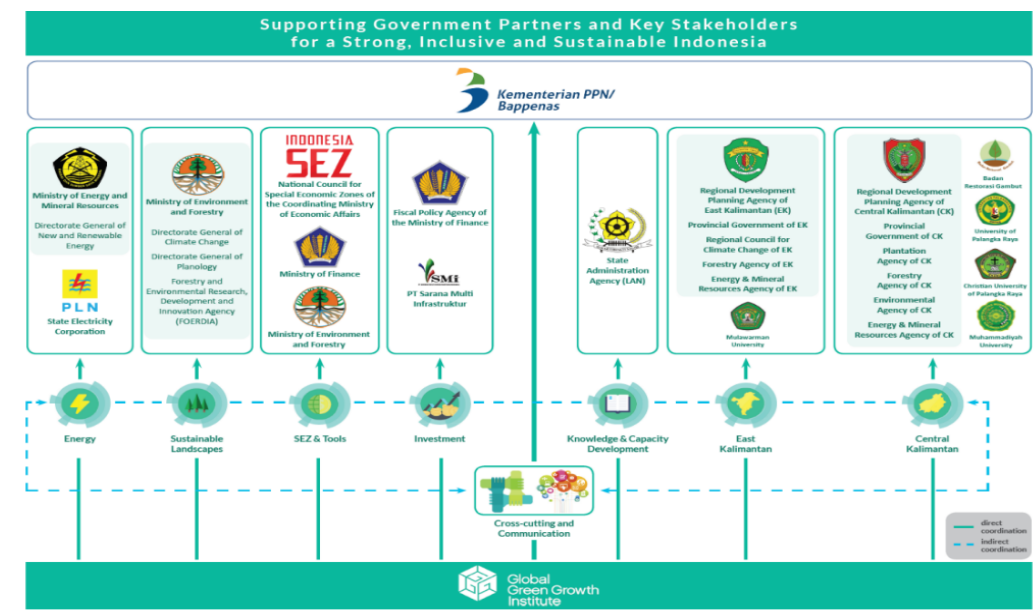
Indonesia's national mid-term development plan (RPJMN) 2019-2024 is a part of the RPJPN 2005-2025. Covering a period of five years, it is the end policy in the sequence of policies within the RPJPN 2005-2025. In RPJMN 2020-2024, GE is one of the policies and pillars of implementation deployed in Indonesia to attain the SDGs. In the Indonesian RPJMN 2020–2024, for instance, GE is a national priority agenda and a strategic priority project. There are arrangements for GE targets in the Indonesian RPJMN 2020-2024 which make GE a National Need Motivation and Vital Need Venture. In expansion, there is the Green Development Program started by the Worldwide Green Development Founded by the Global Green

Growth Institute (GGGI) through the Indonesian National Advancement Organization. It aims to realize many aspects of GE development at the local and national levels (Bappenas RI, 2019; Georgeson et al., 2017).

3.1.3 Green Growth Program

The Green Growth (GG) Program is designed to increase gross domestic product (GDP) and living standards sustainably. It aims to reduce pollution, build clean and resilient infrastructures, and use resources more efficiently. The final goal is to support the national economic development in such a way that it has positive impacts on human well-being (Everett, et al: 2010). Indonesia is projected to emerge as one of the top five world economies by 2050 (in terms of GDP Purchasing Power Parity) (Pricewaterhouse Coopers, 2017). The Government of Indonesia is implementing the Indonesian GG Program jointly with the Ministry of National Development Planning (BAPPENAS), the Global Green Growth Institute (GGGI), and a number of national and sub-national governments.

[Figure 2-2] The Green Growth Framework of Indonesia



Source: Bappenas (2017).

To realize GG in Indonesia, the GGGI and the Government of Indonesia have been working closely with a large policy portfolio and bankable project initiatives for the past five years.

With the support of the World Resources Institute-New Climate Economy (WRI-NCE) and the GGGI, Indonesia is now developing the sectoral implementation action plans to achieve its mitigation and adaptation goals and the aspirations set out in its Nationally Determined Contributions (NDC). Moreover, Indonesia is embarking on its low carbon development initiative, which explicitly integrates climate action into all development policy planning processes in order to establish an equitable and low-carbon Indonesian economy sumber (Bappenas, 2017).

Indonesia's Green Growth Framework (GGF) is a guide, rather than a blueprint or detailed plan. As such, the framework is intended to complement and support policy initiatives such as the Green Planning and Budgeting Strategy (GPB) launched by the Ministry of Finance, and the Ministry of Environment and Forestry's Sustainable Consumption and Production Efforts (Bappenas, 2017). The following four outcomes are expected from the GGF: 1) continuous economic growth; 2) inclusive and equitable growth; 3) social, economic, and environmental resilience for a healthy and productive service provider ecosystem; 4) reduction of greenhouse gas emissions (Bappenas-GGGI, 2015).

3.1.4 The G20 Development Working Group

Indonesia is a member of the G20 Development Working Group (DWG). The G20 provides recommendations for crafting an inclusive national GG policy framework. Recommendations are prepared by the co-facilitator of inclusive green growth (IGG) and international organizations such as the Asian Development Bank (ADB), the Organization for Economic Cooperation and Development (OECD), the United Nations (UN), and the World Bank. It presents a set of policy options to assist developing countries in identifying and addressing barriers and obstacles to their efforts to realize inclusive green growth. This policy is related to environmental, economic and social issues. Furthermore, investment, infrastructure, and technology-innovation are included in this policy (Harisman, 2012).

Several members of the G20 countries have created a framework for long-term GG development strategies. South Korea, for example, has devised a thorough GG strategy, while other countries have set up national GG development strategies which are positively

integrated into the fields of economics, sociology, and environment. Some governments have adopted environmental policies in such areas as climate change, resource productivity, and biodiversity to tackle green growth and achieve SDGs (OECD, The World Bank, The UN, 2012).

3.1.5 Circular Economy Programs

Circular economy (CE) is part of the GEG policy aimed at achieving SDGs (Giampietro, 2019). The Government of Indonesia focuses on five priorities of the CE program, i.e. food and beverages, textiles, construction, wholesale and retail trade, as well as equipment electronics. In 2019, those sectors contributed as much as one-third of Indonesia's GDP and also accounted for a large percentage of its employment rate, employing more than 43 million people (BAPPENAS, 2021a). CE is trusted and can provide significant benefits for Indonesia by 2030, when viewed from the perspective of the 3P approach: Profit (economy), which takes the form of additional GDP of Rp 593–638 trillion; Planet (environment), i.e. reducing waste by as much as 52% in the five potential sectors and reducing GHG emission by up to 126 million tons, which is equivalent to 9% of the current rate of emissions; and People (society), i.e. by creating 4.4 million new jobs of which 75% will be for women (BAPPENAS, 2021b).

[Table 2-2] Circular Economy Programs Initiated by the Government of Indonesia

Programs	Goal	Law
Green Industry Standard	A guide for companies to run efficient and environmentally- friendly production processes. From 2017–2021, the Green Industry Center certified 44 companies.	• National Act No.3/2014 concerning Industry. • Law of the Minister of Industry of the Republic of Indonesia No. 39/2018 concerning the Green Industry Certification Procedure. • Law of the Government of the Republic of Indonesia No. 28/2021 concerning the implementation of the Industrial Sector.
Waste Reduction Roadmap by Manufacturer	Prohibition of plastic disposables, including straws and plastic cutlery and containers made of	• Law of the Ministry of Environment and Forestry Regulation No. P.75/2019. • Law of the Ministry of Environment and Forestry No. P.75/MENLHK/SETJEN/KUM.01/

Programs	Goal	Law
	Styrofoam, and single-use plastic bags.	10/2019 concerning the Roadmap for Waste Reduction by Producers.
Green Building Policy	Certification of green buildings based on six assessment criteria (appropriate site development, energy efficiency and conservation, material resources and cycle, indoor health and comfort, water conservation, and building environment management).	The Ministry of Public Works and Public Housing has issued regulations related to the development of environmentally-friendly infrastructures (green buildings) in order to minimize the volume of carbon emissions generated by the construction sector. There are two interrelated regulations, namely, Ministerial Regulation (Permen) PUPR No. 9 of 2021, which concerns the Guidelines for the Implementation of Sustainable Construction, and Minister of Public Works and Public Housing No. 21 of 2021, which concerns Performance Assessments of Green Buildings.

Source: Bappenas (2021b)

Bappenas supervises CE policies from the upstream to the downstream process. This is the first CE business model to embrace many areas such as the use of renewable energy, the selection of biologically-based materials, or the design of products that can be recycled. This policy involves diverse actors including the government, the private sector, start-ups, state-owned enterprises, communities, and non-governmental organizations (NGOs). They are included in many fields such as textiles, transportation, construction, food and beverages, wholesale and retail, electronics and electricity, and waste management among others (BAPPENASa, 2021).

3.1.6 Green Economy Index (GEI)

Bappenas launched the Indonesian Green Economy Index (GEI) in August 2022, during a series of three meetings of the Development Working Group (DWG). The GEI provides Indonesia with a set of benchmark indicators for the development of measures for striking a sound balance between economic welfare and social equality in the wider community, as well as mitigating the risk of environmental damage. Indonesia's GEI provides an overview of Indonesia's achievements and scores in its transformation to the green economy (LCDI,

2022).

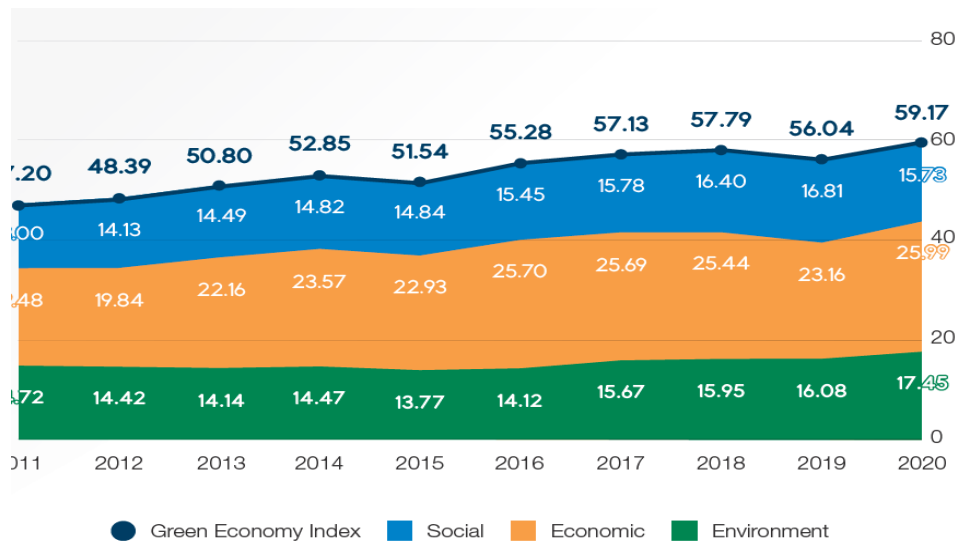
[Figure 2-3] The GEI's 15 Indicators



Source: BAPPENAS (2022).

The GEI has fifteen indicators that cover the links between the three pillars of sustainability, i.e. environment, social, and economy. Looking at the progress of the various indicators from 2011 to 2020, the economic indicators saw the greatest advance, with a larger area in 2020 than in 2011, especially final energy intensity, which obtained a score of 34 points in 2011, rising to 74 in 2020. As shown in Figure 7, four indicators obtained very high scores above 75: forest cover, waste management, industrial labor productivity, and life expectancy (BAPPENAS, 2022).

[Figure 2-4] Indonesia's GEI, 2011-2022



Source: BAPPENAS (2022).

Furthermore, the findings of Indonesia's GEI Report will be reflected in both the National Medium-Term Development Plan (RPJMN) 2025-2029 and the National Long-Term Development Plan (RPJPN) 2025-2045.

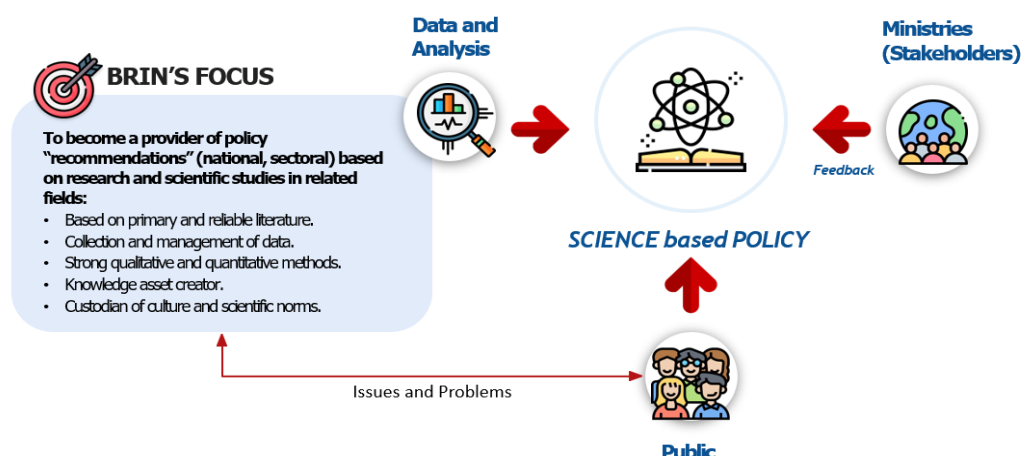
3.2 BRIN's Profile as an STI Policy Think Tank in Indonesia

The specific functions of BRIN (National Research and Innovation Agency) in the STI field are very broad. Aside from acting as an implementer of R&D, BRIN also functions as a policy maker and an evaluator. Its mandate is clearly stated in the Presidential Regulation No. 78/2021.

The Agency covers the whole range of R&D activities from upstream to downstream, starting from basic research, applied research, and innovation. Basic and applied R&D activities fall under the duties of twelve research organizations (ORs), while STI policies are the responsibility of seven deputies. BRIN's other function is to produce STI policies and deliver them to stakeholders (henceforth, it will be referred to as a think tank). As a think tank, BRIN identifies national issues and then translates them into policy recommendations for policymakers based on the main and trusted literature, data collection and management,

the use of strong qualitative and quantitative methods, creators of knowledge assets, and custodians of cultural and scientific norms.

[Figure 2-5] BRIN as a Policy Think Tank



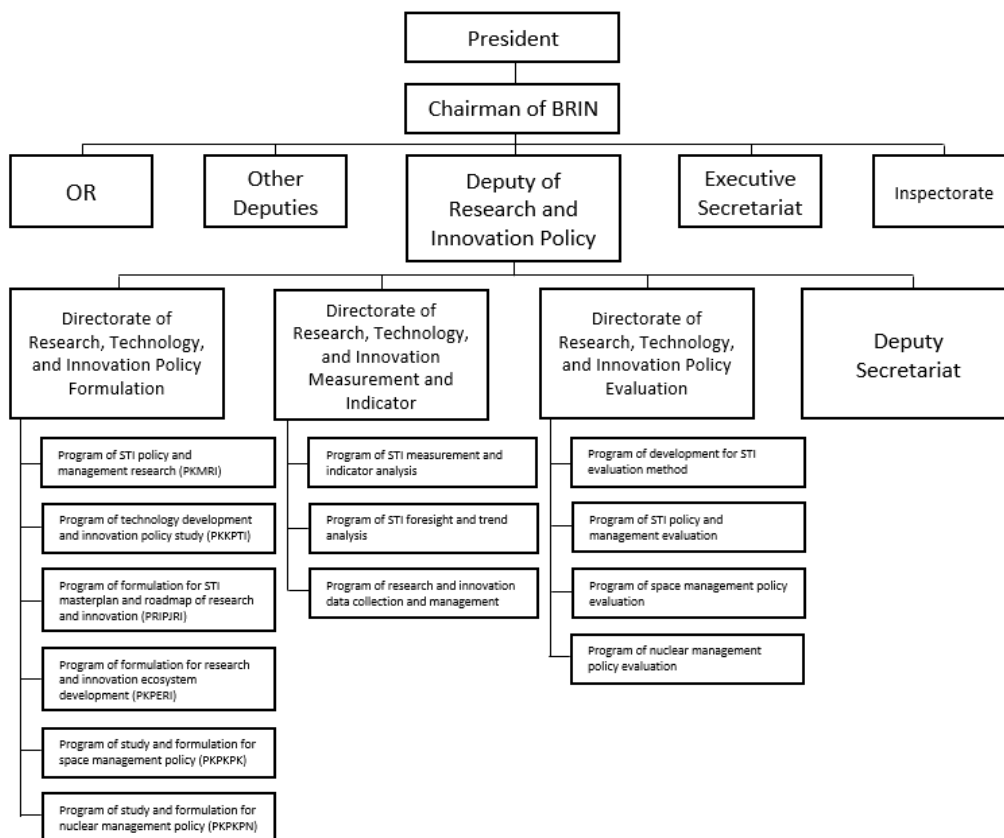
Source: BRIN (2022b).

BRIN has seven deputies under three different positions who serve as think tanks, namely the Deputy of Development Policy (DKP; for national development policies), the Deputy of Research and Innovation Policy (DKRI; for national STI policies), and the Deputy of Local Research and Innovation (DRIDA; for local STI policies). According to Presidential Law No. 78/2021 on BRIN, the DKRI has the following tasks: carrying out R&D on STI policy and management and formulating STI policies (master plan and roadmap for STI advancement). The DKRI has three directorates, each of which supports BRIN's role as an STI policy think tank. First, the Directorate of Research, Technology, and Innovation Policy Formulation (DPKRTI), whose duties consist in formulating national STI policies. Second, the Directorate of Research and Innovation Measurement and Indicator (DPIRTI), whose duties consist in measuring indicators, conducting foresight, and analyzing trends of research and innovation, as well as collecting and managing primary data of research and innovation. Lastly, the Directorate of Research, Technology, and Innovation Policy Evaluation (DEKRTI), whose remit is to evaluate research and innovation policies and develop methods of evaluating R&D.

The role of the STI policy think tank in the DKRI is at the DPKRTI due to its function as an STI policy formulator. The DPKRTI is expected to bridge STU knowledge to STI policy makers.

This directorate operates six programs (equivalent to Echelon 3) to produce STI policy studies and recommendations. The function of formulating STI policy is firmly attached to the three main programs: 1) PKKPTI focuses on formulating the draft versions of presidential regulations concerning Artificial Intelligence; 2) PKPERI focuses on formulating the drafts of government regulations concerning the implementation of S&T; and 3) PRIJPRI focuses on formulating the drafts of government regulations concerning the master plan for S&T advancement, and the draft of the presidential regulation on botanical gardens. Other functions include the following: 4) PKMRI focuses on supporting and supplying knowledge and evidence on STI policy and management. Therefore, the core function of the think tank at the DPKRTI was actually empowered by the PKMRI's support for evidence-based policies. The other two programs, i.e. 5) PKPKPK and 6) PKPKPN, focus on space and aeronautics and on nuclear policy, respectively, but PKPKPN's function does not work due to no activities inside.

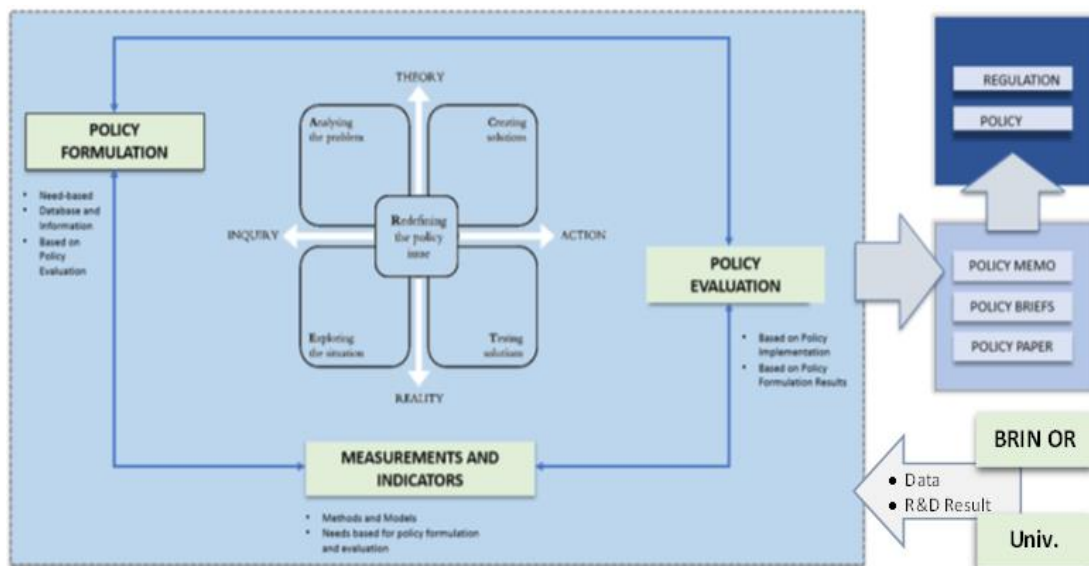
[Figure 2-6] Organizational Structure of the DKRI



Source: Modified from BRIN Regulation No 1/2021.

The STI policy does not exist in a vacuum as there is always an intersecting wedge between STI and other development policies (Triyono et al., 2020). In carrying out its tasks, the DPKRTI requires support from the other directorates, for example, primary data on research and innovation from the DPIRTI, or the results of evaluations of research and innovation policies from DEKRTI, as input materials for making STI policy recommendations. In addition, to obtain more comprehensive data and deeper knowledge in formulating STI policy, the DPKRTI needs to build connections with researchers from the DKP, research centers at the OR, and universities.

[Figure 2-7] Business Process of the DKRI



Source: Deputy of Research and Innovation Policy (2022), elaboration based on Andrews (2015), MindLab (2017), and Kumar (2012).

Transforming knowledge into policy is a complex phenomenon. The world's academics are considered too perfect and remote in terms of modelling reality; thus, the many recommendations they propose are ultimately considered not to be in reality (Banister, 2011; Daviter, 2015; Marble, 2006; Papanastasiou, 2019). On the other hand, the characteristics of policymaking as a political process require input that is rapid in terms of time, accurate in momentum, simple but content-specific, and applicable (Scartascini and Tommasi, 2012; Stoutenborough et al., 2015; Thapa, 2014). Many policy-oriented researchers then develop methodologies that seek to link theory to practice. The DKRI's

business process for the STI policy think tank was adopted from the design science research model (Andrews, 2015; MindLab, 2017; Kumar, 2012). This model, which is also known as “React”, starts off by redefining the policy issues and exploring the situation, analyzing the problems for the inquiry process, and creating solutions based on theory and scientific data. The model then continues by taking action, i.e. by re-exploring the situation, testing the solution and, finally, checking the reality after the solution has been implemented.

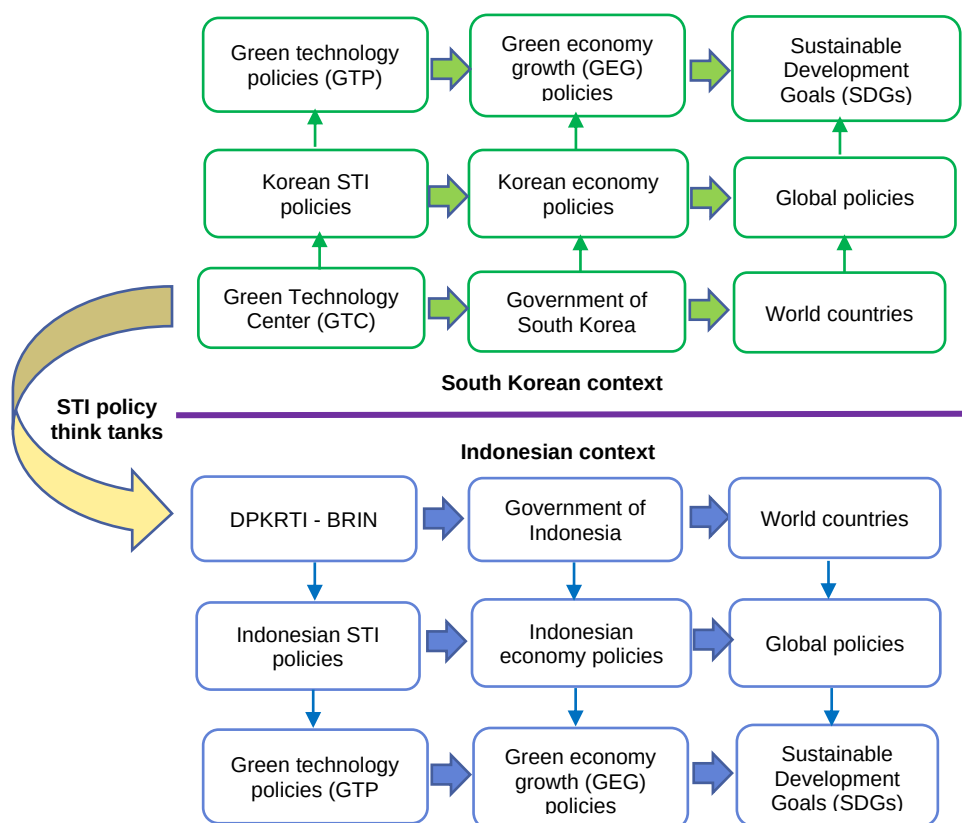
React at the DKRI is closely connected to six other deputies, an executive secretariat, twelve ORs in BRIN's internal environment. In addition, the DKRI strongly cooperates with ministries, local government agencies, universities, firms, non-governmental organizations (NGOs), and other partners within BRIN's external environment. It is strongly suggested that overseas networks should be included in the DKRI's business process in producing knowledge for policy-making and in disseminating policy to the related stakeholders.

4.

Analysis Framework and Methodology

4.1 Analysis Framework

[Figure 2-8] Analysis Framework



Source: Author's elaboration

This study concerns lesson learned from the practice of South Korea's Green Technology Center (GTC), which has made a significant contribution to economy policies in South Korea. As an STI policy think tank, the GTC focuses on producing and delivering green technology policies (GTP). GTP is a representative feature of Korea's STI and economic policies. The GTP policies are designed to support Korea's green economy growth (GEG) policies. In turn, GTP has contributed to achieving the country's sustainable development goals (SDGs), which are common goals shared by developed and developing countries alike. Each country implements a specific policy to achieve its SDGs, and Indonesia is one of the developing

countries that is expected to implement GTPs similar to those of South Korea. As such, this study proposes that the Directorate of Research, Technology and Innovation Policy Formulation (DPKRTI) – National Research and Innovation Agency (BRIN) should aim to become like the Green Technology Center.

4.2 Data Collection and Method

This study uses qualitative research to derive narrative explanations of real phenomena in various fields. Qualitative methods can be used to explore real and hidden facts that cannot be exposed using quantitative methods. The issues surrounding GEG and STI are the main topics of this research. This study aims to obtain comprehensive information in areas of GEG policy and STI policy by learning from South Korea's practices and experience. Thus, similar policies could be applied to the Indonesian context. In this respect, South Korea is represented by the GTC, while Indonesia is represented by the DPKRTI-BRIN. Furthermore, we identify the three main elements, i.e. resources, strategy, and institution governance.

This research used primary and secondary data collected between April and October of 2022. In the first phase, April-June 2022, data were collected from secondary data sources such as scientific books, scientific journals, scientific proceedings, government documents and reports, reliable mass media articles, working papers, and globally-published documents. In the second phase, July-October 2022, data were collected from primary and secondary data sources. The primary data sources consisted of in-depth interviews conducted via zoom (online) or in the form of direct, face-to-face dialogues (offline).

[Table 2-3] List of Key Informants

No.	Institution	Special Agency	Number	Representative
1	National Research and Innovation Agency of Indonesia (BRIN)	Directorate of Research, Technology, and Innovation Policy Evaluation (DEKRTI)	1	Academician
2	National Research and Innovation Agency of Indonesia (BRIN)	Directorate of Research, Technology, and Innovation Policy Formulation (DPKRTI)	1	Government

No.	Institution	Special Agency	Number	Representative
3	National Research and Innovation Agency of Indonesia (BRIN)	Nuclear Polytechnic of Yogyakarta	1	Academician
4	University of Brawijaya (UB)	Faculty of Administrative Science	1	Academician
5	University of Gadjah Mada (UGM)	Faculty of Economics and Business	3	Academician
6	University of Gadjah Mada (UGM)	Program for Master of Public Administration	1	Academician
7	Local Government of Malang Regency (Pemkab-Malang)	Research and Innovation Agency	3	Government
8	Center for Innovation and Policy Governance (CIPG)	Supervisor	1	NGOs
9	SMERU Institute	Researcher	1	NGOs
10	Indonesian Environment Organization (WALHI)	Staffs	2	NGOs
11	Waste-processing industry in Bandung	Manager	1	Business

Source: Author's elaboration

In Indonesia, we interviewed sixteen key informants from eleven different working units, and divided them into four representative groups: government (four informants), academician (eight informants), non-government organization (three informants), and business (one informant). They are represented of quadruple helix model contributing to the innovation system (IS). Additionally, we attended many conferences held by Bappenas, the Ministry of Industry, the Ministry of Environment and Forestry and the Ministry of Finance because these ministries are the main players in formulating and implementing GEG policy in Indonesia. The participation of researchers as the research instruments was indispensable to this study. In particular, BRIN's researchers and policy analysts served as research instruments in this study.

In this study, data triangulation, which is generally used to validate the findings and reduce the bias of researchers, was conducted in three ways: First, a search for green policy documents, especially from Bappenas, the Ministry of Industry and the Ministry of

Environment and Forestry, was conducted to reinforce and upgrade the latest information and policies launched by the ministries; second, literatures on STI policy, think tanks, the green economy (GE), and green growth (GG) were traced; and third, South Korean cases involving expert(s) related to the STI policy think tank in supporting GEG policy is necessary for in-depth research.

5. Green Growth Policy and System in South Korea

5.1 Korean Green Growth Strategy

In August 2008, the sixtieth anniversary of the establishment of the Republic of South Korea, the President of South Korea announced "Low Carbon Green Growth" as a new vision to guide the nation's growth for the next sixty years. The vision sets out South Korea's aim of becoming a global leader among low-carbon societies based on its green growth policy. Furthermore, it declares Korea's aim of rising to the rank of fifth "Green World Power" by 2050 and seventh by 2020. The government announced the Framework Act on Low Carbon Green Growth in 2009, and renewed it in 2010 and 2021 in an effort to meet its lofty objectives. The government also established the presidential committee for sustainable development in 2007, and provincial committees in 2021.

The milestones of the National Green Growth Strategy, which primarily consists of three objectives and ten policy directions, are defined in the legal instrument. Given that 96 percent of South Korea's principal energy sources are imported, improved energy independence and climate change mitigation constitute the first goal. Meanwhile, the development of green technology and the promotion of the industry's green structure are elements of the second goal, which is to produce new catalysts for economic growth. Lastly, the third goal is to promote a sustainable approach to living, green transportation infrastructure, and the use of land and water.

The South Korean government originally unveiled the Five-Year Action Plan 2010-2015 to promote technological development and assist the implementation of the Green Growth Strategy.

Each year South Korea invests roughly 2% of its GDP in R&D, in addition to the range of green development initiatives outlined in the plan. The government created a plan for the implementation of the K-SDGs (Korean Sustainable Development Goals) in 2018, which included 119 specific goals and 236 indicators. As shown in the table below, several avenues for communication and feedback, such as public forums for transmission, the sharing and

promotion of visions, and active cooperation with local governments, would be established to support the execution of the K-SDGs.

[Table 2-4] Main Indicators of K-SDGs

Area	Main Indicators	2030 Targets Compared to 2017
Society	• Income gap between the upper and lower classes	36.8%* → 31.0%
	• Elderly poverty rate	65.9% → 85.5%
	• Suicide rate per 100,000 people	46.5% → 31.0%
	• Occupational fatality rate per 10,000 people	24.3 → 11.9
	• Utilization rate of public kindergartens	0.52 → 0.22
	• Ratio of households below minimum housing standards	24.0% → 44.0%
	• Ratio of households below minimum housing standards	5.9% → 4.6%
Environment	• Restoration rate of major endangered species	74.3% → 90.0%
	• Area of restored tidal flats (km ²)	0.2 → 6.0
	• Ratio of certified organic farming areas	4.9% → 10%
Economy	• R&D expenditure relative to GDP	4.23%** → 4.29%
	• Rate of successfully commercialized R&D projects	51.6% → 52.9%
	• Recycling rate of industrial waste	75.8%** → 95.4%
	• Number of eco-friendly cars	97,000 → 8.8 million

Source: MOE (2020).

Korea's green growth policy is being promoted in connection with sectoral plans such as climate change, new and renewable energy, sustainable development, and urban and regional development based on the National Green Growth Strategy and the Five-Year Action Plan. Each ministry promotes policy measures for green growth. For example, the Ministry of Environment, Ministry of Trade, Industry and Energy, Ministry of Oceans and Fisheries, and Ministry of Science and ICT are all involved in promoting the development of green technologies.

5.2 Korean Think Tank for Green Growth: Green Technology Center

The Paris Agreement specifies that each party shall increase its cooperation on technology development and transfer, with "technology" in particular being emphasized as the key to overcoming climate change. As a research organization for green and climate technology policy in Korea, the Green Technology Center has focused on establishing its basis and expanding its functions since its foundation in 2013.

By promoting national climate change technology plans and actively engaging in international cooperation projects to support domestic institutions and small and medium-sized businesses, it aims to become one of the leading institutions in the area of green and climate technology both domestically and internationally. In accordance with its goals, the Green Technology Center now plans to emerge as a think tank for climate-change response technologies and policies that can drive inclusive and innovative growth on a global scale.

The Green Technology Center actively supports international technology negotiations as well as tailored cooperation based on its experience in climate technology cooperation and multinational and multi-field networks, drawing on the knowledge it has accumulated in establishing the national policy on green technology R&D and its role as an information analysis institution.

The figure below explains the vision and role of the GTC. The vision focuses on think-tank identity in the field of climate change, while the mission addresses green climate technology for the domestic climate industry as well as global efforts to mitigate climate change. To that end, the mission is dependent on the following two roles: formulating policies and establishing strategies, as shown in the figure below.

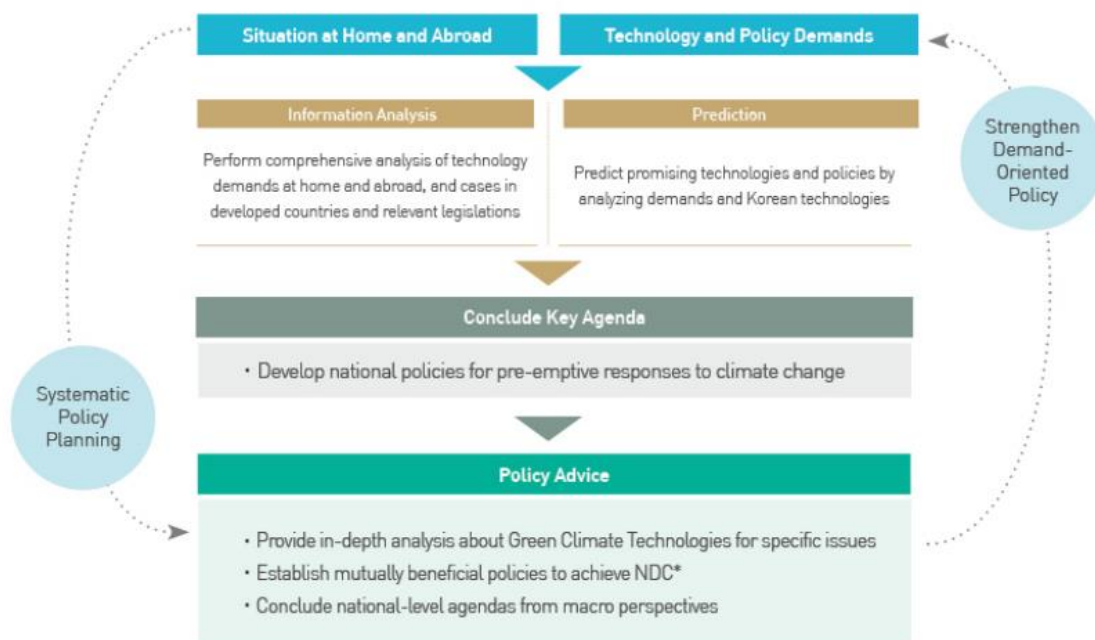
[Figure 2-9] The Vision-Goal System of the GTC



Source: GTC website (www.gtck.re.kr).

The roles of the GTC are divided into four areas: research on green climate technology policies; production of information on green climate technology; creation of cooperative strategies; and technology cooperation. First, as shown in the figure below, regarding green growth policy, the GTC advises the Korean government on ways to develop the most practical policies for promoting green climate technology. To do this, on the one hand, the Center performs comprehensive analyses of technology demands at home and abroad, and success cases in developed countries and the relevant legislation. On the other hand, it tries to predict promising technologies and policies by analyzing demand and Korean technologies. Based on these analyses, it puts forward key items of agenda related to the development of national policies and measures for a pre-emptive response to climate change. Moreover, the policy contains three elements: in-depth analysis of green climate technologies for specific issues; establishment of mutually beneficial policies to achieve NDC (Nationally Determined Contributions); and proposals of national-level items of agenda from macro perspectives.

[Figure 2-10] The First Research Area of the GTC



Source: GTC website (www.gtck.re.kr).

Second, in terms of the information system on green climate technology, the GTC creates customized information according to the policy demands of customers at home and abroad. The GTC also provides reliable information via a robust delivery system that includes comprehensive data analysis. In order to understand the customers' demands, it carries out the three stages of the customization process as follows: First, the production of databases, including statistics on the climate technology industry, climate technology classification, and the 'green climate technologies (from surveys on current status). Second, in order to process information, the GTC provides an overview of hub countries for technology transfer as well as the technologies, markets, and policy developments of major countries. Third, with regard to creation of the platform named CTiS (Climate Technology Information Systems), the GTC analyzes trends in policy, technology, and industrial developments. The CTiS includes an online/offline supply-demand matching system and updated issues at international organizations.

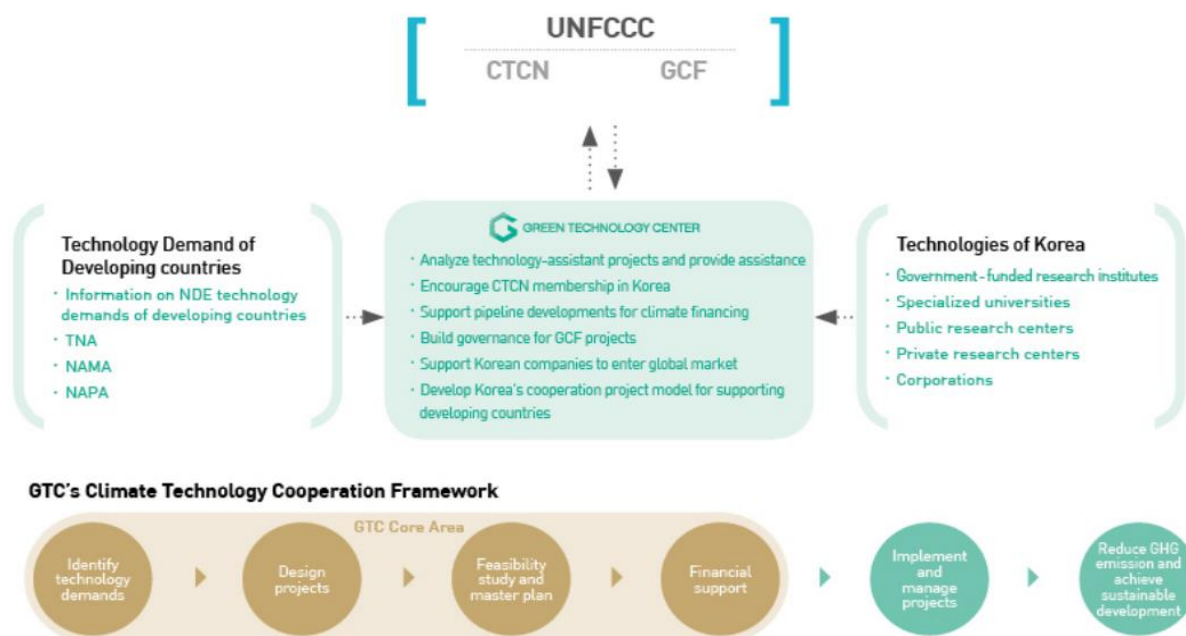
[Figure 2-11] The Second Research Area of the GTC



Source: GTC website (www.gtck.re.kr).

Third, the GTC accelerates the transfer of Korea's climate technology to developing countries by establishing a matching process that includes the full cycle of demand analysis financing and project implementation. In order to do this, it identifies technology demand in developing countries based on information on the NDE (National Designated Entity) technology demands of developing countries, as well as Korean technologies from government-funded research institutes, specialized universities, public research centers, private research centers, and corporations. After identifying and examining these parties, the GTC analyzes technology-assistant projects and provides assistance, encourages CTCN (Climate Technology Center and Network) membership in Korea, supports pipeline developments for climate financing, builds governance for GCF projects, supports Korean companies' entry to enter global markets, and develops a Korean cooperation project model for supporting developing countries.

[Figure 2-12] The Third Research Area of the GTC

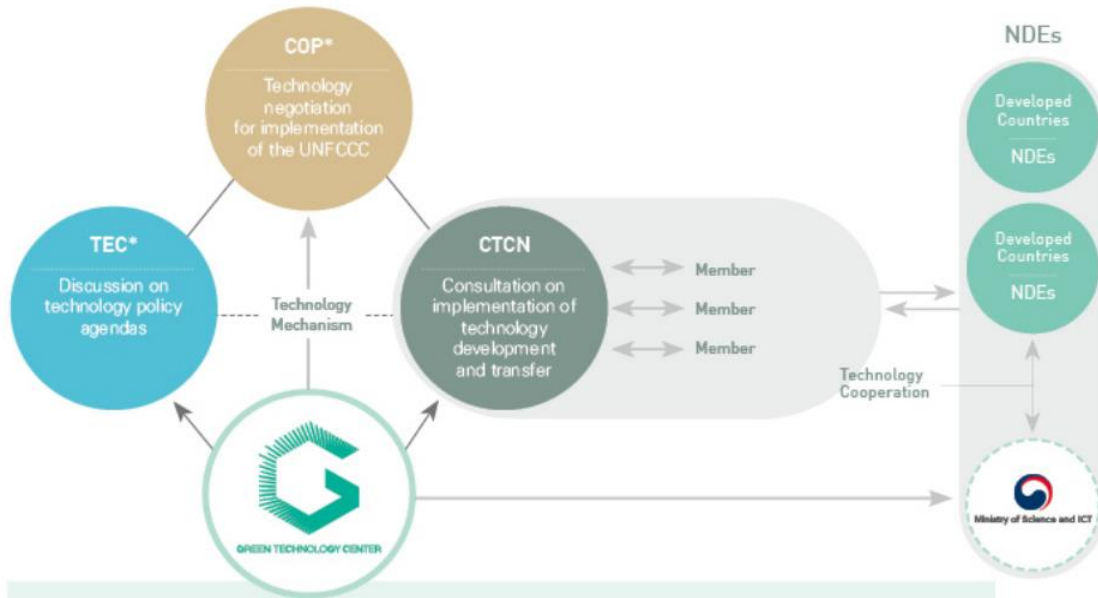


Source: GTC website (www.gtck.re.kr).

Finally, the GTC explores research methodologies for accelerating global cooperation through the technology mechanism of the UNFCCC (United Nations Framework Convention on Climate Change). It also suggests strategies for cooperation on climate technology as a designated institution under the Ministry of Science and ICT, the Korean NDE assigned by the UNFCCC. To this end, the GTC analyzes international negotiation trends and policies of the technology mechanism of the UNFCCC, publishes reports on the agenda and policy analysis of the TEC (Technology Executive Committee) and CTCN board meetings, and establishes a network-based support system (e.g. an interdepartmental committee) for the MSIT.

While fulfilling these roles, the GTC received an achievement award from CTCN in recognition of its efforts to lead the revitalization of climate technical cooperation, such as obtaining the first and largest orders of technical support projects for CTCN. Moreover, the GTC is involved in efforts to tackle social problem and address local development issues, while supporting sustainable development by opening the Green Technology Partnership Initiative (GTPI) in Indonesia.

[Figure 2-13] The Fourth Research Area of the GTC



Source: GTC website (www.gtck.re.kr).

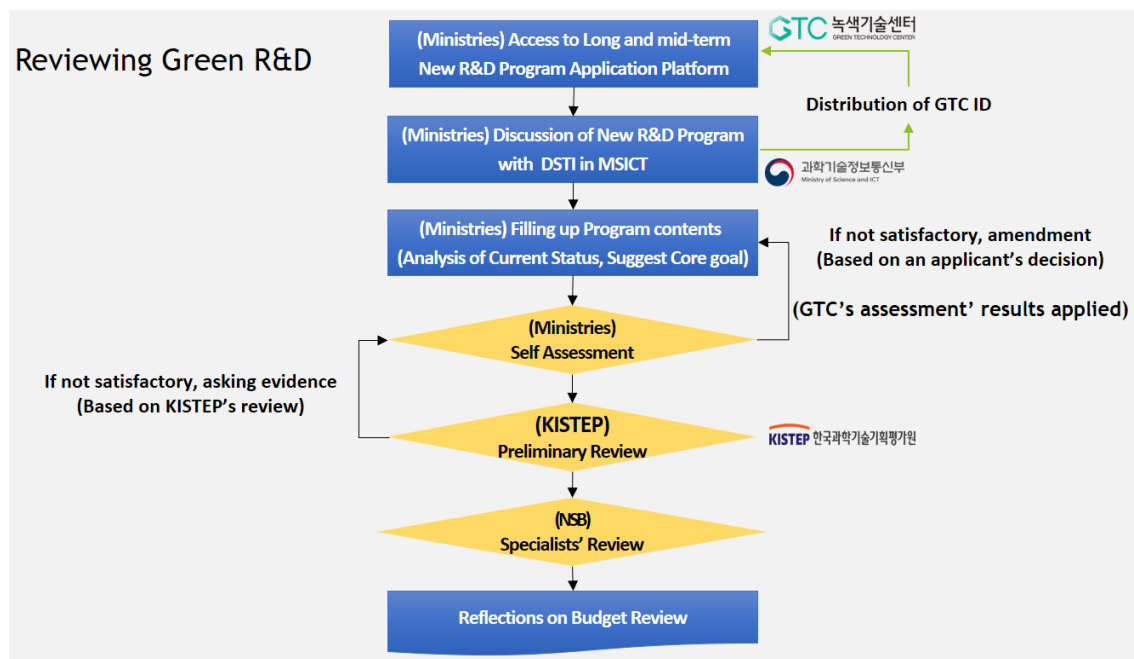
5.3 The GTC's System for Managing Green Growth Technology

The review process for green R&D has been designed by the GTC, as shown in the figure below. First, each ministry accesses the platform and uploads new R&D program applications. Regarding its technological and economic validity, this program is discussed with personnel of the Division of Science and Technology Innovation (DSTI) under the Ministry of Science and ICT (MSIT). The GTC is particularly involved in small and medium-sized green R&D program applications planned by various ministries. Second, based on the interaction between the GTC and various ministries during the previous stage, the program contents are filled by the ministries in the platform. The contents include such matters as the current status of technologies applied, R&D goals, performance indicators, and so on.

Third, each ministry conducts an internal assessment of the details of its own programs, and if the results of the assessment are unsatisfactory, it changes and revises the contents. In this process, the GTC provides specialized advice on such matters as whether there are any duplications with other green R&D programs or whether the programs are technologically feasible and economically effective.

Fourth, after conducting an official preliminary review of the program, the Korea Institute of Science and Technology Evaluation & Planning (KISTEP) passes the revised program applications on to the National Science & Technology Board (NSTB), which is staffed by R&D specialists among others. Ultimately, the program applications submitted by various ministries are passed on to the regular budget deliberation process of the National Assembly.

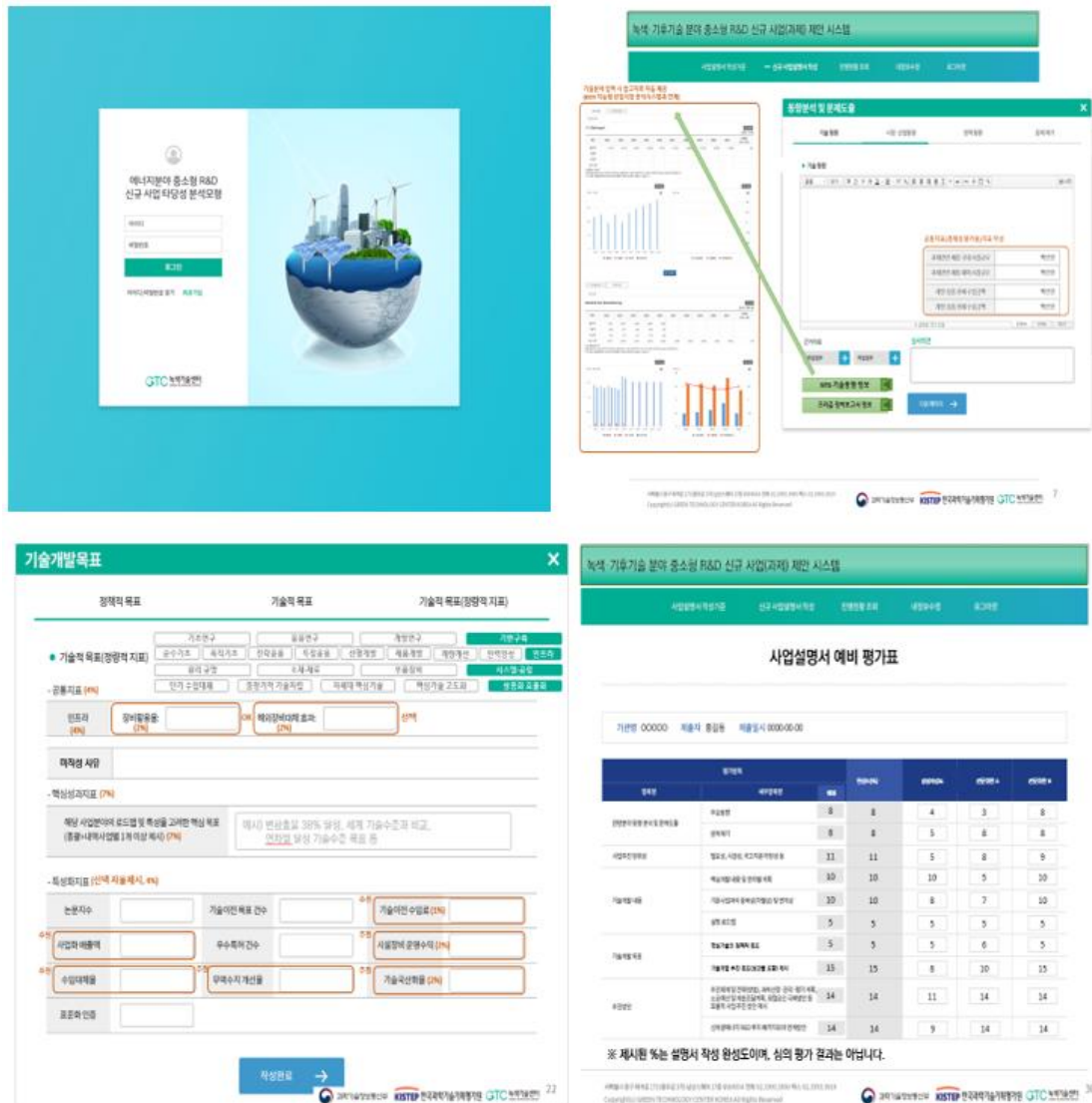
[Figure 2-14] Review Process of Green R&D



Source: Author's elaboration

The website interfaces of the system for evaluating green R&D programs are shown in the four figures below. First, the top-left figure shows the log-in page for submitting R&D programs to the platform system, for which each ministry receives a GTC ID to gain access to the platform. Second, the top-right image shows the website with the present status of the chosen technology, market needs, and general indications for R&D targets filled in. Third, the bottom-left image has to be updated with more detailed target information based on the core and designated indicators. Finally, the bottom-right figure shows the results of the preliminary evaluation.

[Figure 2-15] On-line System Interface for R&D Program Applications



Source: GTC website (www.gtck.re.kr).

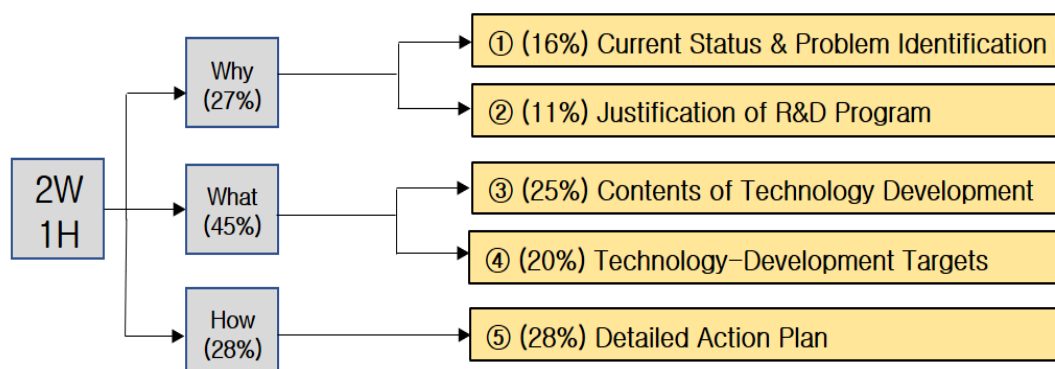
Before the process commences, as mentioned above, each ministry is requested to fill out the R&D plan it is supposed to execute, as shown in the bottom-left part of the above figure, and each one has to propose 2W1H (i.e. two “why” and one “how” questions), as shown in the figure below. Regarding the "why" question, which makes up to 27% of the total weight, the program is supposed to contain the ‘current status’, ‘identification of problems’, and

‘justification of the program’. Current status includes quantitative data on technology (e.g. patent), market, and industry, as well as government policies. The rationale of problem identification is supported by the negative effect or impact that occurs if the program suggested is not implemented. The justification of the program is based on the urgency of and need for governmental intervention.

The "what" question, which accounts for 45% of the total weight, refers to the goals and substance of technical development. The primary technical description, the development schedule, the novelty of the technology, and the commercialization strategy are the components of the necessary technology development. Technological goals can be understood in terms of their relationship to governmental policy and specific indicators, such as the general, core, and specified indicators which are covered in more depth below.

Detailed action plans can provide a response to the "how" question, which might weigh up to 28%. The actors and their responsibilities in terms of implementation, selection and evaluation of applicants, financial strategies, and risk management are all covered in these plans. Furthermore, the connection to significant public expenditure on R&D needs to be explained.

[Figure 2-16] Main Contents of the Green R&D Plan Requested on the Platform

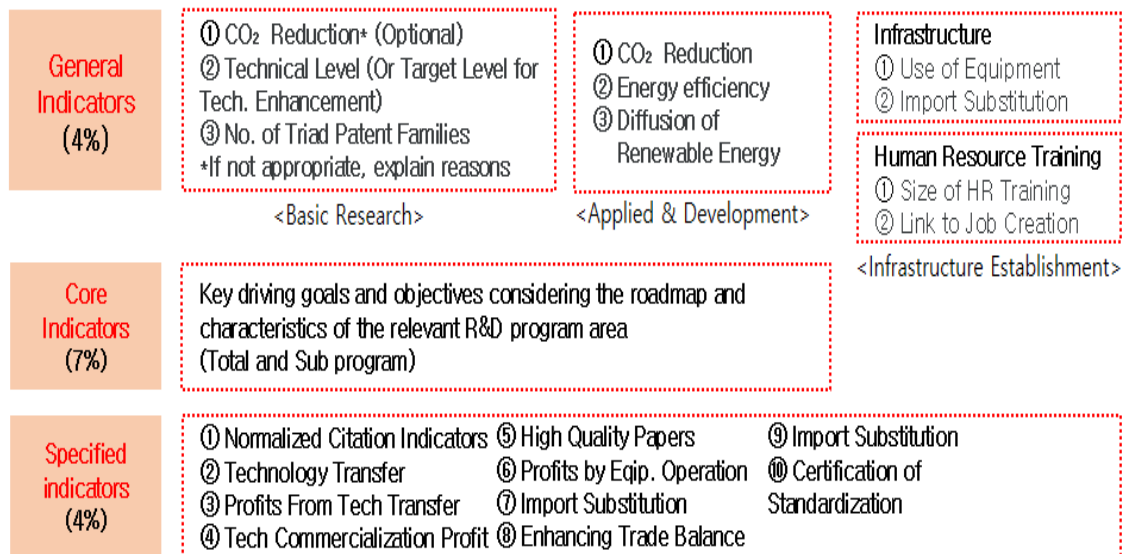


Source: Author's elaboration

When ministries describe the technological aims, as shown in the figure below, three categories of performance indicators are introduced, including general, core, and specific indicators. First, the general indicators are selected based on the traits of the R&D programs,

including basic research, applied and in development, and infrastructure development. For instance, if the program belongs to the fundamental research area, we recommend using metrics such as the number of triad patent families, the intended technical level, and carbon dioxide reduction. Second, the core indicators include both overall and individual sub-program driving aims and objectives that take into account the roadmap and the characteristics of the relevant R&D program area. Third, while the first two indicators are mandatory, the applicants themselves choose or construct the specified indicators. In other words, the applicants must select at least two indicators, such as the number of technologies, nominalized citation indicators, etc. If the existing indicators do not match the requirements of the program being used, alternative indicators might be developed and suggested in line with the programs' context.

[Figure 2-17] Performance Indicators for Technology Targets



Source: Author's elaboration

6. BRIN's Role in Green Economy Growth in Indonesia

The National Research and Innovation Agency (BRIN), Indonesia's largest R&D institution, is a public R&D agency that plays a major role in creating and supporting the National Innovation System (NIS). As mentioned previously, BRIN has seven deputies and twelve research organizations (OR). Each deputy oversees a number of directorates whose role is to support the main tasks of the deputy chairman in formulating and implementing policy in the STI fields. Also, each OR has research centers that support the main tasks of its head in performing research and development (R&D) activities according to specific research areas. Both basic and applied R&D on green technology and green economy growth (GEG) are included within each OR's structure.

This study focuses on the Directorate for Policy Formulation on Research, Technology, and Innovation (DPKRTI) under the Deputy of Research and Innovation Policy (DKRI) at BRIN. The DPKRTI within the BRIN structure (hereinafter, DPKRTI-BRIN) has the dual functions of STI policy formulator and STI policy-management knowledge producer, which it must carry out simultaneously. Unlike this directorate, the other deputies and ORs within the BRIN structure do not perform this role. Conventionally, the DPKRTI-BRIN plays the role of a STI policy think tank in that it focuses on formulating STI policy based on the results of R&D on STI policy management. Therefore, it is important to enhance the role of the DPKRTI-BRIN as a modern STI policy think tank in order to respond to the global climate change issue.

6.1 Organizational Characteristics of the GTC and the DPKRTI-BRIN

The GTC's practices can serve as lessons learned for the DPKRTI-BRIN. Both entities may be characterized as public organizations that function as policy think tanks. It is necessary to examine the organizational characteristics of the GTC and the DPKRTI-BRIN in order to identify significant differences and shared similarities. The similarities and differences between two or more organizations constitute basic information required to study and analyze best practices.

[Table 2-5] Generic Differences between the GTC and the DPKRTI-BRIN

	GTC	DPKRTI-BRIN
Country category	Developed countries, Newly Industrializing Economies (NIEs), and OECD member states	Developing countries, Non-NIEs, and non-OECD members
Headquarters	Seoul, South Korea	Jakarta, Indonesia
Foundation date	2013	2021
Purpose of establishment	To respond to global and national climate change policies (The Paris Agreement).	To address weak coordination in the STI policy formulation process. (It is difficult to establish an NIS.)
Ownership status	A government affiliated research institute under the Ministry of Science, ICT and Future Planning (MSIP)	Governmental agencies (Echelon II under the BRIN structure)
Government level	National level	National level
Vision	A leading think tank for inclusive and innovative global growth in response to climate change	The DPKRTI's vision is formally attached to BRIN's vision: To realize a reliable, professional, innovative BRIN that has integrity in serving the President and Vice President, and to realize the President's Vision and Mission of "An advanced Indonesia that is sovereign, independent, and has personality based on mutual cooperation".
Mission	To promote the advancement of the national climate industry and contribute to the global effort to respond to climate change by establishing green climate technology policies and assisting domestic and international cooperation on the development of climate technologies.	DPKRTI's mission is formally attached to the BRIN's mission: To provide technical and administrative support and the results of fast, accurate, and responsive analyses to the President and Vice President in conducting research, development, studies, and applications, as well as inventions and innovations; to organize integrated nuclear energy and outer space operations nationally; and to monitor control and evaluation of BRIDA's performance of its duties and functions. To improve the quality of human resources and the infrastructure for research and innovation in the

	GTC	DPKRTI-BRIN
		organization of integrated nuclear energy and outer space operations nationally; and to foster the performance of BRIDA's duties and functions. To organize effective and efficient services in the fields of supervision, general administration, information, and institutional relations.
Type of organization	Research and academic institution	Policy formulation and research institution
Role	To conduct research on national green climate technology policies in order to realize a sustainable society comprising: 1) research on future-oriented green climate technology policies, 2) production of integrated green climate technology information. To formulate research strategies for leading global climate technology diffusion comprising: a) research on a full cycle-model for climate technology diffusion linked with the technology-finance-carbon market system; b) assistance with the construction of global climate technology cooperation systems; c) research on climate technology integration strategies for localization	To prepare and formulate policies in the field of research, technology, and innovation, which includes the master plan for S&T advancement, as well as the research and innovation roadmap. To research and formulate nuclear policy and space policy through research and innovation.
Research areas	1) Green climate technology policy 2) Production of information on green climate technology 3) Creation of cooperative strategy 4) Technology cooperation	1) STI policy and management 2) Formulation of research, technology and innovation policy 3) Research and formulation of space management policy 4) Research and formulation of nuclear management policy
Focus of yearly activities	Capacity building, collaboration on innovation, knowledge management, and policy and planning	Capacity building, collaboration on innovation, knowledge management, and policy and planning

**Chapter 2. Designing the STI Policy Think Tank for Green Growth Policy Formulation:
Lessons Learned from South Korea's Practices and Experience**

	GTC	DPKRTI-BRIN
Enablers	Capacity building and training, communication and awareness, governance and planning	Capacity building and training, communication and awareness, governance and planning
Expertise	Adaptation (early warning and environmental assessment, marine and fisheries, and water) Mitigation (energy efficiency, renewable energy, and waste management) 3. Disaster risk reduction (cross-cutting issues)	STI plans and policies STI management Research, technology, and innovation ecosystems Space management policies Nuclear management policies
Division of research	1) Division of technical supervision 2) Division of policy research 3) Division of climate change cooperation 4) Division of national climate technology cooperation center	-Program for policy and management of research and innovation. -Program for technology development and research on innovation policies. -Program for formulation of the master plan and roadmap for research and innovation. Program for formulation for research and innovation ecosystems development -Program for research on and formulation of space management policies. -Program for research on and formulation of nuclear management policies.
Relationship with the United Nations	Member of United Nations (UN) on Climate Technology Center & Network (CTCN)	The UN-CTCN network in the Director General of Climate Change – Indonesian Ministry of Environment and Forestry (not in the DPKRTI-BRIN)
Global networks	Twenty-two organizations in such areas as green economy, environment, socio, economy, investment, finance research, technology, and education	One organization in STI policy area
National networks	Thirty-two organizations in such areas as green economy, environment, socio-community, economy, investment, finance research, technology, education, business, infrastructure, energy and mineral resources	Multiple private and public organizations (but not specifically identified)

	GTC	DPKRTI-BRIN
Working areas	Identification of technology demands, project design, feasibility studies, project financing	Redefining of policy issues related to STI, exploration of situations, problem analysis, creation of solutions through the preparation of policy recommendations, formulation of policies to solve problems or encourage the strengthening of STI policies
Outputs	Publications and policy papers	Publications and policy papers
No. of personnel	80	70

Source: Author's elaboration

Table 2.5 above shows that the GTC and the DDPKRTI-BRIN share some similar features, although most of them are totally different in many ways. As state-owned think tank institutions at the national level, each conducts R&D activities in the policy field, produces policy papers and publications, runs internal education and training programs for human resources, and plans and governs daily activities based on governmental laws. Besides these similarities, however, there are important differences in their purpose of establishment, vision and mission, main roles, research areas, expertise, division of research, relationship with the UN, global and national networks, working areas, and number of personnel.

As history and country status, the GTC goes ahead of the DDPKRTI-BRIN. The GTC is more senior and is supported by an advanced economy ecosystem, while the DDPKRTI-BRIN is newly formed (as of the end of 2021) within a developing economy ecosystem. The GTC has global and national networks widely dispersed around the world encompassing universities, firms, government agencies, and communities, while the DDPKRTI-BRIN has one global organization and multiple national organizations that are not specifically identified. Strikingly, the GTC has a specific focus on research fields and expertise related to green technology, green policy, and climate change management. All its research areas are related to STI policy and its cooperation on sustaining green economy growth (GEG) in Korea and overseas.

The purpose of establishment of the GTC is very clear and specific: to respond to global climate change policy. The GTC's vision and mission is to become a leading policy think tank that provides information and recommends policies concerning green technology to the Government of South Korea or whoever needs such information, whether in South Korea or overseas. In turn, it impacts specific research and cooperation divisions related to green technology and climate change policy.

Unlike the GTC, the DPKRTI-BRIN was established due to the weak coordination in formulating STI policy in Indonesia, which made it difficult to establish the NIS. The DPKRTI-BRIN conducts a wide range of STI research, though not specifically on green technology. Its research topics include the formulation of STI policy, management of STI talents and human resources, the dynamics of R&D institutions, aeronautics policy, renewable energy, research on major plans, and innovation ecosystems.

6.2 Incorporating Key Elements of the GTC into the DPKRTI-BRIN

The section above titled "Scope of research" examines the practices of South Korea's GTC and identifies three key elements, namely, resources, strategy, and institutional governance, which can be adopted by the DPKRTI-BRIN in Indonesia. As with developing countries, Ryszawska (2017) & Sulich (2021) revealed that a lack of financial resources and institutional capacity are the main elements for determining governmental strategies when developing GEG policies for those countries. Practically speaking, Indonesia has similar issues related to the development of national GEG policies.

Responding to the problem of developing GEG policies in developing countries, Ansell and Gash (2007), Wahab (2008), Anderson (2011), Shafritz et al. (2013) all stated that government strategies, existing resources, and institutional governance are indispensable elements for public organizations engaged in the policy-making process. As such, these elements should be present when applying the GTC's practices to the DPKRTI-BRIN.

However, incorporating key elements from the GTC into the DPKRTI-BRIN is not simple in that they cannot be transmitted fully and directly from the GTC to the DPKRTI-BRIN. Indeed, such a process requires adjustments, or perhaps incubation, before being carried out. The

political and governmental systems of South Korea, a developed country, are totally different for those of Indonesia, a developing country. They can influence how a public think tank works and sustain. Besides, history, values and culture are factors that influence the modus operandi of think tanks in both countries.

The GTC of South Korea and the DPKRTI-BRIN in Indonesia are both cases of public R&D institutions that provide STI policies based on the results of R&D. The GTC can provide practical lessons for the DPKRTI-BRIN. For instance, establishing a world-leading think tank requires relatively abundant resources, specific and long-term policies or strategies, and appropriate institutional governance. These elements are elaborated upon in the following section.

6.2.1 Government Strategy

Government strategy is part of the detailed policy of a government (Wahab, 2008). The use of a government strategy is similar to that of a public policy. The emergence of the GTC was very much influenced by the long-term policy of the South Korean government in responding to the Paris Agreement in the global arena. In 2009, the president of South Korea announced “Act of Low Carbon Green Growth” as a long-term policy goal, through which Korea aimed to become the fifth “global green powerhouse” by 2050. This is the general green growth (GG) policy that prevails in the provinces of South Korea. The National Green Growth, which consists of the three objectives and ten policy directions defined in the legal instrument, aim to promote sustainable industrial growth in diverse sectors with a positive impact on human well-being.

The long-term GG policy is translated into the medium-term GG policy, which is called the Five-Year Action Plan, to accelerate the implementation of the GG policy. Each action plan consists of instruments, achievements, and indicators. The government of Korea assumes that the GG policy is not separated from policies for supporting green industry. The GTC was established by the government to respond to two issues simultaneously: i.e. 1) contributing to the global need to reduce GHG emissions by using green products and technologies, and 2) reinforcing the national green technology capability to satisfy national and global

demands.

In the context of the DPKRTI-BRIN, the government's strategy for achieving the SDGs is influenced by both national and global policies. Currently, the DPKRTI-BRIN refers to 1) Indonesia's national long-term development plan (RPJPN) 2005-2025, 2) Indonesia's mid-to long-term development plan (RPJMN) 2020-2024, and 3) the G20 Development Working Group. From the national perspective, this institution should respond more fully in the future to completed national GEG policies for green growth (GG) programs, the circular economy (CE) programs, and the green economy index (GEI); whereas, from the global perspective, it should respond to the global green development and inclusive green growth (IGG) programs initiated by the GGGI, ADB, OECD, UN, and World Bank.

Indonesia's GEG strategies involve Bappenas as the main player (RPJPN, RPJMN, GG program, GEI, CE, G20 Working Group), while the Ministry of Industry plays a role along with Bappenas and other public agencies. Meanwhile, the Ministry of Environment and Forestry, the Ministry of Finance, Bappenas, and local governments arrange the green growth program. The DPKRTI-BRIN, along with other public agencies including Bappenas, is involved in the G20 Working Group. Therefore, it is necessary to reinforce the strategic planning of the national green technology development initiated by the DPKRTI-BRIN by involving those ministries and other public agencies. In addition, the DPKRTI-BRIN pays more attention to such issues as global green technology and climate change in order to connect the national strategies with global strategic plans.

6.2.2 Resources

The GTC has abundant resources for its daily activities. Its internal resources include human capital (researchers and non-researchers), funding, physical infrastructures, R&D and non R&D facilities, accumulated knowledge, and updated information, while its external resources include experts or researchers, funding support, up-to-date information, and laboratories derived from national and global networks. Furthermore, the GTC's customers include external GTC partners in such fields as R&D, policy consultation, policy advocacy, technology cooperation, and the production of climate change information.

Especially, to fulfil the global demand for green technology, the GTC is strongly pushing national and global networks to acquire more resources to support R&D activities. Business networks, academic networks, community networks, and government networks are all external resources that can accelerate R&D commercialization in industry and advocate GEG policy to stakeholders and communities. Accordingly, the GTC's resources are prepared to meet national and global demand.

In the context of the DPKRTI-BRIN, STI resources are divided into the following five categories: human capital, funding, physical infrastructures, hard and soft facilities, and networks. 1) STI human resources include policy and management researchers, policy analysts, engineers, R&D outcome analysts, and technocrats. 2) Funding is fully sourced from BRIN's governmental budget. 3) Physical infrastructures include office buildings and vehicle parks. 4) Hard and soft facilities include desks, computers, Internet websites, software, and subscriptions to global journals. 5) Networks include internal BRIN sources (7 deputies and 12 OR) and external BRIN sources (all ministries and local government agencies). Furthermore, the DPKRTI-BRIN is closely connected to an overseas STI organization, namely, the Science and Technology Policy Institute (STEPI) of South Korea.

There is no specific resource for incorporating GEG policy into the DPKRTI-BRIN. Existing resources should be moved to specific resources in supporting green technology development. STI research and funding should be directed so as to specifically contribute to green technology, green policy, GE, GG, and other SDG issues. The networks should be widened by involving global climate change and green technology organizations, countries affiliated with green economy goals, SDG research centers, and multinational green industries. At the national network level, the involvement of public agencies (ministries and local government agencies), industries, R&D research centers, universities, and NGOs should be focused on two aspects in particular: STI that contributes to green technology development and STI that contributes to GE development.

6.2.3 Institutional Governance

The GTC concentrates on its identity as a think tank in the field of climate change, with the

mission of addressing green climate technology for the domestic climate change industry as well as global efforts to mitigate climate change. To that end, its mission is dependent on two roles, namely, policy-making and the establishment of strategies. As a think tank, the GTC focuses on four areas: research on green climate technology policies, production of information on green climate technology, creation of cooperative strategies, and technology cooperation. Regarding these four areas, the GTC addresses the current demand for green technologies based on a comprehensive analysis process and predicts future demand for green technologies in South Korea and overseas.

The results of the GTC's comprehensive analyses and future predictions for green technology are coined into a GG policy agenda initiative and delivered to stakeholders in South Korea and overseas. The GTC arranges GG policy recommendations through strengthening policy document planning resulting from feedback from users such as academics, business, government, and communities (ABGC). Also, the GTC deploys its information system to facilitate information management, as information is dispersed among, and derives from, numerous sources. Information is managed in three steps: production of a database, processing of information, and establishment of a platform. This information system is connected with the accomplishment of SDG global cooperation networks and the GG support policy.

The GTC is partially funded by the South Korean government and regulated by its laws. Uniquely, it is an autonomous organization that manages internal and external resources independently, including decisions on whether to form cooperative relationships with many partners at the national and global levels. The GTC is not separated from the directive policies of the government of South Korea. On the one hand the GTC must conform to Korea's laws, whereas, on the other hand, it is able to accommodate green technology demands from overseas countries whose laws are different from South Korea's. Globally, the GTC is able to build robust governance among the South Korean government, the United Nation Framework Convention on Climate Change (UNFCCC), the Climate Technology Center and Network (CTCN), and multiples countries.

Again in the context of the DPKRTI-BRIN, BRIN's vision and mission concern national research, technology, and innovation development, while green technology and climate

change are just small parts of its duties. In the DPKRTI-BRIN, establishment of the “national green technology policy program” (NGTPP) is required. At BRIN, the program of an organizational structure whose function is to operate specific tasks at the level of top management (the deputy at BRIN). Like the GTC, the NGTPP’s main tasks are to conduct policy and management research for green technology development, offer policy consultation and advocacy on green technology issues, promote cooperation with national partners dealing with green technology and climate change issues, and produce public information materials on STI’s contribution to GEG. In the early stage, the NGTPP will be asked to respond to and meet the demand for green technology in Indonesia.

The NGTPP is expected to produce such outputs as publications, pilot projects, and policy papers (including policy recommendations) based on the strategic planning of national green technology development (long-term green technology policies). Such resources as human capital, funding, and networks will be mobilized and directed to support the NGTPP’s functions. Meanwhile, the outputs of the NGTPP should be well documented and connected to stakeholders or users, such as the ‘ABGC’, by utilizing the information system. For the NGTPP, the GTC’s production of a database, processing of information, and establishment of a platform are good examples of how to produce open access information. Importantly, it is connected to national networks concerned with green technology and climate change issues. In the future, it will be widely connected to global networks in the same field.

The DPKRTI-BRIN is fully funded by the governmental budget, regulated by governmental laws, and managed by bureaucrats. This means that all the activities within the DPKRTI-BRIN are adjusted to the rules of game as determined by the government. The emergence of the NGTPP in the DPKRTI-BRIN should have diverse governance under BRIN’s system. BRIN should shed on semi-autonomous authority for the NGTPP in managing resources, in involving external experts of BRIN, in determining the priority research areas, in communicating with stakeholders, and in collaborating with domestic and overseas partners. As for the NGTPP, it should focus on the Indonesian style of governance of public organizations.

6.3 Enhancing the Think Tank Functions of the GTC and Transmitting them to the DPKRTI-BRIN

The functions of a policy think tank in an open innovation system (OIS) are twofold: 1) to facilitate the establishment of hard and soft infrastructures related to improving the innovation system (IS) and monitor the system, and 2) to organize and maintain multiple networks widely (funders, experts/academician, society, other partners) (Seo et al., 2021). As a policy think tank, the GTC focuses on the core areas of agenda setting (i.e. identifying technology demand), policy formulation (project design, feasibility study, and master plan), and financial support.

The GTC facilitates the establishment of hard and soft infrastructures in order to improve the use of green technology across countries based on three aspects, namely, 1) discussion of the GG policy agenda with stakeholders, 2) negotiations about the implementation of technology policy with the UNFCCC, and 3) consultation on the development and application of green technologies with CTCN members and developing countries. The GTC does not carry out technical assistance projects to build such infrastructures; however, it can be a partner for many counterparts in diverse countries in discussing GEG policies and their instruments based on its accumulated information and knowledge.

The GTC organizes and maintains multiple networks in the national and global arenas, and has the following divisions: 1) Technical Supervision, 2) Policy Research, 3) Climate Change Cooperation, 4) National Climate Technology Cooperation Center, 5) Administration, 6) Strategic Planning, and 7) Audit. These divisions involve diverse partners such as academicians, researchers/experts, government/stakeholders, business/shareholders/practitioners, donor/funding agencies, communities/NGOs, and related partners from South Korea and overseas. The GTC has the main duty of keeping all the networks going in order to support its functions.

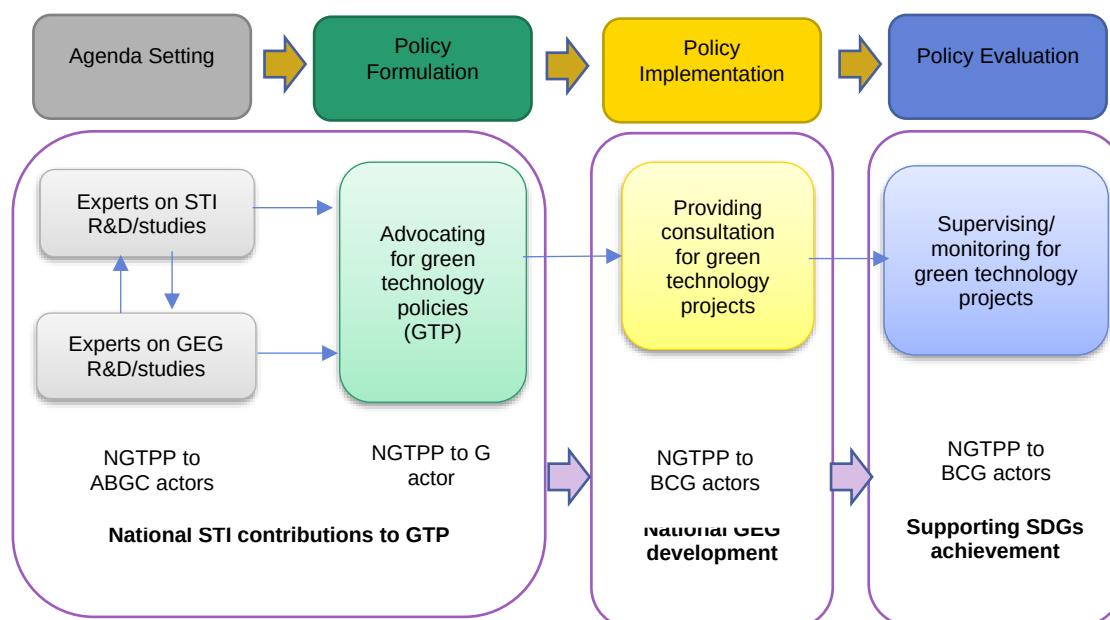
The GTC requires support from the UNFCCC, CTCN, and developing countries. Also, it is closely connected to the Technology Executive Committee (TEC) and the Ministry of Science and Information Communication Technology (ICT) in South Korea. Korea's National Designated Entities (NDE) are a trigger for the GTC to develop the national green technology

capability and then to transmit the results to developing countries. The GTC is dedicated to strengthening national R&D capacity in order to develop green technology and to responding to global green technology demands. In turn, the GTC, through the UNFCCC, is allowed to collaborate with developing countries in developing a green technology partnership initiative (GTPI) with Indonesia.

The GTC's function is limited to setting the agenda and formulating policy on green technology, while implementing and managing a green technology project lies outside its core area. As an ideal policy think tank in the OIS, a think tank plays a role in the entire policy-making process encompassing agenda setting, formulating, implementing, monitoring, and evaluating. Policy entrepreneur is more acknowledged than merely policy advocate (Seo et al, 2021). Referring to Seo's study, on the one hand the GTC is categorized as a conventional think tank, while on the other hand it is evolving into an ideal think tank by opening as many national and global networks as possible in the climate change and green technology fields. Therefore, it could be very helpful as a best practice for the DPKRTI-BRIN.

In the DPKRTI-BRIN context, a think tank is perceived as "bridging or translating the STI knowledge produced by academicians to STI policy produced by stakeholders." Agenda setting and policy formulation are core areas of a think tank's activities. Referring to Seo et al. (2021), it is a conventional think tank because STI policy think tanks should play a role in the implementation and evaluation areas. The emergence of the NGTPP is an opportunity to deliver and enhance the STI think tank at BRIN. The NGTPP is expected to become the leading think tank in Indonesia because of its new working area. It will play a role in four areas at once, namely, the role of an expert in the agenda setting phase, an advocacy role in the policy formulation phase, the role of a consultant in the policy implementation phase, and the role of a supervisor/monitor in the policy evaluation phase.

[Figure 2-18] The Four Working Areas of NGTPP



Source: Developed by the authors with reference to Seo et al (2021).

During the agenda setting phase, the NGTPP interacts with academician, business, government, and community (ABGC) actors. In the policy formulation phase, it engages with government actors. During the policy implementation and at the policy evaluation phase, it engages with business, government, and community actors. During the agenda setting and policy formulation phases, it produces STI contributions to green technology policies (GTP). In the policy implementation phase, it develops national green economy growth (GEG). Lastly, during the policy evaluation phase, it supports sustainable development goals (SDGs) achievements.

As part of BRIN's organizational structure, the NGTPP has broad advantages in operating its working areas. First, BRIN provides great opportunities for its researchers to collaborate with outside partners from around the world. Second, BRIN provides co-working spaces, allows work from anywhere (WFA), has flexible working spaces that can be easily accessed by all its personnel. Third, BRIN reduces the hierarchy-bureaucracy ladders to simplify decision making within the BRIN environment. Fourth, BRIN has large resources in human capital (researchers and non-researchers), funding, national and global networks,

laboratories, and hard/soft infrastructures and facilities. Fifth, BRIN is connected to all national stakeholders concerned with the application of STI. And, lastly, BRIN supports GE development in Indonesia to respond to the global climate change issue.

The four working areas and advantages described above will support the tasks and functions of the NGTPP in practice. In the beginning, the NGTPP will have at least two functions i.e. facilitating and monitoring the establishment of public infrastructures in supporting the green technology system (GTS) in Indonesia and reinforcing and maintaining multiple national and global networks in supporting the creation of GTS. Such functions are not technical or administrative routines as the NGTPP's tasks (See Institutional Governance above). Indeed, the NGTPP's functions include policy consultation, policy negotiation, policy advocacy, and information-sharing activities concerning green technologies and products in Indonesia.

In the four working areas, the NGTPP can play various roles ranging from strategic planning and project design to funding approval, implementation, and evaluation. Nevertheless, not all roles are played by the NGTPP, and using the NGTPP's resources. As a think tank, the NGTPP can collaborate with the ABGC in performing its tasks and functions. From this point of view, the key strengths of the NGTPP are that it has many interwoven networks and that its networks contribute to GTS.

6.4 Designing a Think Tank for Green Technology Policy in the DPKRTI-BRIN

The think tank for green technology policy (GTP) at the DPKRTI-BRI is the National Green Technology Policy Program (NGTPP), as discussed above. The emergence of the NGTPP within BRIN's organizational structure was inspired by the emergence of the GTC under the Korean Ministry of Science, ICT, and Future Planning (MSIP). Hierarchically, the NGTPP and the GTC are governmental agencies dedicated to meeting the green technology demands of their respective countries. The difference is, that the GTC aims to meet green technology demands in Korea and overseas countries, whereas the NGTPP is first expected to meet green technology demand only in Indonesia.

Designing the NGTPP within the structure of the DPKRTI-BRIN is not a simple task because the DPKRTI is also a new organization under BRIN. The integration of BRIN has brought about significant changes in various areas, including research on nomenclature and its functions, the mobilization of S&T human resources (researchers, policy analysts, engineers, etc.), the R&D funding mechanism, R&D facilities and infrastructures (like laboratories), and priority research topics. Amid the current conditions, BRIN is in the phase of consolidation and adaptation, and is establishing its business process. Nationally, it has a direct impact on R&D collaboration schemes with external partners in both domestic and global networks.

The discussion with experts from the University of Brawijaya, the University of Gadjah Mada, and BRIN itself has revealed practical findings related to how to design the NGTPP in the DPKRTI-BRIN. There are five sequential stages for designing the NGTPP within the structure of the DPKRTI-BRIN: 1) identify and re-collect the S&T human resources in BRIN; 2) create shared values of the actors involved in a new program; 3) prepare and improve potential funding sources and infrastructures; 4) enhance and strengthen national and global networks; and 5) propose a feasible organizational structure for the NGTPP.

6.4.1 Identifying and Re-collecting S&T Human Resources in BRIN

BRIN is the largest and the only public R&D institute in Indonesia. All R&D activities are integrated under BRIN's structure, including R&D activities previously handled by ministries or public agencies at the national level. This means that R&D activities related to STI policy and management, green technology, climate change, economic development, environment, and green energy all fall within the scope of BRIN's research topics. Currently, the researchers of these topics are dispersed among several research centers (PR) in the research organization (OR) as well as among several directorates under the deputy of development policy (DKP) and the deputy of research and innovation policy (DKRI).

[Table 2-6] S&T Human Resources on Research Topics of Green Development in BRIN

No.	Research Center/Directorate	Research Organization/Deputy
1	Directorate of Environment, Maritime, Natural Resources, and Nuclear Policy	Deputy for Development Policy
2	Directorate of Economy, Employment, and Regional Development Policy	Deputy for Development Policy
3	Directorate of Policy Formulation for Research, Technology, and Innovation	Deputy for Research and Innovation Policy
4	Research Center for Climate and Atmosphere	Research Organization for Earth Sciences and Maritime
5	Research Center for Behavioral and Circular Economics	Research Organization for Governance, Economy, and Community Welfare
6	Research Center for Public Policy	Research Organization for Governance, Economy, and Community Welfare
7	Research Center for Environmental and Clean Technology	Research Organization for Life Sciences and Environment
8	Research Center for Biomass and Bioproducts	Research Organization for Life Sciences and Environment
9	Research Center for Sustainable Production System and Life Cycle Assessment	Research Organization for Energy and Manufacture
10	Research Center for Nuclear Fuel Cycle and Radioactive Waste Technology	Research Organization for Nuclear Energy
11	Research Center for Environmental Archaeology, Maritime Archaeology, and Cultural Sustainability	Research Organization for Archaeology, Language, and Literature

Source: BRIN Internal resource

The names of the research centers and directorates represent the researchers, policy analysts, engineers, and experts in green development and STI policy. First, the DPKRTI-BRIN identifies and re-collects them by mapping potential researchers and non-researchers who are interested in supporting the NGTPP. Second, the DPKRTI-BRIN initiates the creation of an informal discussion group to accommodate discourses related to green technology issues. Third, the DPKRTI-BRIN invites them to join research collaborations, workshops, and seminars that deal with topics relevant to green development.

6.4.2 Creating Shared Values of Actors Involved in a New Program

Efforts to create shared values of the actors involved in a new program show an organization's readiness to change. Simpson (2009) states that organizational readiness for change (ORC) is determined by three layers of factors: 1) the individual level is the lowest layer in which each individual works for the sake of his/herself; 2) the group level is the middle layer in which individuals work and interact with other individuals in the same groups; and 3) the organization level is the highest layer in which all groups work and interact together in accomplishing an organization's goals. Thus, according to the three layers of ORC, creating shared values works on three levels: individual, group, and organization.

Creating shared values at the individual level is related to how each individual is able to gain benefits in responding to GEG and SDGs issues. Each individual tends to maximize personal benefits for him/herself, during which different values in individual interests are likely to emerge. In this case, the DPKRTI-BRIN should facilitate each individual to make shared values among those individuals related to GEG and SDG goals. Common agreements, individual career development, rewards, broad opportunities to join in national and global seminars, publication support, and closeness to an exclusive group circle are practical examples of attracting individuals to join the NGTPP.

Creating shared values at the group level is related to how the individuals within groups are able to work autonomously when addressing GEG and SDG issues. Each group tends to struggle collectively to show their group's contribution, to be silo and not coordinated each other, and to compete fiercely to attract the attention of the top management. To address this problem, the DPKRTI-BRIN makes a clear division of the tasks for each group, and there are interdependent tasks where multiple groups need to work together. As each group is unique in managing tasks and individuals, the DPKRTI-BRIN can provide valuation, feedback, acknowledgements, and rewards to each group accordingly

Creating shared values at the organization level relates to how all groups are able to collaborate in achieving the organization's goals concerning GEG and SDG issues. Each group has a different aim or intention in contributing to the advancement of an organizational goal. At this level, the involvement of individuals, groups, and top-middle level management is

very much required to achieve one common goal of the NGTPP. It is not easy to unite all the multiple goals of the groups into one within the NGTPP. The role of the DPKRTI-BRIN is to establish a dynamic working system by involving all internal groups in the NGTPP, and also to make them accessible for those who want to join a group, even if not on a regular basis. It means that this working system can be changed periodically according to common consideration and agreement reached among individuals, groups, and top-middle level management.

6.4.3 Preparing and Improving Potential Funding Sources and Infrastructures

BRIN has a special R&D funding scheme and a complete set of R&D infrastructures. BRIN's R&D funding and infrastructures have even become the main reference for all other R&D units in Indonesia. In relation to the NGTPP, BRIN's support in allocating funding and providing infrastructures needs to be specially prepared. The DPKRTI-BRIN provides funding sourced from BRIN and provides infrastructures including rooms, desks, internet, software, and computers; and these are adjusted to the NGTPP's needs. For instance, there is a special room where the NGTPP can discuss GTP and climate change issues. As regards R&D laboratories, they are adjusted to meet the needs of the researchers involved in the NGTPP's projects.

The potential funding sources and their infrastructures also originate from outside BRIN. The NGTPP offers more opportunities to individuals and groups to obtain external funding sources and manage them autonomously. Including R&D infrastructures, there are many potential R&D laboratories in firms and universities that could be utilized to support green technology development in Indonesia. Certainly, an agreement or cooperation mechanism will be needed if the NGTPP uses the funding sources and infrastructures of external partners.

6.4.4 Enhancing and Strengthening National and Global Networks

National and global networks are the main capital and resources required to operate an organization. Networks are a must for those who want to develop their career within an organization. Networks are required by a new program like the NGTPP in order to gain broader recognition from the actors of the ABGC at both the national and global levels. This means that the NGTPP should enhance and strengthen multiple networks to convince the ABGC by proving that what the NGTPP does will also be beneficial for them.

The NGTPP does not search for new networks, as the networks are already tightly closed to individuals or groups who have joined the NGTPP. Thus, BRIN is composed of various S&T human resources including those who are experts in the green technology, policy, and economy areas. Previously, they worked in R&D units, and therefore they are still connected to their networks individually. The NGTPP needs to reinforce the existing networks by operating in a more organizational mode rather than an individual one. Therefore, activities such as R&D collaborations, co-publications, joint patents, co-funding for projects, co-workshops, co-seminars, and joint policy consultations and advocacies to stakeholders are key points for enhancing and strengthening the national and global networks of the NGTPP.

6.4.5 Proposing a Feasible Organizational Structure for the NGTPP

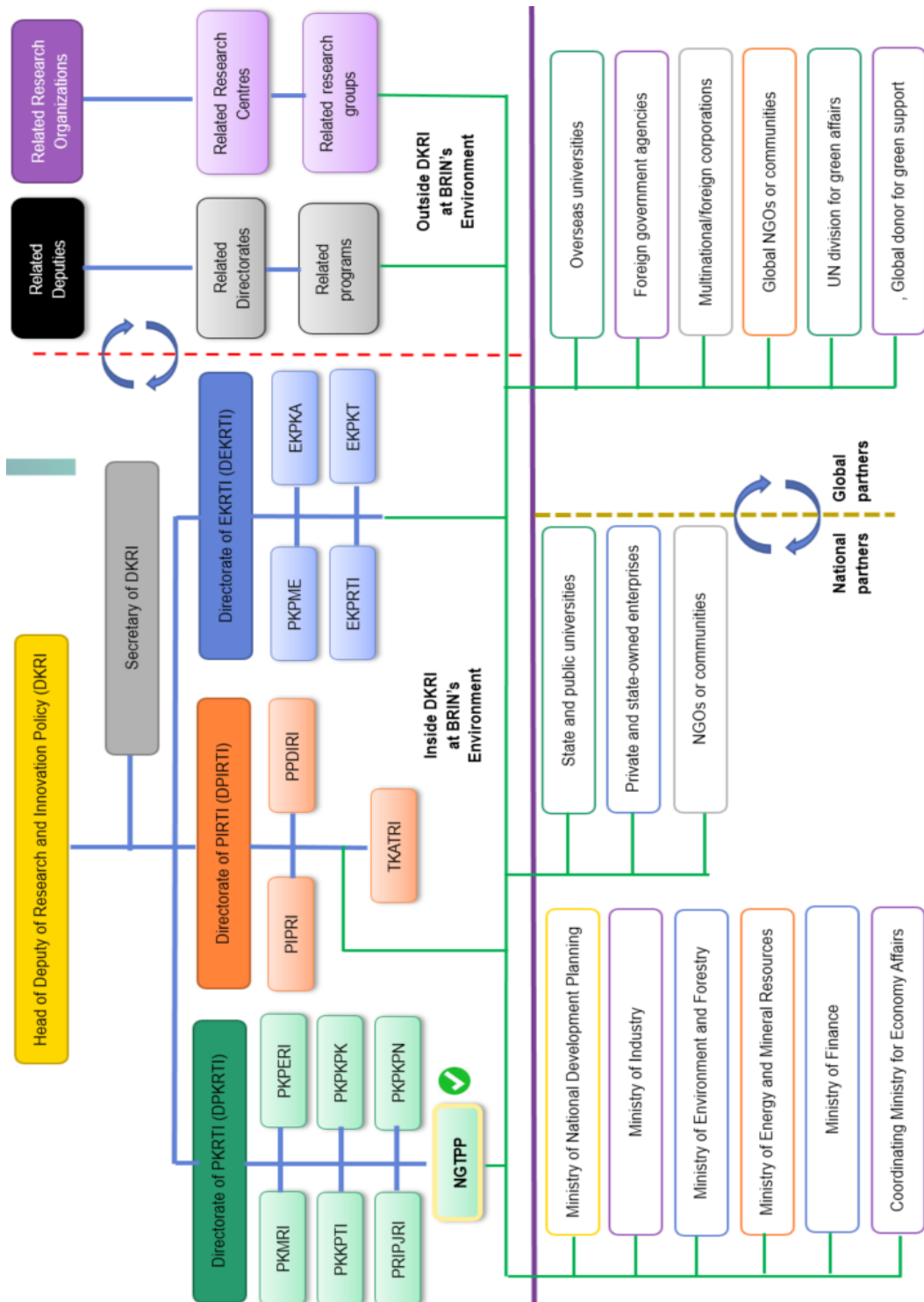
Among the four stages elaborated in detail above, only the last stage discusses ways to propose a feasible organizational structure for the NGTP. This is because, when designing a new organization, it is not sufficient to merely sketch out the organizational structure without first understanding the goals of the organization, the shared values of the individuals and groups within it, and the support resources, as well as with whom the organization will work. We use the term “propose” to point out that this organizational structure design is not fully fixed, but rather is subject to change according to the current conditions.

Proposing a new organizational structure for the NGTPP under the DPKRTI-BRIN involves creating a new institution, a new policy, and new governance. Creating the NGTPP as a new program will not change the hierarchical structure of the DKRI and BRIN wholly. Program(s)

can be added or removed at the DPKRTI-BRIN level because they are not part of the echelon structure. Therefore, creating the NGTPP under the hierarchical structure of the DPKRTI-BRIN is feasible. The NGTPP will be led by a program coordinator accountable to the director of the PKRTI-BRIN. Also, the NGTPP will be connected to other programs within the internal and external DKRI environments (other deputies and research organizations at BRIN).

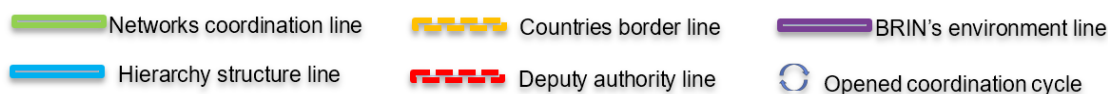
The NGTPP is expected to be broadly connected to national and global networks related to GEG and SDG issues. Such networks can become partners for consultation, or become policy users, project users, authors of publications, donors, regulators, discussion partners, informants, respondents, etc. They are representative from ABGC actors. National partners include the Ministry of National Development Planning, the Ministry of Industry, the Ministry of Environment and Forestry, the Ministry of Finance, the Ministry of Energy and Mineral Resources, the Coordinating Ministry for Economic Affairs, and the Ministry of Finance, as well as state and public universities, private and state-owned enterprises, NGOs, and communities. Global partners are mostly overseas universities, foreign government agencies, multinational/foreign corporations, global NGOs or communities, the UN Division for Green Affairs, and global donors for green support. Nevertheless, coordination and collaboration between the DKRI's environment and other institutions, and between national and global partners, are possible without the NGTPP's direct involvement.

[Figure 2-19] Structure of the NGTPP and Its Networks



Source: Author's elaboration of analysis result

Notes:



Description:

BRIN	: National Research and Innovation Agency
DKRI	: Deputy of Research and Innovation Policy
DPKRTI	: Directorate of Research, Technology, and Innovation Policy Formulation
PKMRI	: Program of STI Policy and Management Research
PKKPTI	: Program of Technology Development and Innovation Policy Study
PRIPJRI	: Program for Formulation of the STI Master Plan and Roadmap of Research and Innovation
PKPERI	: Program for Formulation of Research and Innovation Ecosystem Development
PKPKPK	: Program of Study and Formulation for Space and Aeronautics Management Policy
PKPKPN	: Program of Study and Formulation for Nuclear Management Policy
NGTPP	: National Green Technology Policy Program
DPIRTI	: Directorate of Research and Innovation Measurement and Indicators
PIPRI	: Program of STI Measurement and Indicator Analysis
PPDIRI	: Program of Research and Innovation Data Collection and Management
TKATRI	: Programs of STI Foresight and Trend Analysis
DEKRTI	: Directorate of Research, Technology, and Innovation Policy Evaluation
PKPME	: Program for the Development of STI Evaluation Methods
EKPRTI	: Program for Evaluation of STI Policy and Management
EKPKA	: Program for Evaluation of Space and Aeronautics Policy
EKPKT	: Program for Nuclear Policy Evaluation
UN	: United Nations
NGOs	: Non-Governmental Organizations

There is no one-size-fits-all approach to implementing green growth; rather, particular policies and activities must react to national priorities and conditions (UNEP, 2014; OECD, 2011). Regarding the NGTPP, it is important is to create an institution and governance that function and perform the tasks of managing resources and networks, as well as responding to governmental policies. NTGPP institution and governance are related to the business processes made by common agreement among individuals, groups, and management. Shared values, resources, infrastructures, and networks are decisive factors in making the business process of the NGTPP work. Furthermore, the provision of policy instruments by the NGTPP is necessary to make this program work.

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CHAPTER 3

Developing STI Human Capital in Indonesia

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Chapter 3. Developing STI Human Capital in Indonesia

1. Introduction

According to Presidential Regulation No. 38/2018 on RIRN 2017-2045, the Indonesian government has set various goals for research development for the period 2015-2045. Each target will be raised simultaneously every five years from 2015 to 2045. In contrast, the Indonesian research goal for 2045 is the same as the Korean scientific achievement for 2015. Consequently, in 2015 Indonesia had a 30-year gap to catch up with Korea. In 2015, Indonesia had 1,071 researchers per million people, whereas Korea now has 8,600 researchers per million. Regarding the government budgets for R&D, Indonesia had a BOARD/GDP ratio of 0.15 percent, while Korea had a BOARD/GDP ratio of 1.26 percent in 2015. Furthermore, Indonesia had just 1,071 researchers per 1 million people, while Korea had 8,600 researchers per 1 million. Regarding the government budget for research and development, Indonesia's BOARD/GDP ratio was 0.15%, compared to Korea's 1.26% in 2015. GERD was another ratio used to calculate government spending on research and development. In 2015, Indonesia had a GERD/GDP rate of 0.20%, whereas Korea's was 5.04%. When comparing graduate and undergraduate students, Indonesia had a 5.6% of graduate students per undergraduate students in 2015, while Korea had a 100% ratio. Furthermore, for research productivity in 2015, Indonesia received a score of 0.02, while Korea received a score of 0.22. Finally, in 2015, Indonesia had a multifactor productivity score of 16.7%, while Korea scored 70%.

Human resources in science and technology (HRST) work in STI fields as researchers, engineers, lecturers, and others, according to Law No. 11/2019. According to the PDDikti, 2020, holders of master's degrees dominated among education-level lecturers in Indonesia in the years 2014 to 2017, i.e. about 62% of lecturers held a master's degree and 27% held

a doctorate, while the remaining 11% only had a bachelor's degree. A similar problem exists regarding researchers in Indonesia. According to BRIN data (2020), the average government researcher in Indonesia holds a master's degree, i.e. 55% of government researchers have a master's degree or higher, while only 24% have a doctoral degree, and the remaining 21% hold a bachelor's degree. In terms of engineers, data from BRIN (2020) show that 46% of government engineers in Indonesia have a master's degree and that 43% have a bachelor's degree, while only 8% have a doctorate, and the remaining 3% have a diploma.

Most HRST workers, namely, researchers, engineers, and lecturers, are educated to the master's degree level. Furthermore, less than a quarter of all HRST employees have a doctorate. Furthermore, many people only have a bachelor's degree, particularly government engineers (43%). As such, HRST workers' skills and abilities must be enhanced through additional formal education, particularly by pursuing a doctoral degree. However, HRST workers must of course be provided with funding to enable them to study. HRST employees must also participate in other informal activities such as training and development.

Other than providing more funding to improve their formal and informal education, government HRST workers also need to improve their skills and knowledge by collaborating with other experts, as such collaboration will also result in knowledge transfer and sharing. Furthermore, it is necessary to develop regulatory support as to enable the government to embody the success of collaborative programs. Also, it is necessary to ease the rigid regulations for government researchers and other government HRST workers in order to increase their flexibility and capability. For example, at present government HRST employees are not permitted to become start-up owners. Therefore, these issues necessitate new government strategies to replace rigid regulations with flexible and adaptable regulations for government HRST workers, as this will enable them to broaden their skills and knowledge.

Long-term HRST improvement could begin by providing Indonesians with access to high-quality education. Regarding education indicators, Indonesia has a system similar to South Korea, comprising six years in elementary school, three years in junior high school, three years in a senior or vocational school, and four years of study at a polytechnic or university.

In terms of the sheer number of universities, Indonesia outnumbers South Korea. However, the population per university is far too high. Indonesia has a population per university ratio of about 154,000: 1, whereas Korea has a population per university ratio of about 59,000: 1. In terms of the top 1,000 global universities, Indonesia has only nine universities in the top 1,000, whereas South Korea has thirty. Furthermore, only 16.7% of Indonesia's population has passed through tertiary education, compared to 50.7% in South Korea. This gap between Korea and Indonesia in terms of the level of education is far too large, and will have a negative impact on the country's competitiveness in science and technology production. Therefore, a new government strategy is required to increase access to qualitative education, as this will influence the development of science and technology and thereby improve the country's competitiveness.

The availability of sufficient human resources in science and technology (HRST), in both quality and quantity, is a crucial component of the development strategy. It is clear that the number of available researchers is tiny compared to the national population, and that productivity is also very low. As a result, this study will look at how Indonesia can develop HRST using existing policies regarding quality and quantity. In this respect, the HRST development process is viewed as an investment in people with expectations of future productivity, so the perspective of HRST as human capital has been adopted.

Human capital can be defined in several ways. Shultz, T.W. (1961) defines human capital as the knowledge and skills that people acquire through education and training. This is similar to the understanding of Denison, E.F. (1962), who links human capital to education, which helps economic growth. This understanding is also consistent with that of Becker, G.S. (1964), Bowman, M.J. (1969), Blaug, M. (1976), Psacharopoulos, G., and Woodhall, M. (1985), Cohn, E., and Geske, T.E. (1990), Becker, G.S. (1993), and Bontis, N (1996). Meanwhile, P.M. Romer (1986, 1987) views human capital as a new growth theory.

Meanwhile, Becker et al. (1990) propose a human capital fertility model. Fitz-Enz, J. (2000) defines human capital as the characteristics that an individual brings to a job, which is consistent with the ideas of David, P. and Lopez, J. (2001), who define human capital as acquired human capabilities that improve performance in socially valued activities.

According to the World Bank (2022)¹, human capital is the sum of the knowledge, skills, and health that enable people to reach their full potential as productive members of society. According to these various definitions, human capital is knowledge, skills, and health obtained through education or training. It is an investment in education that positively affects performance in socially valued activities, jobs, and even economic growth.

¹ (<https://databank.worldbank.org/source/human-capital-index>) Retrieved on June, 2022.

2. Methods

This section will look at the approach that will be used. This study employs a qualitative approach to observe and provide feedback on the most recent policies on human capital development in Indonesia based on South Korea's experience.

1. Understanding the Development of Human Capital in South Korea
2. Understanding the Development of Current Human Capital in Indonesia
 - a. Desk research.
 - b. In-depth interviews with Universities: Binus University, Padjadjaran University, and Gadjah Mada University.
 - c. In-depth interviews with BRIN's representative.
 - d. In-depth interviews with the Ministry of Culture and Education of the Republic of Indonesia.
3. Developing Policy Alternatives in Indonesia based on South Korea's Experience in Developing Human Capital

3. Issues of HRST as Human Capital Development in Indonesia

During his first term, President Joko Widodo focused on growing Indonesia's human capital and its mastery of science and technology. As a result, several changes have already occurred, including a slight rise in Indonesia's Human Capital Index (HCI) from 0.53 in 2018 to 0.54 in 2020.

Despite the HCI Index's growing importance, several broad classical difficulties persisted. High-quality human resources, for example, will help develop innovation and labour productivity, impacting industry growth and the economy. Furthermore, developing highly trained, well qualified human resources makes it easier for employers to absorb the workforce produced by educational institutions, which in turn helps to break the vicious cycle of poverty by allowing previously impoverished individuals to find employment and decent earnings, propelling them toward greater social mobility.

[Table 3-1] HRST Policy Development in Indonesia (1950-2020)

Period	Economy	Research and Education
1950s-1970	• Supporting agricultural exports in the agricultural sector industries (1969-1974) • Boosting agricultural process in the agricultural sector industries (1974-1979)	• Eradication of illiteracy: After independence, the literacy rate of Indonesian society was only 10.8% for males and 2.2% for females. • Lack of Expert Human Resources: Indonesia only had 560 tertiary education graduates after independence, increasing to 21,832 by 1968. • Shortage of researchers in 20 GRI; more than 80% are STEM researchers. • Provision of scientific and technician training for researchers with high school and university backgrounds.

Period	Economy	Research and Education
1970s-1980	• The agricultural sector pursues food self-sufficiency and boosts industries that process raw materials into finished goods. • Emphasis on labour-intensive industries (1979-1984). The agricultural sector continues efforts to achieve food self-sufficiency by improving industries that produce industrial machinery, i.e. both heavy and light industries (1984-1989)	• Training and education for researchers in the agricultural, industry, and mining sectors. • Development of four research information centers. • Increased investment in vocational education to cater to labour-intensive industries.
1980s-1990	• The agricultural sector pursues food self-sufficiency and boosts industries that process raw materials into finished goods. • Emphasis on labour-intensive industries (1979-1983). • The agricultural sector continues efforts to achieve food self-sufficiency by improving industries that can produce industrial machinery, both heavy and light industries (1984-1989).	• Improvement of the competences of researchers to enable them to compete globally. -Establishment of the government's first overseas scholarship program in 1982 (Habibie Awards).
1990s-2000	• The agricultural sector effects the transportation, communication, and education (1990-1994). • Economic sectors related to industry and agriculture: improves the foreign investment climate and boosts the national economy and industry(1994-1998).	• Eradication of illiteracy to 0% • People are required to receive at least 9 years of education.
2000s-2020	• Development in various fields accelerated by emphasizing the establishment of a solid economic structure based on competitive advantages in various areas, supported by qualified and competitive human resources (2020).	• LPDP scholarships for overseas postgraduates (2013) • Twelve years of compulsory education (2015). • Riset-Pro for HRST scholarships and training (2012). • Law 11/2019.

Period	Economy	Research and Education
		• Law Industry 3/2014. • Integration of GRIs under BRIN.

Source: Author's elaboration

On the other hand, efforts in Indonesia to develop high-quality human resources have come a long way. As shown in the table above, the scarcity of qualified human resources in the early independence period made it a priority to develop human resources at the most fundamental level, i.e. meeting basic educational needs and eradicating illiteracy. The government began sending experts on overseas study programs in the 1980s, and real progress began to be made. By the 2000s, Indonesia already had twelve years of compulsory education and more high-quality human resources to compete in the global chain. As a result, in the context of HRST, the macro problem associated with human capital development is the scarcity of highly qualified HRSTs capable of competing globally. This report will now look at the three points presented below.

3.1 Inadequate HRST Productivity

The lack of governmental support for research is also a barrier to higher education in Indonesia. Indeed, Indonesia is near the bottom in this category. According to the World Bank, the Indonesian government spends only 0.08 percent of its GDP on R&D, which is considerably less than neighbouring countries such as Singapore (2.19 percent) and Malaysia (1.29 percent) World Bank, 2015)². Due to insufficient research funding, researchers are more or less compelled to leave Indonesia to seek better career opportunities and well-equipped facilities elsewhere.

The development of research activities in Indonesian research institutions is also low, resulting in a low number of patents issued by Indonesian scientists and engineers. Low GERD also results in low productivity because scientists and engineers need to receive adequate funding for their research activities. Most scientists and engineers working for government research institutions (have) suggested that industries and private companies

² (<https://databank.worldbank.org/source/human-capital-index>) Retrieved on June 2022.

should get involved in order to increase research productivity in Indonesia.

The unfavourable circumstances for HRST development were largely related with structural issues inherent in state universities and research institutions, with most research previously confined to providing technocratic input for the government's development strategies (Hadiz and Dhakidae, 2005; Guggenheim, 2013), which "cut down autonomy and academic freedom" (Guggenheim, 2005, p. 147), thereby contributing to the long-term, systematic suppression of critical thinking. The current bureaucratic and technocratic design limits academics' contributions to national development (Moeliodihardjo et al., 2000; Nizam, 2006; Wicaksono and Friawan, 2013). As such, academics found it difficult to interact with their peers from other areas of expertise due to the rigid separation of disciplines within the Higher Education Institutions (HEI).

The research productivity of universities is also at the forefront of critical issues. When attempting to produce quality research programs that result in international publications, researchers are limited by time constraints. The average time required to conduct research that leads to international publication ranges from nine to twelve months.

There is also the problem of credit-seeking (KUM) academics who take advantage of Indonesia's state university bureaucracy. State employment regulations and policies for civil servants are based on the State Employment Agency's promotion system, which recognizes advancement based on the accumulation of credit points rather than academic performance. These points can be earned by teaching, giving seminars, and other activities, with research and international publication receiving less weight.

In addition, state universities have focused on gaining greater autonomy at the institutional level in order to be able to admit more students and thereby fund their operating costs, resulting in a decline in academic research at state universities. The issue of "inbreeding" has resulted in isolationism because, despite efforts to encourage more joint research, these problems indicate that universities and their faculty are more concerned with their individual research interests. Under these circumstances, Indonesian academics are more likely to pursue advanced degrees at their home institutions, which may have contributed to more "inbreeding" among state universities.

Due to a lack of institutional support, including core funding for academic research, university research centers focused on consultancy and service provider activities rather than basic (or socially embedded) research for critical socioeconomic issues. The incentives for a consultancy orientation worked against developing university research cultures and practices.

Under this system, Indonesians who gained prestigious degrees from local and overseas universities could not readily continue a research career after graduation. The incentives for careers lay elsewhere: academics with a low university salary necessarily looked to improve their family's welfare by taking up more lucrative opportunities off campus. This included, for instance, gaining access to government project funding, becoming an advisor with close links to senior decision-makers, or becoming a high-ranking official. Alternatively, a researcher could gain economic and political resources from international networks and NGOs. Hence, at the micro-individual level, academics gained status and financial rewards by becoming prominent media commentators, policy advisors, and service providers rather than by doing research (McCarthy, 2010).

3.2 HRST Availability Shortage

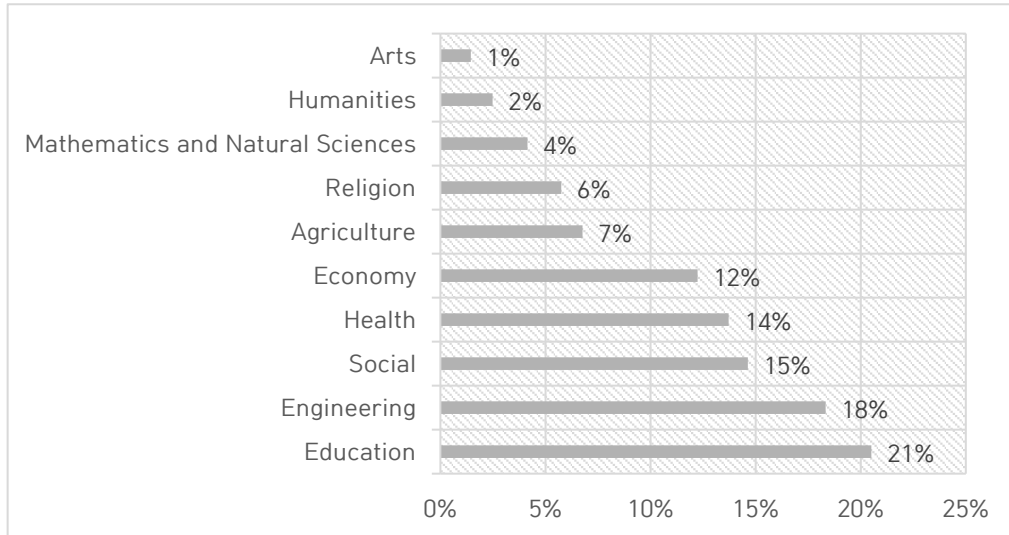
The number of researchers per million people is a proxy indicator of a country's human capital research capabilities. In 2018, Indonesia had 215 researchers per million people, whereas Australia had 4,500, Korea 7,500, and Thailand 1,350 (World Bank, 2021)³. In this regard, Indonesia's competence lags way behind that of its regional neighbours and fellow ASEAN economies.

One of the factors supporting the number of HRSTs is the availability of appropriate study programs. Indonesian universities offer diverse study programs in the arts, humanities, mathematics, natural sciences, religion, agriculture, economics, health, social, engineering, and education. (The number of science study programs in the United States is 29,413.) These programs are distributed across the high education institutions in Indonesian provinces, but each province has different needs regarding its educational programs,

³ (<https://databank.worldbank.org/source/human-capital-index>) Retrieved on June 2022.

resulting in the development of differentiated educational programs for each region.

[Figure 3-1] Number of Study Programs in 2020 by Field of Study



Source: PDDikti (2020).

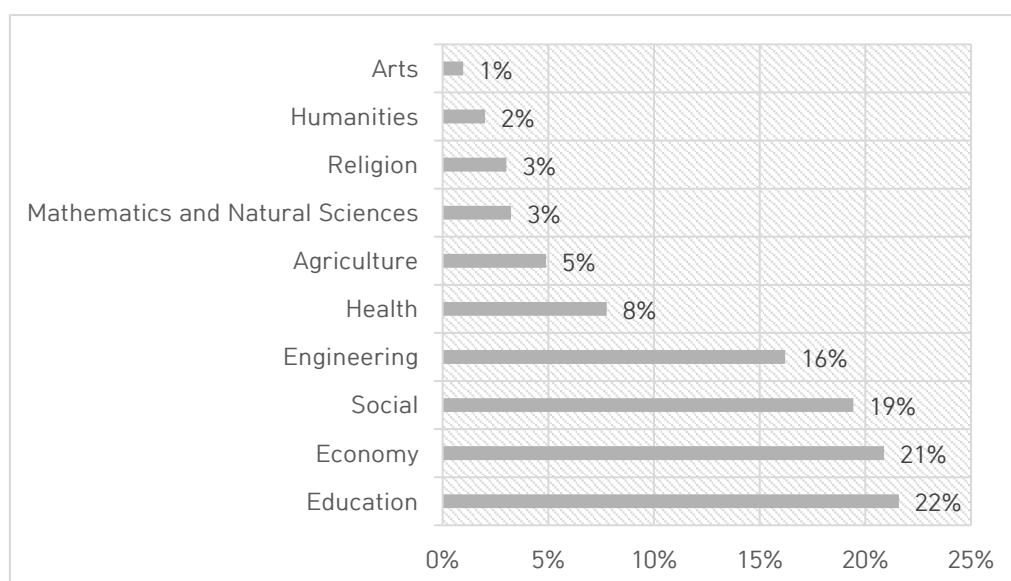
Within five years, every level had improved. In 2015, the number of study programs at public and private universities by level of education was 11,359 for bachelor's degrees, 2,075 for master's degrees, and 497 for doctoral degrees. The number of master's degree study programs grew to 3,252, and that of doctoral degree study programs to 774. A significant increase in bachelor's degree study programs occurred in 2018, with as many as 17,261 programs, and the figure was expected to increase to 18,158 by 2020.

In 2020, educational programs accounted for the highest share of public and private university programs at 21% (6,032 study programs), followed by engineering study programs at 18% (5,390), social at 15% (4,302), health at 14% (4,034), economy at 12% (3,599), agriculture at 7% (1,988), religion at 6% (1,692), mathematics and natural sciences at 4% (1,219), humanities at 2% (731), and arts at 1% (426).

Students in higher education are potential human resources who can contribute to the advancement of science and technology. Higher education balances the maximum number of students in each study program with the capacity of the available facilities and

infrastructure, lecturers, education staff, and other educational services and resources. New student admissions for each study program can be made through the national PTN student admission pattern or the PTS regulations by each institution. In 2020, the number of new admissions amounted to 2,163,682, across all provinces. There were 1,875,337 new registrants at academic and vocational colleges, with 1,875,337 new students for the academic type of program and 176,022 for the vocational college type.

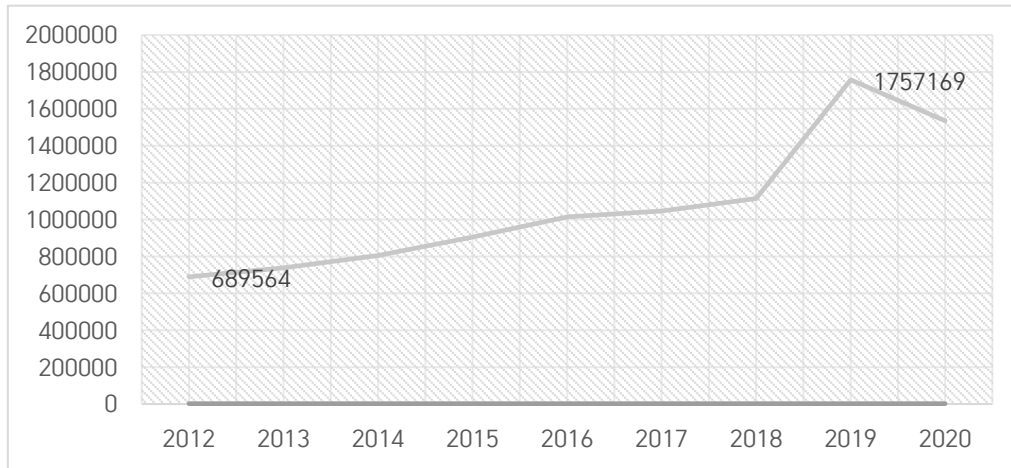
[Figure 3-2] Total Number of Registered Students by Field of Study



Source: PDDikti (2020).

In 2020, some 8,483,211 students were enrolled on higher education courses in diverse fields of knowledge at Indonesia's universities. The largest group were interested in the field of education, accounting for as many as 1,831,746, or 22%, of the total number of registrants, followed by economics at 1,771,864 applicants (21%); social at 1,648,219 applicants (19%); engineering at 1,374,528 applicants (16%); health at 658,633 applicants (8%); agriculture at 414,890 applicants (5%); mathematics and natural sciences at 274,693 applicants (3%); religion at 256,044 registrants (3%); humanities at 170,221 applicants (2%); and arts at 82,373 registrants (1%). These prospective students then applied and became new students in universities throughout Indonesia.

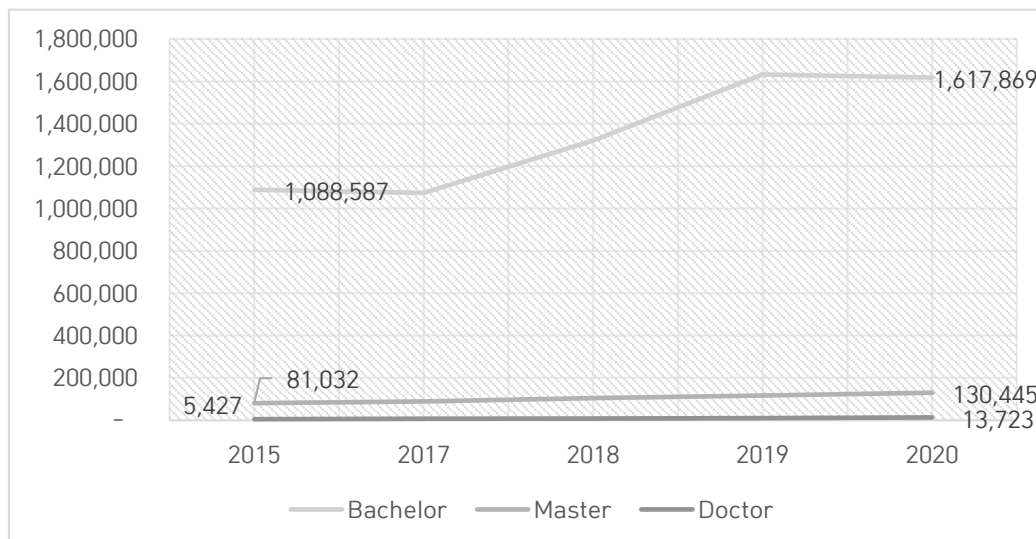
[Figure 3-3] Trend of New Entrants in Higher Education by Education Level, 2015-2020



Source: BPS (2016); Kemenristekdikti (2017, 2019); PDDikti (2018, 2020).

Over a period of five years, the number of new students in higher education continued to increase. In 2015, there were 1,088,587 new bachelor's level students at public and private universities, 81,032 master's degree students, and 5,427 doctoral degree students. By 2020, the numbers had risen to 1,617,869, 130,445, and 13,723, respectively.

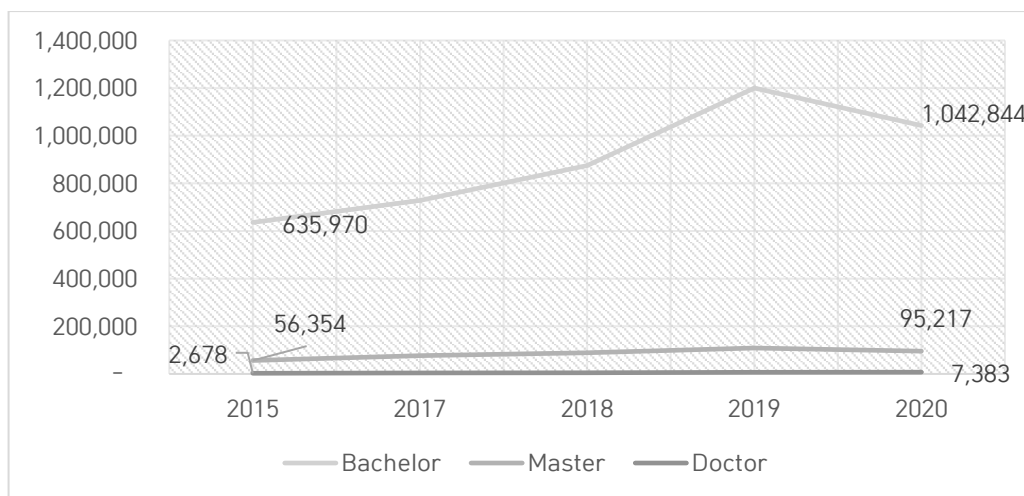
[Figure 3-4] Trend of Higher Education Graduates



Source: BPS (2016); Kemenristekdikti (2017, 2019); PDDikti (2018, 2020).

Over a period of eight years, this trend remained positive. In 2012, 689,564 students graduated at various levels of education, including diplomas. A significant increase occurred in 2019, with 1,757,169 graduates, while a slight decrease occurred in 2020, with 1,535,074 graduates.

[Figure 3-5] Trend of Higher Education Graduates based on Education Level, 2015-2020



Source: BPS (2016); Kemenristekdikti (2017, 2019); PDDikti (2018, 2020).

The number of bachelor's degree graduates was 635,970 in 2015, but this was expected to rise to 1,042,844 by 2020. The same happened at the doctoral level, with the number of graduates increasing from 2,678 in 2015 to 7,383 in 2016. By way of contrast, there were 56,354 master's level graduates in 2015, which rose to 108,665 graduates in 2019, before falling slightly to 95,217 graduates in 2020.

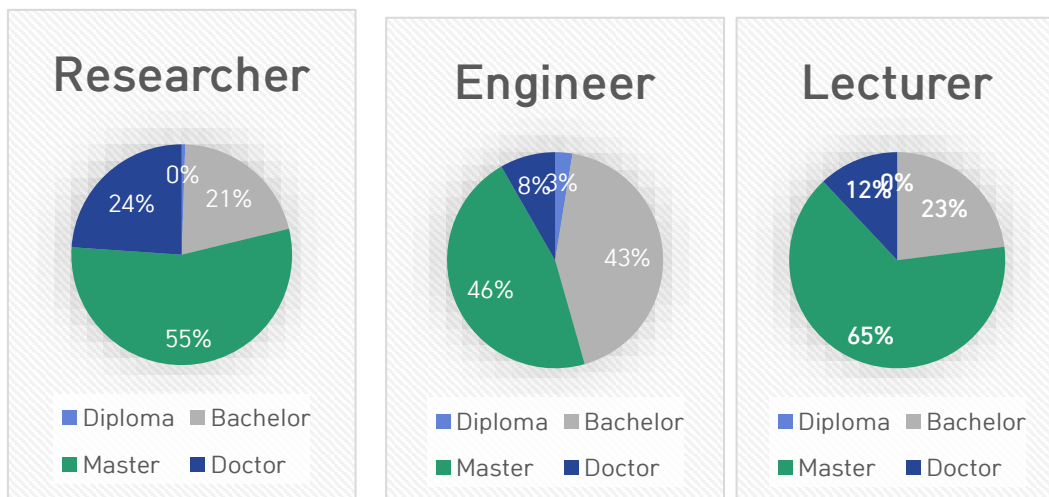
3.3 Scarcity of Highly Qualified HRST

We divide the HRST of government institutions into researchers and engineers who work in the field of science and technology. According to LIPI regulation no. 20 (2019), a researcher is a civil servant who is assigned authority, rights, responsibilities, and obligations by the competent authority to carry out the full range of duties including technical tasks related to science and technology research, and development or study at government-owned research and development organizations. Meanwhile, an engineer is a civil servant who is

assigned authority, rights, responsibilities, and obligations by authorized officials to carry out the full range of duties related to engineering activities in a functional working group in applied research, development, engineering, and operation, according to BPPT regulation no. 105 (2013). According to data collected in 2020, 6,797 researchers were expected to be working in government institutions by 2022, while 2,894 people would be working as engineers in government institutions by 2022.

Most HRSTs (researchers, engineers, and lecturers) are educated to the master's degree level. However, the proportion of doctoral holders is still tiny, with researchers accounting for 24%, engineers accounting for 8%, and lecturers accounting for 12% (See [Figure 3-6]).

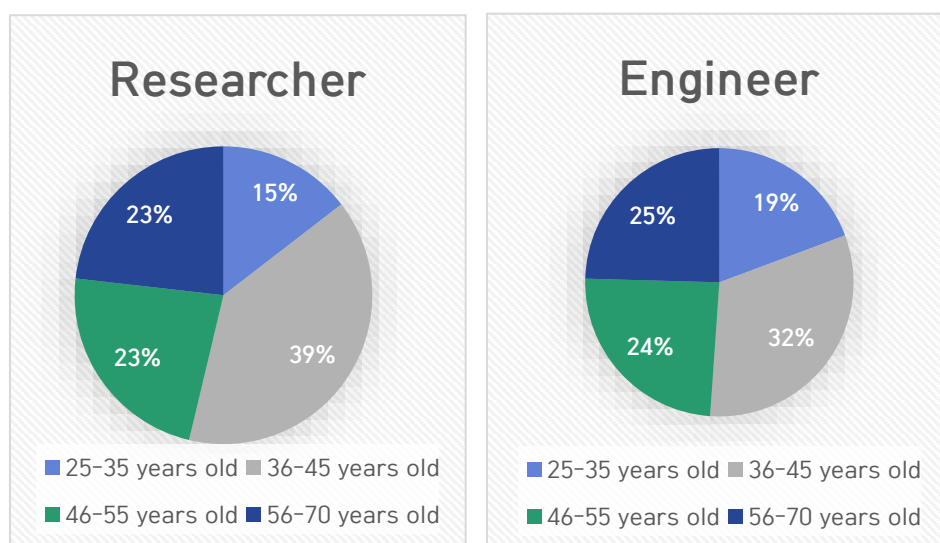
[Figure 3-6] Proportions of HRST based on Educational Background



Source: Data Processed by Researchers from the LIPI Database (2021)

Master's degrees are the most common educational background for researchers, accounting for or 55% (3,729) of the total population of researchers in government organizations. This is followed by researchers with a doctorate 24%; 1,626) researchers, researchers with bachelor's degrees (1,407) and researchers with diploma degrees (35). Furthermore, 46% (1335) of all engineers at government institutions hold a master's degree, followed by those holding a bachelor's degree (43%; 1,244), those holding a doctorate degree (240), and those holding a diploma degree (75).

[Figure 3-7] Proportions of HRST by Age



Source: Data Processed by Researchers from the LIPI Database (2021).

[Figure 3-7] compares the ages of researchers and engineers working at government institutions. Researchers aged 36 to 45 years old (2,662) accounted for the largest share, followed by those aged 56 to 70 years old (1,577), those aged 46 to 55 years old (1,569), and, finally, those aged 25 to 35 years old (989). In addition, HRSTs aged 36 to 45 years old accounted for the largest share of engineers working at government institutions (32%; 920), followed by those aged 56 to 70 years old (25%; 711), those aged 46 to 55 years old (704), and those aged 25 to 35 years old (559).

4.

Developing STI Human Capital in Korea

4.1 S&T Human Resources Statistics of Korea

Korea's S&T human capital has been continuously expanding ever since the 1960s. This study examines 1) changes in the number of R&D manpower, 2) changes in the number of science and engineering manpower, 3) the status of doctoral degrees in science and engineering personnel trained at home and abroad, 4) the trend of nurturing human resources in elementary and secondary education, and 5) the current status of the workforce in the field of industrial technology.

4.1.1 R&D Manpower

[Table 3-2] shows the changes in the number of researchers in terms Korea's R&D manpower. Looking at the size, total R&D manpower amounted to just 2,962 researchers in 1966, whereas there were more than 550,000 by 2020. As Korea's economic and industrial development progressed, the primary research manpower shifted from public research institutes to private companies. Up until the mid-1970s, public research institutes accounted for the most significant proportion, but from the early 1980s on, the number of researchers at universities increased significantly. Moreover, since the mid-1980s, the number of researchers from government-invested institutions and private companies has increased sharply, accounting for the most significant proportion, at 71.9%, in 2020.

[Table 3-2] Trends of Researchers in Korea by Research Conductor (1966-2020)

(unit: headcount, person)

Year	Public Institutes	Enterprises	Universities	Total
1966	2,286	220	456	2,962
1970	2,458	1,159	2,011	5,628
1975	5,308	2,655	2,312	10,275
1980	4,598	5,141	8,695	18,434
1985	7,542	18,996	14,935	41,473
1990	10,434	38,737	21,332	70,503
1995	15,007	68,625	44,683	128,315
2000	13,913	94,333	51,727	159,973
2005	15,501	154,306	64,895	234,702
2010	26,235	226,168	93,509	345,912
2015	35,550	317,842	99,870	453,262
2020	41,005	401,116	115,924	558,045

Notes:

- 1) "Researcher" refers to a person who has a bachelor's degree or higher, or to a person with specialized knowledge in an equivalent degree or higher who participates in R&D projects.
- 2) Public research institute: National and public research institutions, government-funded institutions, local government-funded institutions, other non-profit institutions, national and public hospitals, private hospitals.
- 3) University: national/public universities, private universities.
- 4) Enterprise: government-invested institutions, private enterprises.

Source: National Science & Technology Information Service(www.ntis.go.kr/rndsts/selectStatsDivIdctVo.do)

4.1.2 Human Resources in Science and Engineering

[Table 3-3] shows the development of Korea's science and engineering manpower produced by higher education over a twenty-year period spanning 2000 to 2021.

The total number of human resources in science and engineering studies, i.e. combining bachelor's, master's, and doctoral degrees, increased by around 200,000 every year, with more personnel being promoted in engineering departments than in natural science

departments.

During the same period, the proportion of those who obtained doctoral degrees rose significantly compared to the trend of graduates with bachelor's and master's degrees. The number of persons who received a doctorate in natural sciences increased by 2.4 times, while the number of those who received a doctorate in engineering increased threefold. This increase in the number of Ph.D. graduates was the result of the policy to expand the supply of high-quality human resources.

[Table 3-3] Trends in Science and Engineering Graduates (2000-2021)

(unit: person)

Degree	Field	2000	2005	2010	2015	2020	2021
Total	Natural Science	61,497	61,419	56,211	61,887	60,021	59,261
	Engineering	145,595	159,401	122,608	138,930	152,279	152,230
	Total	207,092	220,820	178,819	200,817	212,300	211,491
Bachelor's	Natural Science	54,797	53,679	48,103	53,004	50,762	50,215
	Engineering	131,736	143,793	107,177	122,121	134,326	134,468
	Total	186,533	197,472	155,280	175,125	185,088	184,683
Master's	Natural Science	5,543	6,209	6,278	6,601	6,514	6,268
	Engineering	12,328	13,470	13,123	13,477	13,435	13,226
	Total	17,871	19,679	19,401	20,078	19,949	19,494
Doctorate	Natural Science	1,157	1,531	1,830	2,282	2,745	2,778
	Engineering	1,531	2,138	2,308	3,332	4,518	4,536
	Total	2,688	3,669	4,138	5,614	7,263	7,314

Note 1) Bachelor's: Graduates from a college (two years or more) or university.

Source: HRST Policy Platform (www.hrstpolicy.re.kr/kistep/kr/main.html), National Science & Technology Information Service (www.ntis.go.kr/rndsts/selectStatsDivdctVo.do)

4.1.3 Current Status of Doctoral Degree Students in the Science and Engineering Fields at Home and Abroad

[Table 3-4] shows the status of Ph.D. degree holders at home and abroad by year. In the 1960s, i.e. the early stages of Korea's industrial development, there were more overseas doctorates. As a result of the government's policy to foster highly skilled manpower, however, the number of domestic Ph.D. graduates increased every year from the 1970s on.

However, since the mid-2000s, the number of doctoral degree holders returning to Korea after obtaining doctoral degrees overseas has declined.

[Table 3-4] Number of Persons Obtaining Doctorates at Home/Abroad by Year (1965-2021)

(unit: person)

		1965	1970	1975	1985	1995	2005	2015	2021
Domestic doctorate	Natural Science	4	29	82	197	647	1,531	2,282	2,778
	Engineering	2	18	102	238	1,125	2,138	3,332	4,536
	Total	6	47	184	435	1,772	3,669	5,614	7,314
Overseas doctorate	Natural Science	6	21	30	68	206	202	88	26
	Engineering	5	9	22	119	366	417	191	62
	Total	11	30	52	187	572	619	279	88

Note: Holders of overseas degrees are those who return to Korea and report their acquisition of a degree.

Considering the number of doctors who did not return to Korea and did not report after returning to Korea, it is estimated that there could be more Korean doctorates who obtained doctoral degrees abroad in science and engineering.

Source: Statistical Yearbook of Education (Ministry of Education, Korea), Foreign Doctorate Registration (the National Research Foundation of Korea).

Those who have obtained a Ph.D. have contributed to universities' research on and training of human resources. In particular, in the early stages of Korea's industrial development, Ph.D. holders who studied abroad contributed significantly to the development of science and technology and nurtured the next generation in academia.

As education aimed at fostering the next generation of academia was promoted, the number of domestic doctorate holders among college, university, and graduate school

faculties increased. The proportion of doctoral degree holders who studied abroad was over 60% in the 1960s, but the figure has decreased markedly since the 1980s, dropping to 31.4% by 2021([See Table 3.5]).

[Table 3-5] Proportion of Foreign Degree Holders among University Faculty (All Fields)

(unit: person, %)

Category	1965	1970	1975	1985	1995	2005	2015	2020	2021
Ph.D. holders (Total)	703	1,440	2,807	9,291	29,159	49,257	70,523	72,512	73,389
Domestic	263	926	1,938	5,775	18,545	32,493	46,435	49,165	50,314
Overseas	440	514	869	3,516	10,614	16,763	24,088	23,347	23,075
Overseas degree holders (%)	62.6%	35.7%	31.0%	37.8%	36.4%	34.0%	34.2%	32.2%	31.4%

Source: Statistical Yearbook of Education (Ministry of Education, Korea).

Notably, in science and engineering, the proportion of Ph.D. holders who studied abroad is higher than the average for all fields ([See Table 3.6]). Based on the National Research Foundation's report on university research activities, among full-time four-year university professors in the science and engineering fields, the number of professors with overseas degrees is about 40%.

[Table 3-6] Proportion of Doctoral Degree Holders among Full-Time Faculty in the Science and Engineering Fields (2016-2019)

(unit: person, %)

Category		Domestic degree		Overseas degree		Total
		No. of faculty	Proportion (%)	No. of faculty	Proportion (%)	No. of faculty
2016	Natural Science	3,928	52.3	3,586	47.7	7,514
	Engineering	8,544	57.5	6,305	42.5	14,849
	Total	12,472	55.8	9,891	44.2	22,363
2017	Natural Science	3,988	52.9	3,545	47.1	7,533

Category		Domestic degree		Overseas degree		Total
		No. of faculty	Proportion (%)	No. of faculty	Proportion (%)	No. of faculty
	Engineering	8,750	58.4	6,243	41.6	14,993
	Total	12,738	56.5	9,788	43.5	22,526
2018	Natural Science	4,009	53.6	3,470	46.4	7,479
	Engineering	8,834	58.9	6,154	41.1	14,988
	Total	12,843	57.2	9,624	42.8	22,467
2019	Natural Science	4,040	54.8	3,338	45.2	7,378
	Engineering	9,172	60.4	6,026	39.6	15,198
	Total	13,212	58.5	9,364	41.5	22,576

Source: Survey Analysis Report on University Research Activities (National Research Foundation of Korea, 2018-2021).

4.1.4 Human Resources Development in Elementary and Secondary Education

During the 1980s and 1990s, the Korean economy made a paradigm shift from imitation to creative innovation. As such, a change in science and technology education was necessary to reflect this transition.

The government established the first science high school in 1983. Science high schools aim to provide research-oriented education to selected students with an excellent academic record. Science high schools have been established in every province since 1983, and some twenty science high schools were in operation as of 2021.

In 2000, the Law for the Promotion of Education for Gifted and Talented Students was enacted to provide the most talented students with the opportunity to receive special education, including science and mathematics. As of 2021, eight high schools for the academically gifted were operating across the country. In addition, various science and engineering talent nurturing institutions, such as the Education Centers for Gifted Science Students, attached to universities have been established, training around 10,000 talented

people each year ([See Table 3-7]).

[Table 3-7] Number of Educational Institutions for Gifted Science Students and No. of Recipients (2010-2020)

	No. of educational institutions for the science-gifted			No. of recipients of education for the science-gifted		
	Science High School/ School for the Science Gifted	University-affiliated Education Centers for Gifted Science Students	Total	Science High School/ School for the Science Gifted	University-affiliated Education Centers for Gifted Science Students	Total
2010	21	25	46	6,004	4,093	10,097
2011	23	25	48	6,045	4,497	10,542
2012	24	27	51	5,790	4,905	10,695
2013	25	27	52	5,649	5,263	10,912
2014	26	27	53	5,643	5,695	11,338
2015	27	27	54	6,023	5,614	11,637
2016	28	27	55	6,668	5,490	12,158
2017	28	27	55	6,777	5,531	12,308
2018	28	27	55	6,831	5,288	12,251
2019	28	27	55	6,864	4,874	11,738
2020	28	27	55	6,881	4,841	11,722

Source: National Science Gifted Information Service (<http://143.248.7.55/statistics/condition?reportSeq=1040>)

4.2 Special Act on Support for Scientists and Engineers for Strengthening National Science and Technology Competitiveness

4.2.1 Background to and Purpose of Enactment of the Special Act

In Korea, in the early stages of national economic growth, the role of scientists and engineers who returned to Korea after completing doctoral programs was significant. They studied in advanced countries such as the United States and worked in the industries of advanced countries to build their careers. These advanced human resources in the fields of science and engineering made a highly significant contribution to nurturing excellent human resources for master's and doctoral degrees in universities in Korea.

In the late 1990s, the following series of problems were raised (Kang et al., 2003: 30-36) in relation to the imbalance in the supply and demand of S&T human resources and the use of human resources in the science and engineering fields: 1) the lack of high-quality professionals in the science and engineering fields arising from a failure to respond to industrial demand; 2) the inadequate establishment of a career-path development system for scientific and technological personnel; 3) talented youth's avoidance of science and engineering subjects due to poor socioeconomic treatment compared to professional occupations such as law and medicine; 4) the problem of science and engineering researchers and graduate students switching majors to non-science and engineering fields; 5) the lack of relevant and comprehensive science and engineering statistical data and human resource databases; 6) the relatively high rates of unemployment among high-level human resources with master's and doctoral degrees after the IMF economic crisis; and 7) the lack of science and engineering personnel in public offices.

Among these problems, the avoidance of science and engineering majors by outstanding talents was given particular emphasis. As such, it is necessary to propose a legal basis for inducing excellent students to choose university majors in science and engineering and to support human resources in science and engineering. Accordingly, in April 2004, the Special Act on Support for Scientists and Engineers for Strengthening National Science and Technology Competitiveness (hereinafter referred to as the "Special Act") was enacted.

4.2.2 Overview of the Special Act

The Special Act covers all aspects of human resources in the science and engineering fields, from nurturing to promoting the use and improving the status of science and engineering personnel and building the necessary infrastructure. At the time of its enactment in 2004, the Special Act consisted of four chapters and twenty-five articles ([See Table 3.8]). Looking at the main contents, Chapter 1, as a general rule, stipulates the reason and purpose for enacting the law and the state's responsibility. Chapter 2 defines the contents of the establishment and implementation of the master plans for nurturing and supporting scientists and engineers. Chapter 3 presents ways to foster human resources in the science and engineering fields, and stipulates scholarship support for outstanding science and engineering students (Article 9), support for strengthening linkages between industrial, academic and research circles (Article 10), fostering and supporting research-oriented universities (Article 11), and re-education and retraining for science and engineering personnel (Article 12). Chapter 4 aims to promote the use of scientists and engineers and improve their status. This chapter also stipulated expansion of the appointment of scientists and engineers as public officials (Article 13), financial support for companies seeking to recruit scientists and engineers (Article 16), support for universities that operate programs to strengthen research and education in the science and engineering fields (Article 21), and the establishment of R&D job placement centers to assist scientists and engineers with employment or reemployment (Article 22).

[Table 3-8] Main Contents of the Special Act on Support for Scientists and Engineers

Chapter	Key Contents	Article
1. General Provisions	Responsibilities of the State	3
2. Master Plans For Nurturing and Supporting Scientists and Engineers	Master Plans and Annual Implementation Plans	4, 5
	Establishment and Utilization of a Comprehensive Information System for Scientists and Engineers	6
	Surveys on the Current Status of Scientists and Engineers	7
3. Nurturing Scientists and Engineers and Upgrading Their Quality	Provision of Information concerning the Promotion of Entry to Science and Engineering Universities	8
	Expansion of Opportunities for Outstanding Students to Obtain Scholarships and Restitution of Research Grants	9, 9-2
	Strengthening Linkages between Industrial, Academic, and Research Circles	10
	Nurturing and Supporting Research-Centred Universities	11
	Re-education and Retraining of Scientists and Engineers	12
	Establishment of Improvement Measures to Increase Appointments of Scientists and Engineers as Public Officials	13-14
	Promotion of Use/Employment of Scientists and Engineers through R&D Projects	15
4. Facilitation the Use/Employment of Scientists and Engineers and Improvement of their Status	Support for Enterprises' Use/Employment of Scientists and Engineers	16
	Cooperation and Expansion of Human Resource Exchanges between Industrial, Academic, and Research Circles	17
	Provision of Research Subsidies to Core Scientists and Engineers	20
	Support for Supply and Demand Programs of Scientists and Engineers	21
	Establishment of Research Career Development Centers	22
	Support for Broadcasting Programs engaging Science and Technology	23
	Support for Organizations concerning engaging in S&T	24

Source: Republic of Korea (2004).

4.2.3 Major Contents of the Special Act

Among the contents of the Special Act, this study focuses on 1) the establishment of a master plan for nurturing and supporting scientists and engineers, 2) an actual survey on the status of human resources in the fields of science and engineering, 3) policy tools for nurturing scientists and engineers and improving their qualifications, and 4) measures to promote the utilization/employment of human resources in the science and engineering fields.

First, the government should establish a master plan every five years. The contents of the master plan include

1. Nurturing of and support for scientists and engineers; development of a system of utilization during the whole life cycle.
2. Expansion of opportunities for scientists and engineers to enter public office and improvement of their treatment.
3. Support for the performance of R&D and technology transfer.
4. Expansion of exchanges of scientists and engineers between enterprises, universities, research institutes, and the central and local governments.
5. Development and use of an information system for scientists and engineers.
6. Qualitative improvement of the education provided at science and engineering universities and graduate schools; strengthening of the system linking industrial, academic, and research circles.

Second, the government should survey the current status of the nurturing, utilization, and treatment of scientists and engineers in universities, research institutes, and enterprises. Such a survey can be divided into the following three types:

1. An institutional survey of universities, research institutes, and companies that recruit and use human resources in the science and engineering fields
2. An individual survey designed to identify the work experience of those who have obtained a doctorate in the science and engineering fields.

3. A survey designed to create an index of the present situation concerning the inflow/outflow of scientists and engineers to and from other countries (Index of the balance of scientists and engineers).

The Korea Institute of Science and Technology Evaluation and Planning (KISTEP) has conducted institutional and individual surveys since 2006. The institutional survey includes employment status, job fields, salary level, welfare level, recruitment methods, education and training systems, and technology and research achievements. The individual survey includes career development, treatment and job satisfaction, intention to leave, and post-retirement plans. The survey contents may change depending on the timing and purpose of the survey.⁴

Third, the government should introduce various policy tools to foster human resources in the science and engineering fields and to improve their qualifications. First, the government should strive to increase opportunities for excellent university (graduate) students in science and engineering to obtain scholarships. However, a provision also stipulates that the government may retribute all or a part of the amount paid or cease payment. The research subsidy may have to be refunded if a person who has received a research grant 1) gives up studying or switches to another field that is not science and engineering or 2) fails to engage in industrial, academic, and research circles for up to twice the period funded by the research grant. Second, the government should provide support for fostering research-oriented universities. Examples include support for attracting foreign scholars (e.g. the World-Class University Project) and support for creating a research environment for graduate schools (e.g. the Brain Korea 21 Project). Third, the government should support the re-education and retraining of science and engineering manpower. Examples include educational program support for the Korea Institute of Human Resources Development in Science and Technology (KIRD) and the establishment of a career development center for scientists and engineers.

Lastly, the government should provide policies to facilitate the use or employment of

⁴ Statistics Korea, Statistical explanation DB (accessed on May 28, 2022)
<http://www.narastat.kr/metasvc/svc/SvcMetaDcDtaPopup.do?confmNo=105005&inputYear=2020>

scientists and engineers and to improve their status. The central and local governments should establish and promote measures to expand the appointment of scientists and engineers as public officials. The recruitment ratio of senior civil servants and civil servants with backgrounds in science and engineering has been raised. In addition, to support the recruitment of science and engineering manpower, the government subsidizes labor costs for small and medium-sized enterprises whenever they hire new science and engineering manpower.

Furthermore, since the enactment of the Special Act in 2004, the S&T policy environment has changed due to the expansion of industrial technology, changes in the labor market, the acceleration of digitalization, and convergence between technologies. As a result, a number of new issues, such as the importance of 1) strengthening education for elementary and middle school students, 2) expanding the targets of employment and start-up support, 3) devising a utilization plan for highly-experienced scientists and engineers who are retired or about to retire, and 4) establishing a lifelong learning system for science and engineering personnel, are all being discussed (Lee et al., 2020: 19-65).

[Table 3-9] Summary of the Major Contents of the Special Act

Category	Contents
5-year Master Plan	1. How to nurture scientists and engineers 2. Expansion of opportunities to enter public office and improvement of the treatment of scientists and engineers 3. Support for the performance of R&D and technology transfer 4. Expansion of exchanges of human resources 5. Development of an information system 6. Qualitative improvement of education, strengthening of industry-university-research cooperation
Survey of Current Status	1. Institutional survey of universities, research institutes, and companies 2. Individual survey to identify the work experience of those who have obtained a doctorate in the science and engineering fields 3. Survey to create an index of scientists and engineers
Policy Tools	1. Provision of scholarships 2. Support for fostering research-oriented universities 3. Support for re-education and retraining of science and engineering manpower

Category	Contents
Utilization of Manpower	1. Establishment and development of measures to promote the appointment of scientists and engineers as public officials 2. Subsidizing of labour costs for small and medium-sized enterprises
New Issues after the Enactment of the Special Act	1. Strengthening of education for elementary and middle school students 2. Expansion of the targets of employment and start-up support 3. Drafting of a utilization plan for highly-experienced scientists and engineers who have retired or are about to retire 4. Establishment of a lifelong learning system for science and engineering personnel

Source: The authors.

4.3 Korean Cases of Human Resource Development

The Korean government has made considerable efforts to nurture high-quality human resources with the focus on universities and graduate schools in science and technology. This study will examine the following three Korean cases related to the development of high-quality human resources in S&T: 1) a case in which the Korean government directly established an institution for research and education (KAIST); 2) a case in which field experts were nurtured through industry-university linkage using government-funded research institutes (UST); and 3) a case in which the government strengthened the research competitiveness of public and private universities by providing them with financial support (BK 21 Project).

[Table 3-10] Characteristics of Korean Cases of Nurturing S&T Human Resources

	KAIST	UST	BK 21 Project
Legal Basis of Establishment	KAIST Law	Act on the Establishment, Operation and Fostering of Government-funded Science and Technology Research Institutions, etc.	Special Act on Support for Scientists and Engineers for Strengthening National Science and Technology Competitiveness
Jurisdiction	Ministry of Science and ICT	Ministry of Science and ICT	Ministry of Education
Target	Gifted college students, graduate students (Master's and Ph.D.)	Graduate students (Master's and Ph.D.)	Graduate students, post-doctoral students, new research personnel
Characteristics	• Established by the government (high amount of government subsidies) • Established a comprehensive education system (high schools, undergraduate and graduate schools) • Research-oriented university specializing in Science and Engineering • Admission as a single department • (undergraduate school) • Support for undergraduate course tuition fees • Payment of living expenses of master's and doctoral students (research scholarships)	• Joint operation of 32 GFRI (use of the GFRI as a campus) • Utilization of GFRI experts (employment of faculty) • Field-oriented education considering the demands of the nation and industry • Flexible creation and change of majors (responding to industry needs, current national S&T issues) • Labour contracts with research institutions (paid to all students) • Participation in national R&D projects conducted by GFRI	• University (public/private) selection for the long-term financial support project • Reorganization of the university educational system, (reform of the curriculum, emphasis on industry-university cooperation) • Reinforcement of international competitiveness of universities by fostering research-oriented universities • Research scholarships for graduate students, support for labour costs for new research personnel • Improvement of performance goals: World university ranking (QS ranking),

	KAIST	UST	BK 21 Project
	• Operation of a research support program for undergraduate/ • graduate students • Operation of 150 research institutes within schools		number of SCI papers, SCI Citation Index, Paper Impact Factor
Note	• Faculty: 646 • Staff: 944 (2021) • Budget: KRW 1.015 Trillion (2001) ; Government Subsidies 23%, Research Grant 45%, Other Income (Donations, etc.) 32% • 5 Colleges, 7 Schools, 13 Graduate Schools, 27 Divisions	• Faculty: 1,262 • Staff 72 (HQ) (2021) • Budget: KRW 25,031 million (2021) • Government Funds: 66.6% • Carry over balance 15.8% • Income 17.6% • 43 majors	-

Source: The authors.

4.3.1 Government established Top Research University: KAIST

Established in 1971 as Korea's first research-oriented university, the Korea Advanced Institute of Science and Technology(KAIST) has produced a total of 69,388 graduates (as of 2021), including 19,457 bachelor's, 35,513 master's, and 14,418 doctoral degree students.⁵ It has established itself as an excellent university, ranking 41st in the QS World University Rankings in 2022 and 16th in Engineering in the 2021 QS World University Rankings (KAIST, 2021a). KAIST primarily focuses on history, education, and research (KAIST, 2021b; KAIST, 2021c).

⁵ KAIST (<https://www.kaist.ac.kr/kr/html/kaist/01.html#0111>). Accessed on August 24, 2022

■ History

The idea for a new graduate school of science and engineering surfaced in the late 1960s, along with the rising demand for skilled engineers to enhance the nation's industrialization. KAIST was established in February 1971 as the country's very first research-intensive university under the name of the Korea Advanced Institute of Science (KAIS). KAIS was established with the technical and financial assistance of the U.S. Agency for International Development (USAID) with the full endorsement of the Korean government, and was mandated by a special law. Under the jurisdiction of the Ministry of Science and Technology, KAIS concentrated on fostering science and engineering human resources with considerable autonomy. Furthermore, with full backing from the government, KAIS was able to provide support, such as teachers' salaries at the level of advanced countries and tuition waivers for full-time students, as well as substituting mandatory military service with research.

In January 1981, KAIS was merged with Korea Institute of Science and Technology (KIST, Korea's first comprehensive government research institute created in 1966), resulting in a change of name to KAIST. The government revised the Special Law on KAIST in order to establish an undergraduate college and established the Korea Institute of Technology (KIT) in 1985. In July 1989, KAIST was separated from KIST and merged with KIT, which made KAIST a full-fledged university able to provide every level of degree course including bachelor's, master's and doctoral programs.

In 1996, the Korea Institute for Advanced Study (KIAS), Korea's first research institute devoted purely to theoretical studies on the basic sciences, was established as an affiliated institution of KAIST. In 2009, the Information and Communications University (ICU), established in 1997 under the Ministry of Information and Communication, was absorbed into KAIST. As a result, KAIST reached a critical mass for securing global competitiveness in the IT field. Also in that year, Korea's first school of science for gifted students, the Korea Science Academy(KSA), became a part of KAIST. As a result, KAIST has established a comprehensive education system for nurturing gifted science and engineering students from high, undergraduate, and graduate schools.

■ Education

Since KAIST was established and operated under the KAIST Law, it has enjoyed a relatively high degree of autonomy in student selection compared to other public and private universities regulated by the Ministry of Education.

The characteristics of the student selection process are as follows. In addition to the written test, selection is made on the basis of creativity tests and in-depth interviews. In particular, to induce high school students at the top of the science and engineering fields to enter the university, high school students are allowed to apply for the entrance examination before graduating from high school graduation, provided that they meet certain conditions. All new students are admitted into a single department, and may only choose their majors and departments freely at the end of their first year. In the case of the graduate school, admission without a written test was introduced in 1994 for the doctoral program and in 1998 for the master's program. The written test may be taken according to the policy of the department.

KAIST runs its departments by reflecting government policies in its graduate programs. It has established the Graduate School of AI and the Graduate School of Future Strategy, and also runs a specialized master's program for nurturing industrial human resources. In 2016, it introduced a master's program in entrepreneurship and innovation comprising a one-year training period in order to nurture talents with an entrepreneurial spirit.

KAIST also has a well-equipped scholarship system for nurturing talented people in S&T. In the case of its undergraduate programs, tuition fees and a small portion of students' living expenses are subsidized. In the case of its graduate programs, tuition fees are subsidized differentially according to each individual student's admission conditions. However, KAIST provides support of at least 700,000 won for master's students and one million won for doctoral students. If the allowance that graduate students receive through such activities as employment as research and teaching assistants is less than the amount specified above, KAIST provides them with additional grants to guarantee them a minimum standard of living.

■ R&D system

KAIST has been carrying out R&D projects ever since its foundation. Established by a special law, KAIST has dual status as a university and a research institute, so it has different characteristics from other public and private universities under the Ministry of Education in terms of the funding it receives for research projects. KAIST receives a large amount of support from the government through national R&D projects. Therefore, government subsidies account for a higher amount of its total budget than is the case for other universities.

KAIST provides basic research funds to all its faculty members to create an environment in which they can immerse themselves in their research. KAIST supports the research projects not only of its faculty members but also those of its graduate and undergraduate students. For example, the Undergraduate Research Participation Program (since 2006) and the Master's and Doctoral Adventure Research Program since (2012) are operated to provide students with opportunities to present creative ideas and conduct research by themselves. The annual budget is around 800 million won for undergraduate research programs, and 400 million won for master's and doctoral research programs.

KAIST has also established and oversees about 150 research centers, including the Research Center of KAIST, the Institute of Natural Sciences, the Institute of Mechanical Technology, the Institute of Industrial Management, the Institute of Applied Science, the Institute of Information Electronics, the Institute of Techno Management, and the Institute of Humanities and Social Sciences.

4.3.2 Government-Funded Research Institute Joint Graduate School: UST

■ History

The University of Science and Technology (UST) was jointly founded by twenty-two government-funded research institutes in 2003. The UST has fostered talented individuals in S&T by utilizing its outstanding experts, research knowledge and experience, and research facilities. As of 2022, thirty-two government-funded research institutes are participating. The legal basis for UST's foundation is set forth in Article 33(1) of the Act on the

Establishment, Operation and Fostering of Government-funded Science and Technology Research Institutions, etc., which is under the jurisdiction of the Ministry of Science and Technology. The main goal of the UST's educational programs is to cultivate practically-minded, creative human resources to lead development and innovation in the field of core and comprehensive technologies and industrial technologies based on field-centered education and research on new interdisciplinary and converging technologies (UST, 2020; UST, 2022).

■ Characteristics of the UST

First, there are thirty-two government-funded research institutes dedicated to fostering master's and Ph.D. students, with around 1,000 of their 13,000 researchers serving as professors. In particular, students are guided by competent doctoral-level researchers who serve as academic advisors, together with a group of experts in the field. Students can use the latest research equipment during their field-centered education. Student mentoring is very strong, with a professor-to-student ratio close to 1:1.

Second, it has the flexibility required to create and change majors. The curriculum is reorganized every 2-3 years according to changes in the R&D environment. Majors are opened according to the professional fields of government-funded research institutes related to high-tech industries and public services. The UST's majors are broadly divided into majors for national strategies and those for resolving current issues. The majors for national strategies are long-term majors aimed at developing and cultivating experts in S&T related to national strategy, whereas the majors for resolving current issues are supported in the mid- to long-term perspective with the aim of fostering S&T experts who can respond to current social demands and governmental issues. The latter majors are related to the following issues: infectious diseases; ICT fields such as artificial intelligence and machine learning; environmental fields such as particulate matter; and safety fields such as earthquake, radioactivity, materials, parts, and equipment. In addition, the UST operates an Industry Co-Operation Research Education (I-CORE) course based on an agreement it signed with industries and national research institutes to create and operate a new major's course for the recruitment of industrial field personnel and the career development of experienced workers. The course is divided into two main tracks: the recruitment type, in which industry

recruits' persons who have obtained a degree in the curriculum established by the UST to reflect industrial demand, and the re-education-type, in which industry requests the UST to train and improve its employees' job competences.

Third, the UST provides an environment in which students can immerse themselves in research and study. Students participate in national R&D projects conducted by research institutes, and sign an employment contract with the research institute to which they belong, which naturally leads to subscription to the four major insurance schemes (occupational health, employment, health, and national pension), thus protecting their rights as employees. In return, students should comply with the service guidelines. Master's students are guaranteed a minimum of 1.2 million won per month, while doctoral students are guaranteed 1.6 million won per month. In addition, the UST operates a variety of student support programs. For example, the Young Scientist Research Program provides research grants to the UST's doctoral students to help them develop their research competencies by themselves and grow as world-class researchers while handling the entire R&D project process. The Overseas Training Program supports students' participation in overseas research projects such as joint research, visiting research, and internships at renowned overseas universities or research institutes. Students also receive financial support for participation in international academic conferences through the International Presentation Support Program.

4.3.3 University Research Capability Reinforcement Project: Brain Korea 21

The Brain Korea 21(BK 21) project was introduced in 1999 to produce top-quality research outcomes while educating highly skilled individuals in advanced science and technology. As of 2022, the fourth phase of the project is in progress. This project supports graduate students' research scholarships and labour costs to enable the next generation of academics to focus on their studies and research (Ministry of Education, 2020). Regarding its budget, BK 21 was the largest government-initiated project in the education sector.

■ BK21 Project Stage 1 : (1999-2005)

This stage of the project aimed to foster world-class graduate schools and nurture international-level R&D human resources. It was promoted in connection with educational

reforms, such as reducing the quota for undergraduate courses, opening doors to graduate schools, and adopting the mandatory publication of SCI-level papers. This part of the project supported 438 project groups(teams) in 72 universities. The first phase of the BK21 project has been evaluated as laying the institutional foundations for establishing a research-oriented university system. The country's global ranking in terms of the number of SCI papers rose from 18th in 1998 (9,444 papers) to 13th in 2004 (18,497 papers).

This stage of the project produced 50,874 master's students, 23,009 doctoral students, 3,196 contract professors, and 4,913 postdoctoral students. However, certain limitations have also been pointed out, such as an over-emphasis on the quantity of SCI papers, ignorance of university-industry collaboration, lack of support in basic and interdisciplinary fields, and a widening gap in research ability between capital and regional universities (Seo, 2019).

■ BK21 Project Stage 2 : (2006-2012)

The second stage of the BK 21 project was designed to foster Korean research-oriented universities by field with the ultimate aim of placing them in the world's top ten. The goal was to achieve 10th place ranking in the national ranking for the number of publications of SCI-level papers by 2012 (13th in 2005). The second stage of the BK21 project emphasized industry-university cooperation compared to the first stage. It aimed to enter the top 10 world ranking in terms of knowledge transfer to industry (21st in 2005) and to improve the rate of university-owned technology transferred to industry. In addition, it also aimed to enhance the research power of regional graduate schools through specialization.

This stage of the project supported 569 project groups(teams) in 74 universities. As a result, the quality of research improved. Furthermore, the impact factor per thesis of the participating professors increased by 35.9% from 2006 to 2011. This stage of the project also attracted foreign scholars with high research capabilities and innovated the education and research climate through joint research (Seo 2019; Ministry of Education, 2020).

■ BK21 Plus project stage 3 : (2013-2020)

The project's third stage was carried out to strengthen the educational and research capabilities of excellent graduate schools and to foster the next generation of academics in

a drive to enhance future national competitiveness.

This stage was carried out with three objectives. The first goal was to foster global research universities. Specifically, it was expected that eleven universities would be selected within the top 200 of the QS university ranking in 2019 (six universities in 2012). The second goal was to cultivate core high-level human resources for each academic field and high-level professionals in convergence fields. The project supported 15,000 outstanding personnel with master's and doctoral degrees and new research personnel per year. The third goal was to improve the quality of domestic university education and research. Universities were requested to develop a creative curriculum. The goal was to reach 20th place by 2019 in terms of the country's ranking in the SCI Citation Index (30th in 2011)⁶.

This stage of the project supported 550 project groups (teams) in 73 universities. As a result, five universities in the top 100 QS universities and seven universities in the top 200 were selected. Regarding human resources training, an average of 1,474 graduates with master's degrees and 5,853 with doctoral degrees were supplied each year in the basic and applied S&T fields, where a shortage of human resources is expected. The number of nurtured human resources was equivalent to about 2.1% of the annual demand for new personnel (Seo 2019; Ministry of Education, 2020).

■ BK21 project stage 4 : (2020-2027)

The fourth stage of the project is being promoted with the aim of nurturing human resources in response to major social changes, including the Fourth Industrial Revolution and demographic changes. Emphasis is placed on nurturing future talents in key academic fields. It supported 578 project groups(teams) in 2020⁷.

The fourth phase of the project has three goals. First, it aims to foster world-class research-oriented universities. The goal is to place seven universities in the world's top 100 QS university ranking by 2027 (five universities in 2019) and seventy departments in the world's top 50 QS department ranking by 2027 (sixty-one departments in 2019). Second, it aims to

⁶ BK 21 Four (<https://bk21four.nrf.re.kr/sub01/sub111/list.do#>) Retrieved on August 26, 2022.

⁷ BK 21 Four (<https://bk21four.nrf.re.kr/sub01/sub111/list.do#>) Retrieved on August 26, 2022.

nurture excellent researchers in each academic and new industry field. This project will support about 19,000 outstanding master's and doctoral students annually for seven years. In addition, 1,500 new research personnel, such as postdoctoral students and contract professors, will be supported yearly. Third, improving the quality of education and research is also emphasized. Universities are required to operate educational programs related to solving industrial and social problems. It aims to reach 10th place in the global ranking of the number of citations for SCI papers by 2027(13th in 2017).

[Table 3-11] Summary of BK21 Projects (1999-)

Category	Stage 1	Stage 2	Stage 3	Stage 4
Background to the project	• Reform of graduate school education	• Nurturing of high-skilled industrial human resources	• Nurturing of human resources to strengthen national competitiveness	• Nurturing of human resources to respond to social changes, such as Industry 4.0
Period	• 1999-2005	• 2006-2012	• Sep. 2013 -Aug. 2020	• Sep. 2020-Aug.2027
Total budget (Korean Won)	• 1.34 trillion	• 1.75 trillion	• 1.9 trillion	• 2.9 trillion
Beneficiary	• 438 project groups in 72 universities (1999)	• 569 project groups in 74 universities (2006)	• 550 project groups in 73 universities (2013)	• 578 project groups (2020)
Result	• The country's ranking in the number of SCI papers increased from 18 th in 1998(9,444 papers) to 13 th in 2004 (18,497 papers).	• The Impact Factor per thesis of participating professors increased by 35.9% from 2006(2.17) to 2011(2.98).	• Five universities ranked among the top 100 QS universities and seven universities among the top 200.	• -
Notes	• Fostering research oriented graduate schools	• Emphasizing industry-university cooperation	• Supplying human resources in the basic and applied S&T fields	• Emphasizing the quality of research

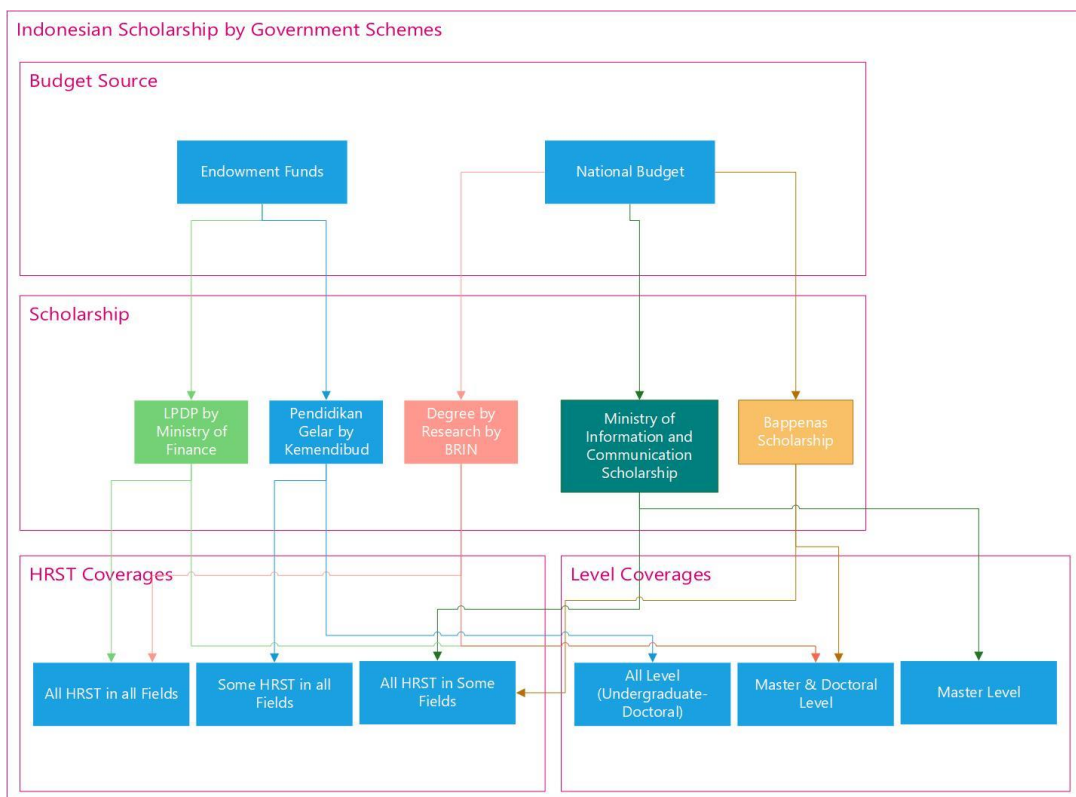
Source: Authors edited from Ministry of Education (2020). pp. 6-10.

5. Recommendations

5.1 Access to HRST Scholarship and Training Programs

Human capital is investing in both formal and informal education and training, which enhances individual productivity through the acquisition of knowledge, skills and attitudes, and the motivation necessary for economic and social development (Psacharopoulos, G and Woodhall, M,1985). In increasing the capacity of research and innovation in the governmental, industrial and higher education sectors, efforts to improve the quality of formal education, especially postgraduate education for science and technology, should be a high priority.

[Figure 3-8] Current Scholarship Program of the Indonesian Government



Source: Author's elaboration

The funds provided under the Endowment Fund scheme originate from two scholarship programs, namely LPDP, which is coordinated by the Ministry of Finance, and the “education by degree” scheme, which is coordinated by the Ministry of Education, Culture, Research and Technology. The source of the funds is the endowment fund that offers this scholarship program to all Indonesian people wishing to take master’s and doctoral education. LPDP covers various universities nationally and overseas, especially the top global and national universities with several selected study programs. To date, LPDP has produced 29,872 award recipients, of whom 61.9% are awardees from government officers. As such, the scholarship provides equal opportunities for HRSTs with diverse competencies.

In contrast, educational scholarships by degree are intended only for science and technology human resources who will work in the education sector in all fields of science and at all levels of education. The main objective of this scholarship program is to increase the ability of the education sector to improve the quality of teaching and research.

Meanwhile, scholarships sourced from the national budget are explicitly offered in specific sectors or are targeted at particular award recipients. Among them are the “Beasiswa Pendidikan Gelar” given by the Ministry of National Development Planning (Bappenas), which is intended only for government employees from planning work units, and “Beasiswa Pendidikan Indonesia,” which is only given to teachers and lecturers. There is also “Beasiswa Kominfo”, which is offered specifically for Information and Communication Technology (ICT) study programs from the Ministry of Communication and Information Technology. The scholarship programs offered by these three ministries are postgraduate degrees from domestic and foreign universities with funding sources from the ministry’s budget (Beasiswa Pendidikan Gelar and Beasiswa Kominfo) and endowment funds (Beasiswa Pendidikan Indonesia).

The latest scholarship program offered by BRIN is “Degree by Research Scholarships”. The program offers scholarships without requiring the recipient to leave their job as a researcher at BRIN. Researchers carry out research activities within BRIN under the supervision of university professors. The fundamental difference of this program is that it allows scholarship recipients to remain in the office and work as usual without having to leave their duties and obligations in order to carry out research and innovation activities within BRIN.

As awardees do not need to leave the work office, they can do their work and research and study at the same time. The scholarship is awarded to government employees who conduct research at BRIN and non-governmental employees who have recommendations from BRIN co-promoters. So the aim is not only to increase the capacity of science and technology human resources at BRIN, but also to increase BRIN's research collaboration with other parties outside BRIN. BRIN cooperates with several universities both at home and abroad. Until now, BRIN has collaborated with nine national public universities and two overseas universities. The program is implemented through a cost-sharing funding scheme between BRIN and the research center under BRIN. Research activities for educational purposes are supervised by two supervisors from the university and BRIN. Since 2019, this scholarship program has welcomed 836 awardees, of whom 76 will graduate in 2022.

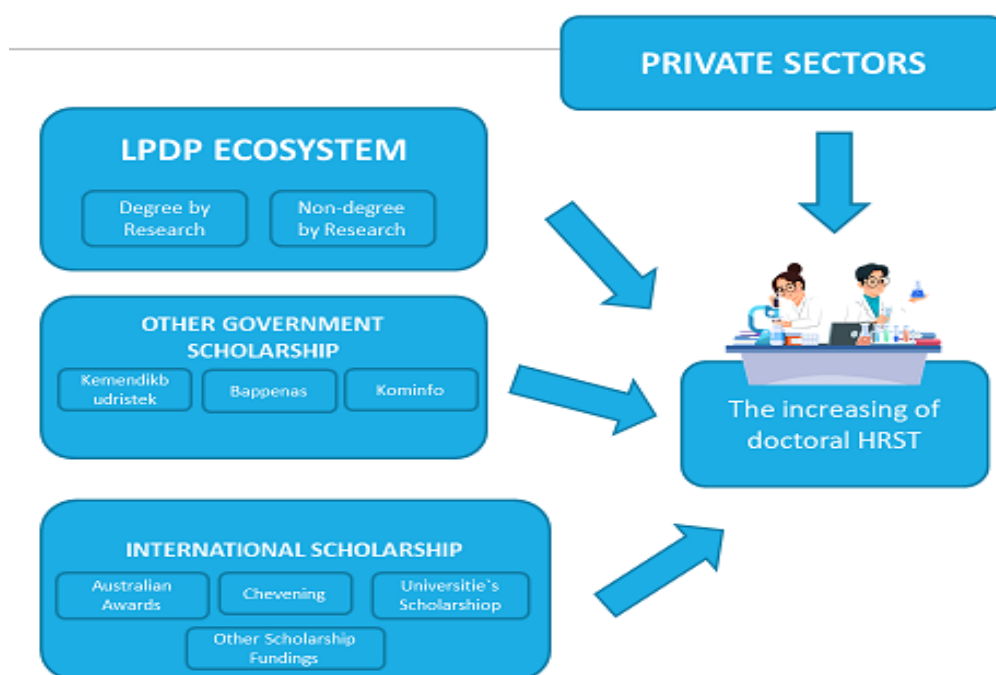
The Indonesian government has made various efforts to increase the capacity and quality of human resources. However, there are several obstacles to implementing the scholarship program, such as the limited number of universities and study programs, and the fact that funding sources are dominated by the government. To improve the quality of research and innovation, the government needs to integrate individual scholarship schemes with national and inter-governmental priorities. Currently there are two popular and appropriate HRST scholarship schemes. The degree by research scholarship is run by BRIN, while the LPDP scholarship is run by the Ministry of Finance. According to the findings of this study, incorporating BRIN's "degree by research" into the LPDP ecosystem could provide effective scholarships for researchers, lecturers, and engineers and thereby help increase the number of doctoral HRSTs in Indonesia.

In addition, BRIN's capacity building and training programs need to be expanded to include, for example, workshops, international training sessions, and conferences among other methods. A 'non-degree by research' program for collaboration between BRIN and foreign institutions could also be attractive. BRIN should support talented students in their efforts to obtain degrees from universities in developed countries. In the early stage of STI human resource development in Korea, doctoral degree holders from foreign universities played an important role in strengthening research capability (See [Table 3-4]). In the Korean case, holders of foreign degrees, upon completing their studies abroad, have returned to Korea

to contribute to research, the transfer of advanced technologies, and the education of domestic doctoral-level researchers.

In addition, the improvement of science and technology human resources is still available in the form of support programs operated by other ministries and international scholarship opportunities. Meanwhile, the private sector, which currently needs to place higher priority on the development of HRST, should increase cooperation, especially with BRIN and universities, as this will improve business processes in the private sector.

[Figure 3-9] Recommendation: Availability of Advantageous HRST Scholarship and Training Programs



Source: Author's elaboration

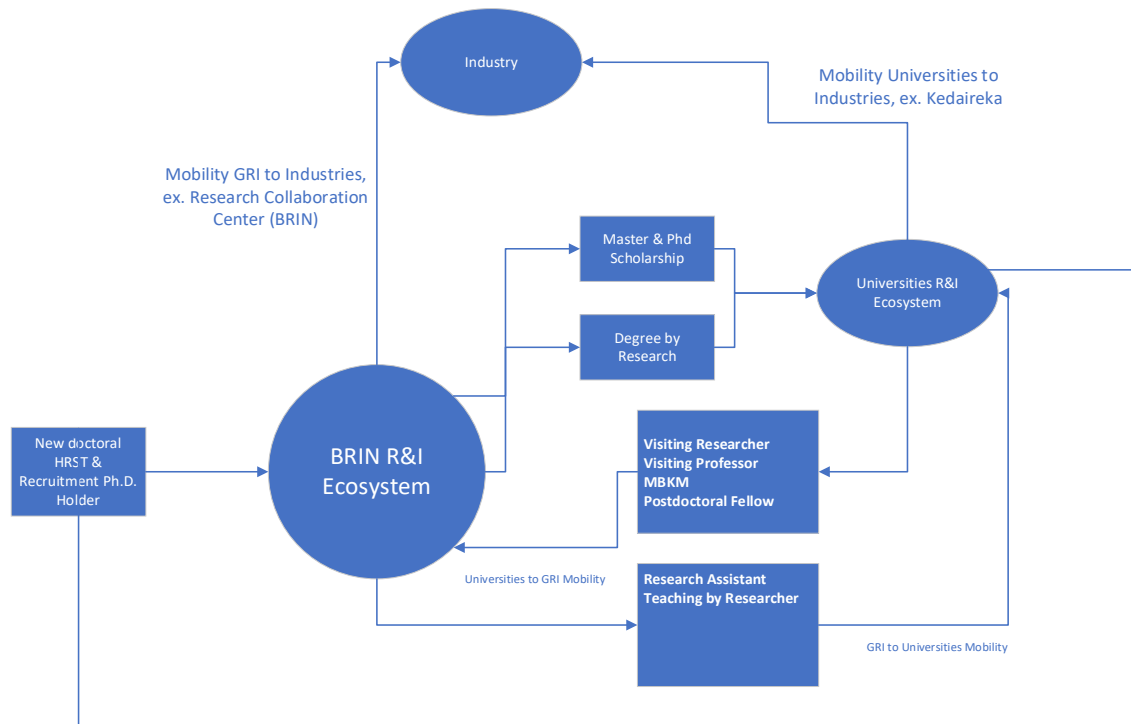
5.2 Optimization of HRST Capacity through Knowledge Transfer by Mobility

A thorough understanding of the value of collaboration also requires appreciation of the mobility of scientists. There are many studies on technology transfer achieved through the mobility of researchers between firms (Levin, Klevoric, Nelson, Winter (1987); Almeida,

Kogut (1999); Cohen, Nelson, Walsh (2002) as cited in Kim (2012: 7). However, the majority of research institutions in Indonesia do not currently prioritize researcher mobility.

To change this practice, some government organizations have developed plans for mobility promotion. Included are the Ministry of Research and Technology and Higher Education, the National Research and Innovation Agency, and the Ministry of Administrative and Bureaucratic Reform. However, the execution of such plans has yet to be optimized. Some regulations are already enforced and running well, but they only apply to the work rotation and transfer procedure.

[Figure 3-10] Current Mobility Scheme between BRIN-University-Industry



Source: Author's elaboration

Moreover, opportunities to work with industrial organizations will help government researchers to build channels for cross-sectoral collaboration. Therefore, the mobility programs will encourage government researchers to perform their duties at universities or in the private sector. University academics also want to have opportunities to transfer from their institutions to the private sector or governmental research organizations. The mobility

programs will ultimately improve the performance of research outcomes, including the development of human resources.

Several previous studies have examined Indonesia's particular circumstances with regard to mobility schemes and pointed out a number of factors that acts as constraints on system-wide research cooperation (e.g. Aminullah, 2020; Siregar, 2020; Ekatjahjana et al., 2019, Hertz et al., 2020; Pellini et al., 2018; and Rakhmani et al., 2020). These studies focus on the following hindrances:

- National research institutions lack resources or do not perform at internationally competitive levels.
- The national system regulates governmental, private, and Islamic institutions separately and has an excessive number of low-quality universities.
- There is a lack of financial incentives for universities to perform research as they believe it is less valuable than teaching.
- State university and government research institutions are staffed by public servants (with all the inherent rigidity and incentive issues that this entails).
- The private sector does not support domestic research capabilities.
- A collection of non-governmental organizations, think tanks, and policy research institutions act as "intermediaries" with variable skills, precarious financing, and a "servicing government" mind-set (e.g. Pellini et al., 2018).

Government researchers are frequently asked to cooperate with the private sector, specifically industrial companies. Though they are still determining whether the private sector will engage in active collaboration and funding research projects, researchers from the government sector see many possibilities for collaboration with the private sector (e.g. CCPHI, 2019: 20).

In our study, participants from the public sector were primarily interested in collaborating with the private sector, and cooperation with industry is in fact an effective way to increase research budgets and gain access to research facilities in the private sector. The participants suggested the "80/20 rule," which stipulates that 80% of the fund should come from the private sector and the remaining 20% from the government.

This rule is based on the understanding that Indonesia is significantly underinvesting in research compared to other countries at present, but that it could replicate the success of countries such as the United States, Germany, South Korea and Japan, where the industrial sector contributes more than 70%. However, it is important to note that these countries have robust industrial sectors, which causes concern in Indonesia when using foreign cases as benchmarks.

In this study, the business participants had considerably more divergent perspectives on research collaboration than did the government participants. They emphasized that cooperation is already taking place, albeit on a small scale, and believed that businesses already have appropriate research and development (R&D) departments to support their operations. However, both parties are concerned when a partnership is suggested because each party's goals differ significantly. The industrial sector typically aims to market the findings of its research. Therefore, it will select research avenues that support this objective. Conversely, many of the initiatives investigated by academics at research institutes or universities lack a high "innovation readiness level (IRL)" or "technology readiness level" (TRL). Participants from the industry argued that the business sector would be uninterested in sponsoring experiments whose outcomes would not be suitable for large-scale manufacturing or commercialization.

The Kedaireka (Indonesian Sovereignty in Design) initiative is a collaboration platform for universities and industry that also runs a flagship program called the "Matching Fund", which is financed by the Indonesian Ministry of Education and Culture⁸. The Kedaireka 2022 Matching Fund program provides more than IDR 1 trillion for higher education and vocational programs, with the aim of supporting collaboration on research and innovation activities between universities and the related industries. The program, which is centered on strategic state and societal issues, encourages academic communities to innovate to address these issues. The Kedaireka initiative seeks to accomplish a close penta-helix partnership in the Tri Dharma of Higher Education, to downstream the outcomes of the Tri Dharma of Higher Education, and to support DUDI (Industrial World Business) through

⁸ <https://www.ui.ac.id/en/encourage-universities-with-industry-ui-holds-kedaireka-funding-socialization/>

education, research agendas and community service in higher education.

It is important to note that Kedaireka is a matching or bridging fund. It provides the financing required for lecturers and researchers to build collaborative partnerships with commercial partners and to lure further financial support from those partners for research projects. Upon completing a scientific article, the research money would still be provided. It is envisaged that this program will pave the way for the continuation of existing research and the commercialization of its findings.

Therefore, the following are the critical notes that must be completed immediately:

1. Adopt the “lessons learned” from Korea in promoting university-industry collaboration by establishing a system for mentoring employees based on the type of collaboration between businesses and universities, as well as retraining and developing human resource capabilities in science and engineering. BRIN, the Ministry of Education, Culture, Research and Technology, and the Ministry of Finance have implemented the Research and Innovation Talent Management program, whose objectives include recruiting Ph.D. holders for research positions, fostering their development as researchers, and facilitating their mobility through Independent Learning-Independent Campuses (MBKM), Research Assistants, Postdoctoral Fellowships, Visiting Researchers, and Visiting Professor programs.

This program involves BRIN, the Ministry of Education, Culture, Research and Technology, Universities, and the Ministry of Finance.

2. Implement the lessons learned from Korea in industry-university collaboration in order to commercialize the developed technologies. The South Korean government provides financing for research institutes or universities to encourage small and medium-sized businesses' participation. In Indonesia, the Kedaireka initiative has become a collaboration hub + enabler for numerous local and international stakeholders, particularly local industries, to enter R&D-based product creations and social engineers. Thus, Kedaireka can encourage researchers from government research institutions to transfer technology to SMEs, industries, and local governments, and to secure matching funds from the private sector.

This program involves BRIN, the Ministry of Education, Culture, Research and Technology, Universities, the Ministry of Finance, Local Government, industry representatives, and SMEs.

5.3 Creation of More Flexible Regulations for Government Researchers

Judging by the results of the data analysis, there are various conditions that are specific to Indonesian researchers. First, researchers and lecturers at state universities in Indonesia have the same personal status as civil servants, which means that they also follow the same rules as civil servants, despite the obvious differences in their work. The rules regarding civil servants are made by KemenPANRB and BKN, and they are also affected by government regulations and other regulations. However, as researchers are professionals who require special competencies in the scientific field, the recruitment process should differ from that of administrative employees.

Second, certain regulations on HR development sometimes conflict with civil servant regulations, such as start-up ownership, mobility, and issues that require flexibility for HRST. Concrete examples in government R&D institutions and universities and the budget management system tend to be rigid, as they are tied to the system's rigidity in the bureaucratic organization, and start-up ownership is limited to researchers. In the past, government R&D institutions such as BPPT, LIPI, BATAN, and LAPAN had incubation institutions which were work units attached to the hierarchical organizational structure. Hence, funding these incubation institutions was still a burden on the state budget.

Meanwhile, the income pattern is based on Non-Tax State Revenue (PNBP), in which the system must deposit with the Ministry of Finance and cannot be directly managed by the incubation institution. The current condition is not much different from that experienced by Pusyantek – BRIN or the university's incubation institute. For this reason, a middle way is needed to bridge the regulatory rigidities, as this will increase the technology transfer pattern.

Technology transfer refers to the transformation of research activities into actual applications (vertical) or the movement of what occurs in one field to another (horizontally) (Grosse, 1996). Based on Law No. 18 of 2002, technology transfer is the ability to use and master science and technology between institutions, agencies and people within the country or overseas. Technology transfer can be realized through a joint venture mechanism or a cross-licensing agreement (Grosse, 1996).

Another form of technology transfer is possible through the Technology Transfer Office (TTO), an intermediary institution that facilitates technology transfer processes by bridging the gap between research and innovation (Ristekdikti, 2017). Through the TTO, government or university R&D can facilitate the formation of start-ups by utilizing products and technology produced by R&D institutions or universities (spill over). They can generate royalties for research inventors, and also generate funding sources, as well as facilitate Intellectual Property (IP) for downstream R&D products.

The TTO can be a Public Service Agency (BLU), i.e. it can be more flexible and not bound by the rigidity of the ASN rules; for instance, in managing the budget autonomously. Regulations regarding the Public Service Agency (BLU) are regulated in Government Regulation No. 72 of 2012. Professional positions in the TTO can be filled by the State Civil Apparatus (ASN) and Non-ASN, including young professionals or entrepreneurs. The portion of non-ASN human resources could be bigger, considering that rigid bureaucratic rules limit the work pattern of ASN.

The TTO could also serve as an autonomous institution that connects R&D institutions with industry (i.e. building corporate relationships with local businesses) because the most important thing is how the R&D results (spill over) can be used by business partners (industry). With the right business partners, R&D institutions and universities could produce the technology products that users need. For this reason, it is vital to have the right basic program and a good program evaluation.

It might be too extreme to convert researchers into third-sector workers (as is the case in Korea and other countries). Nevertheless, regulations that constrain lecturers and researchers need to be made more flexible. In Korea, researchers and civil servants are

separated, with the former situated in the third sector, between private employees and civil servants. Therefore, each GRI has the flexibility to determine the needs, recruitment and pay, and to build an independent staffing system.

From the conditions described above, then, one recommendation would be to construct a more flexible "government researcher" system. This means building a flexible ecosystem for "government researchers" supported by legislation. The results of the SWOT analysis and the recommendations for constructing a more flexible "government researcher" are as follows:

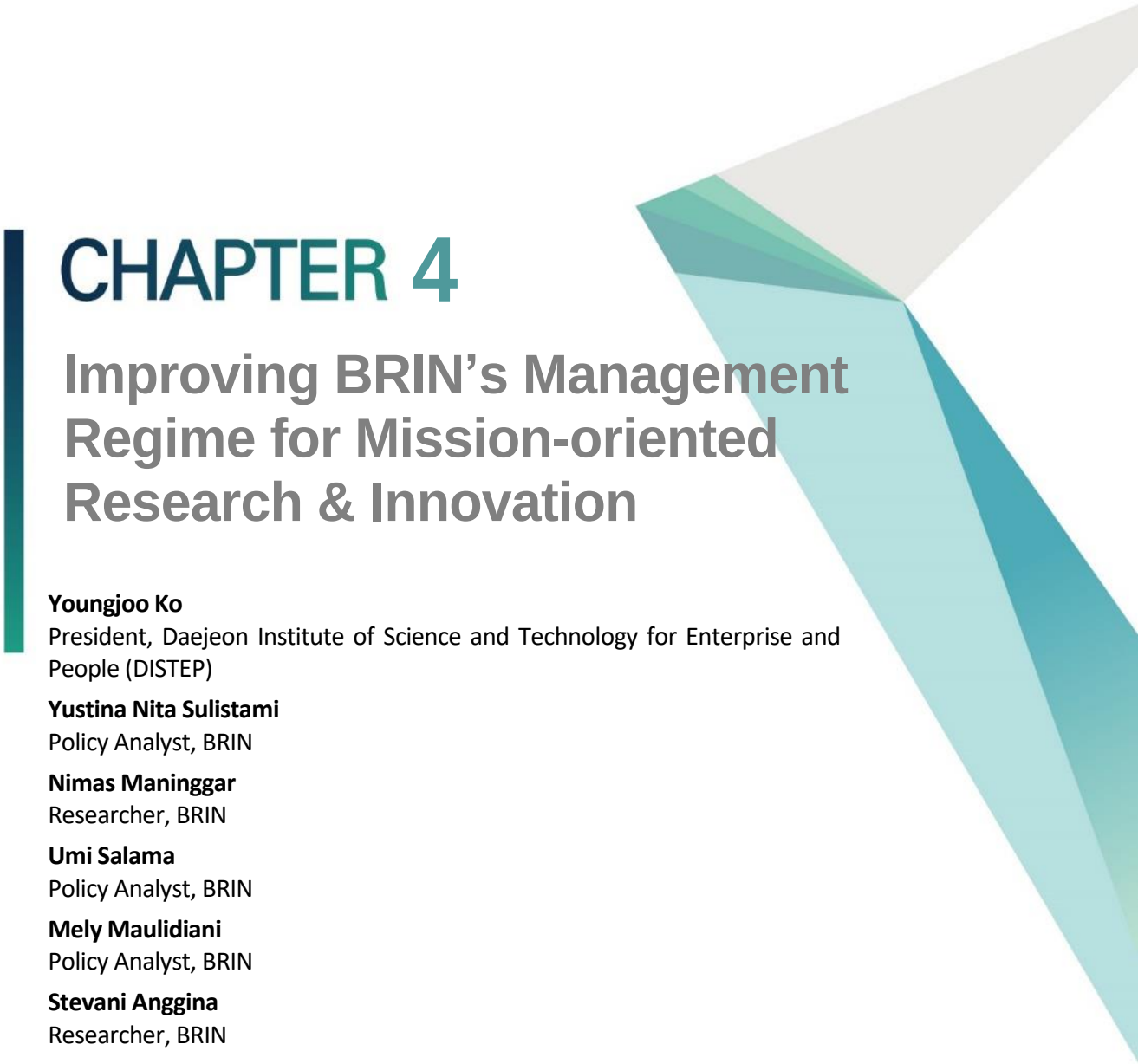
The strength of this recommendation lies in the fact that there is a fairly large number of researchers today and that members of one BRIN Institution can strengthen their position as researchers in order to voice the interests of researchers. Its main weakness, however, lies in the fact that there are still only a few people among BRIN internals who agree with this idea. Proving that the new system will generate more effective research results will be a long process. There have been some comparisons and lessons learned from other countries about applying rules to the position of researchers that differ from those applied to government administrative officials. This can give a good idea of how a country will implement changes in the rules for researchers.

The institutions involved are ones that directly prepare, discuss, and follow up on recommendations from BRIN itself, Bappenas (State Ministry for Development Planning/National Development Planning Agency), BKN, KemenPANRB, DPR, and the President.

6. Conclusion

This study points out that Indonesia urgently needs to take bold steps to promote the development of HRST. More particularly, it needs to increase the number of doctoral researchers as this is essential for increasing the nation's R&D capacity. One approach would consist in coordinating a variety of scholarship programs conducted by different ministries and tailoring them to fit the needs of researchers and graduate school students. Furthermore, it is necessary to upgrade the 'Degree by Research' program offered by BRIN so as to make it a better program that integrates other funding sources, such as LPDP, which is dedicated to HRST.

In addition, collaborations involving public research institutes, universities, and industry are critical to enhancing Indonesia's HRST competencies and capacities. Facilitating mobility by exchanging knowledge and technology currently a major problem for Indonesia. However, mobility can be promoted by targeting funding schemes and by building collaboration platforms. Lastly, a more flexible application of government civil servant regulations to government researchers would facilitate mobility and benefit the development of HRST.



CHAPTER 4

Improving BRIN's Management Regime for Mission-oriented Research & Innovation

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Chapter 4. Improving BRIN's Management Regime for Mission-oriented Research & Innovation

1. Introduction

Innovation is one of the main factors behind the economic growth and competitiveness of nations and industries. Indonesia ranked 87th among the 132 countries included in the Global Innovation Index 2021, falling two places from the previous year.

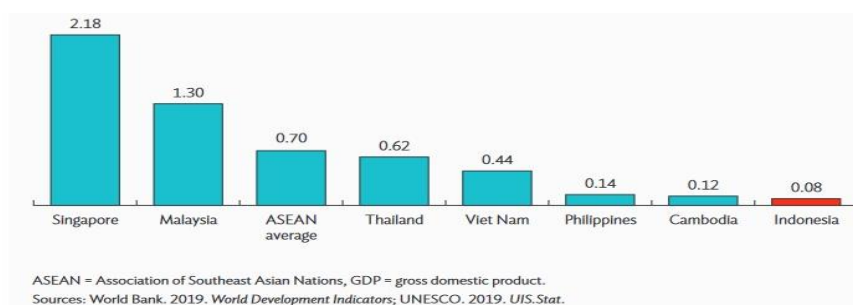
[Table 4-1] Indonesia's GII Ranking, 2019–2021

	GII	Innovation inputs	Innovation outputs
2021	87	87	84
2020	85	91	76
2019	85	87	78

Source: GII (2021).

The figure below shows the R&D expenditure of eight ASEAN countries as a percentage of Gross Domestic Product (GDP) in 2017. With R&D expenditure accounting for just 0.08% of its GDP, Indonesia occupies the last position among these countries (World Bank, 2019).

[Figure 4-1] Research and Development Expenditure in ASEAN, 2017 (% of GDP)



Source: ADB and Ministry of Finance Report (2020).

Referring to the GII indicator in 2021, there are several reviews of the strengths and weaknesses of Indonesia's innovation.

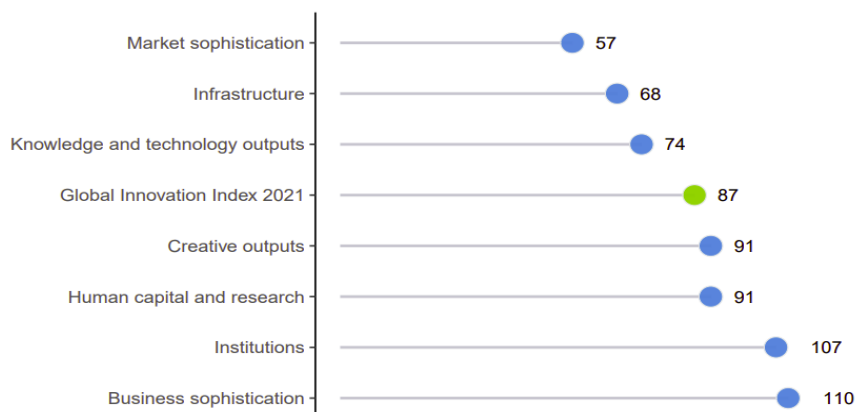
[Figure 4-2] GII Pillars, Indonesia vs Other Countries 2021



Source: GII (2021).

[Figure 4-2] above shows that Indonesia performed below the regional average in all pillars in 2021. Its best performance was in the market sophistication area, while its weakest was in the business sophistication area, and its second weakest was in the institution category with a ranking of 107th (See [Figure 4-3] below).

[Figure 4-3] GII Pillar 2021



Source : GII 2021

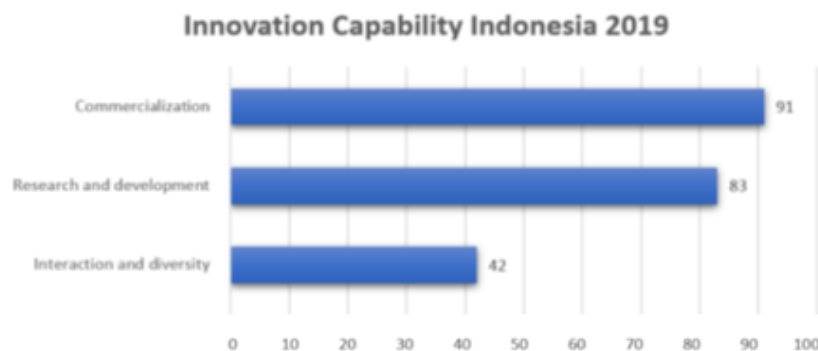
Based on the national competitiveness assessment conducted by the World Economic Forum in 2019, Indonesia's "innovation capability" pillar received the lowest score. This pillar consists of (1) commercialization, (2) research and development, and (3) interactivity and diversity. Of the three indicators, interactivity and diversity are the weakest pillars in Indonesia.

[Figure 4-4] The National Competitiveness Assessment



Source: World Economic Forum (2019).

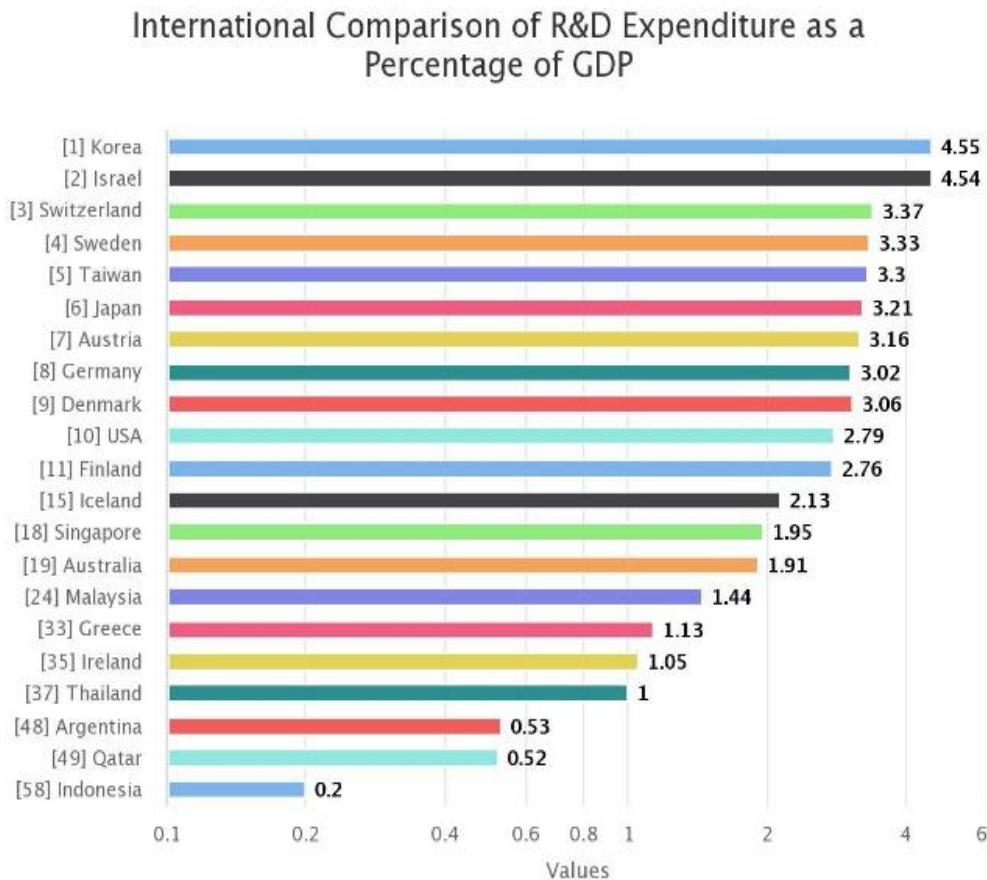
[Figure 4-5] Indonesia's Innovation Capability in 2019



Source: Global Competitiveness Report (2019).

In addition, Indonesia's gross expenditure on R&D was a mere 0.2% of GDP in 2019, which is far lower than the expenditure of other ASEAN countries such as Singapore and Malaysia at 1.95% and 1.44%, respectively.

[Figure 4-6] International Comparison of R&D Expenditure as a Percentage of GDP



Source: <https://mastic.mosti.gov.my> (2019)

According to the GII scores for 2019-2021, Indonesia's three top weaknesses, namely, regulatory environment, cost of redundancy dismissals, and government funding, all belong to the "institution" pillar. In 2019, Indonesia's "institution" indicator scored 53.2, and then decreased slightly in 2021 with a score of 51.2.

[Figure 4-7] Institution score, 2019

 INSTITUTIONS		53.2	99	
1.1	Political environment	53.9	68	◆
1.1.1	Political and operational stability*	66.7	74	
1.1.2	Government effectiveness*	47.5	68	◆
1.2	Regulatory environment	31.1	128	○ ◆
1.2.1	Regulatory quality*	39.1	75	
1.2.2	Rule of law*	37.2	82	
1.2.3	Cost of redundancy dismissal, salary weeks	57.8	125	○ ◆
1.3	Business environment	74.6	49	◆
1.3.1	Ease of starting a business*	81.2	102	
1.3.2	Ease of resolving insolvency*	67.9	33	● ◆

Source: GII (2019)

[Figure 4-8] Institution score, 2021

 Institutions		51.2	107	○
1.1	Political environment	58.5	64	
1.1.1	Political and operational stability*	68.1	74	
1.1.2	Government effectiveness*	54.7	59	
1.2	Regulatory environment	20.4	131	○ ○
1.2.1	Regulatory quality*	41.1	78	
1.2.2	Rule of law*	37.7	82	
1.2.3	Cost of redundancy dismissal	57.8	129	○ ○
1.3	Business environment	74.6	52	
1.3.1	Ease of starting a business*	81.2	108	
1.3.2	Ease of resolving insolvency*	68.1	35	

Source: GII (2021).

Therefore, this study will try to focus on institutional issues with a view to improving Indonesia's innovation system. Indonesia has made some efforts to strengthen its national innovation system in recent years; however, Indonesia's innovation system is still not optimal and has barely improved with regard to the abovementioned GII score; in other words, Indonesia's innovation system is still in the early stage of development. Indonesia's Law No. 11 (2019) on the National Science and Technology System (UU SisNas Iptek) did not provide clear directions regarding the positions and relationships between the actors in the

country's research ecosystem. Therefore, affirmation needs to be made to create and maintain interactions between producers and users and other actors in the knowledge and innovation ecosystem (1) as enablers who provide resources, establish policies, and facilitate actors, and (2) as intermediaries who knit relationships between actors and participate in (3) overseeing the transaction process of knowledge and innovation and in (4) overseeing the related products.

"The new" Law No. 11 (2019) mandates the integration of all public R&D institutions in Indonesia under the auspices of the Ministry of Research and Technology/BRIN to ensure the sustainability of one-way R&D activities in order to increase R&D capacity in national development.

In 2021, a new regulation was passed to form the National Research and Innovation Agency (BRIN). The Indonesian government restructured its STI organizations by integrating the National Research and Innovation Agency (BRIN), including BPPT, LIPI, LAPAN, BATAN, and many other R&D institutes, to reinforce the national innovation capability. Presidential Regulation (PERPRES) No. 78 of 2021, Chapter 4, stipulates that BRIN should perform the functions of (1) implementing research and development on, and promoting the application of, inventions and innovations; 2) conducting nuclear and space operations; and (3) implementing community-based coordination of research and development, assessment, and application of inventions and innovations generated by science and technology institutions. In this case, BRIN is a government research institution that plays a role in, and is responsible for, managing research activities and their results, and is required to manage research activities effectively and efficiently. This can be stated in the objectives, and implemented through strategies, governance, funding, and organization management. Such a transformation will serve to increase national and industrial competitiveness through innovation.

Learning from developed countries is the best way to improve Indonesia's innovation environment. South Korea is one of the countries to which Indonesia could refer due to its high degree of innovation and technological competitiveness. In 2019, South Korea was ranked 13th in the Global Competitiveness Index (GCI), while its gross expenditure on R&D

as a percentage of GDP was 4.81%, which is the highest level in the world.

South Korea, as one of the world's most innovative and technologically advanced countries, ranks 1st among the seventeen economies of South East Asia, East Asia, and Oceania, and ranks 5th among the fifty-one high-income group economies.

[Table 4-2] South Korea GII Ranking, 2019-2021

	GII	Innovation inputs	Innovation outputs
2021	5	9	5
2020	10	10	10
2019	11	10	13

Source: GII (2021).

[Figure 4-9] GII Institution Indicators of South Korea, 2019

INSTITUTIONS		79.7	26	◇
1.1	Political environment.....	77.2	27	◇
1.1.1	Political and operational stability*.....	86.0	21	
1.1.2	Government effectiveness*.....	72.8	28	◇
1.2	Regulatory environment.....	72.4	45	◇
1.2.1	Regulatory quality*.....	71.6	29	◇
1.2.2	Rule of law*.....	77.2	23	
1.2.3	Cost of redundancy dismissal, salary weeks.....	27.4	107	◇ ◇
1.3	Business environment.....	89.4	6	
1.3.1	Ease of starting a business*.....	95.8	11	
1.3.2	Ease of resolving insolvency*.....	83.0	10	

Source: GII (2019).

[Figure 4-10] GII Institution Indicators of South Korea, 2021

Institutions		79.5	28
1.1	Political environment	82.1	18
1.1.1	Political and operational stability*	83.9	13
1.1.2	Government effectiveness*	81.2	21
1.2	Regulatory environment	68.2	57 ◇
1.2.1	Regulatory quality*	71.5	29 ◇
1.2.2	Rule of law*	78.2	23
1.2.3	Cost of redundancy dismissal	27.4	110 ◇ ◇
1.3	Business environment	88.1	10
1.3.1	Ease of starting a business*	93.4	31
1.3.2	Ease of resolving insolvency*	82.9	10

Source: GII (2021).

As regards the “institutions” indicator, South Korea obtained a relatively high score for its “political environment” with 82.1 points in 2021, which is considerably higher than Indonesia’s score of 58.5. South Korea’s “government effectiveness” indicator has increased significantly, rising from 72.8 in 2019 to 81.2 in 2021. South Korea also obtained a notably stronger score in the “rule of law” indicator, i.e. 78.2 points, compared to Indonesia’s score of 37.7 points.

Public sector research organizations have played a significant role in supporting industrial development and technology-related institutions including human resources. There are three types of public sector research organizations (PSROs) in South Korea: government-funded research institutes (GRIs), national/public research institutes (N/PRIs), and non-profit research institutes. Among all PSROs, GRIs have been and remain the core and major R&D performers in Korea.

After the creation of the first GRI, the Korea Institute of Science & Technology (KIST), as a non-profit private contractual research institute operated by grant-in-aid and research contracts with government and industry, GRIs proliferated throughout the 1970s and 1980s with the support and supervision of the related departments and budgetary control by the central government until 1998.

After the enactment of the Act of Establishment, Operation and Fostering of Government-sponsored Research Institutes in January 1999, forty-three of the fifty-two GRIs became members of Research Councils while eight remained under the MOST, supporting or carrying out specific duties related to that ministry (as of 2003).

During the early stage, GRIs tasked with performing economic/social research and humanities were affiliated with the KCESRI and the KCHSRI, respectively, while GRIs responsible for conducting R&D in natural science were organized within three Research Councils, namely the KRCF (basic research), the KORP (public research), and the KOCE (industrial technology), based on the nature and stage of development of their S&T. All five research councils belonged to the Prime Ministers’ office when they were established in 1999. Through several stages of changes of governance, the National Research Council of Science and Technology (NST) was established in 2014 by integrating the RCs and managing

the GRIs in the field of science and technology. The NST now manages twenty-five GRIs with only its Board of Directors.

Korea's NST and GRIs are still evolving towards new missions and objectives to lead technological development, devising strategies for achieving the objectives, and pursuing better governance and funding schemes and more effective management.

Considering the abovementioned problems and development objectives, the fundamental target of the Indonesian STI system should be to learn how South Korea runs its government, institutions, and policy to support science, technology, and innovation (STI). It is very timely to address the following important questions that need to be examined: (1) How does South Korea manage the regime of government funded research institutes (GRIs) managed by the National Research Council of Science & Technology (NST)? (2) How does South Korea manage its institutions related to STI focusing on governance, funding, and technology commercialization? (3) How does the current National Research and Innovation Agency (BRIN), as an institution established to support Indonesia's STI system, achieve its mission and objectives through its current strategic goals in the context of Indonesia's framework, management, strategies, and governance? (4) What are the lessons learned from both countries' strategies for conducting effective research and developing an intermediary system?

The objectives of this research activity are as follows: (1) to evaluate the collaboration model and cross-deputy network of BRIN in acting as an intermediary institution to accelerate the creation of innovation (benchmarking South Korea); (2) to design the BRIN intermediary model based on its evaluation; and (3) to propose policy recommendations for intermediary models to support BRIN in accelerating the creation of innovation.

2. Research Framework and Method

This research covers the research of public research organizations (PRO) in South Korea, such as government-funded research institutes (GRIs), non-profit research institutes (NPRIs), and National or Public Research Institutes (N/PRIs), with a focus on research and effective intermediation (funding, governance, and commercialization of technology), which will serve as worthwhile lessons for BRIN to develop and improve its current management and governance.

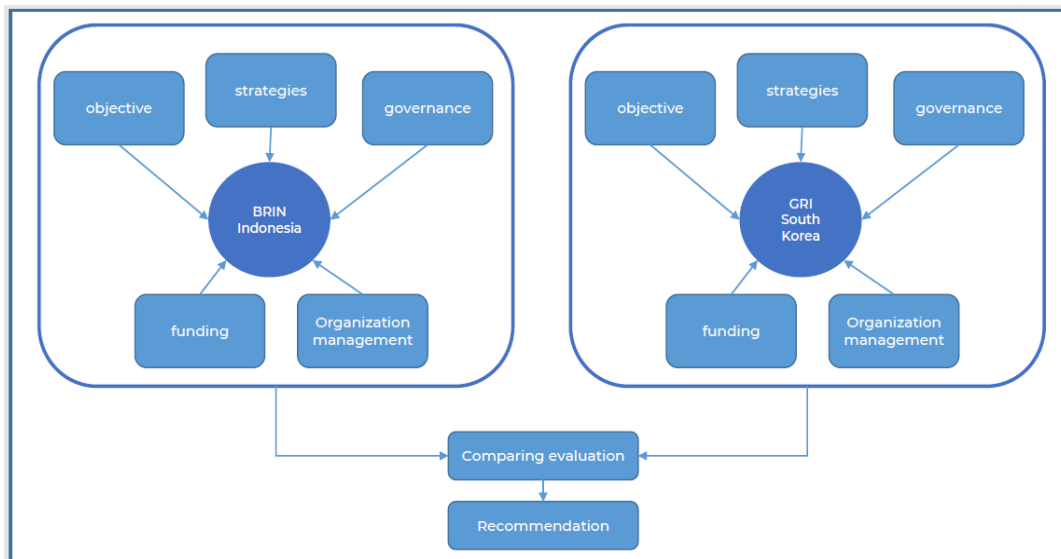
This study has been based on the framework of the management regime. The term “management regime” is commonly used to describe the system, method or way of doing business in an organization. In the scientific realm, this term is often used to describe practices in an environmental ecosystem, such as in agriculture (e.g. fertilization, farming practices, crop circulation), in forestry (e.g. conservation/protection, specie/habitat frequency), or in fisheries (e.g. fish stocks, fishing practices). Focusing on research and innovation, the management regime of such organizations encompasses those aspects that sustain its way of doing business and producing innovations. The five elements of the management regime that will be used in this study are as follows: (1) Objectives, (2) Strategies, (3) Governance, (4) Funding, and (5) Organization Management.

Generally speaking, an “objective” constitutes the main reason or purpose of forming an organization. The objective, in other words, the goal, is something that the organization collectively wants to achieve or realize in the future (Christensen, et al., 2007). Thus, in order to fulfil the organizational objectives (1), strategies (2) are set as a short-term goal, and these are meant to define the organizational objectives in a more operational manner. The objectives and strategies of the organization are then supported by its governance (3), funding mechanism (4), and organizational management (5). The concept of governance refers to the system in which organizations are directed or controlled (Cadbury Report, 1992). The funding mechanism and organization management are more closely related to the technical aspects of management regimes, which explains how an organization manages its funding resources and organizational functions.

This study activity will analyze data from comparative studies of Indonesia and South Korea based on a regime management framework that looks at aspects of objectives, strategies, governance, funding, and organizational management, and draws lessons from the results of these comparisons.

The methodology used in this study is a qualitative and comparative case study with multiple approaches. The cases selected are research institutions in Indonesia and GRIs in South Korea, with a comparative analysis in particular of the management regime of the abovementioned five elements. The comparison of these five factors in the two countries has been used as a material for the preparation of recommendations. The purpose of this study is to draw lessons from the institutional setting and management regime of STI in South Korea. Then, the policy learning can be applied to the institutional setting of STI in Indonesia in order to encourage the growth of competitiveness and innovation.

[Figure 4-11] The Framework and Methodology Used in This Study



We collected a large body of data, information, materials and previous research cases from the two countries and conducted in-depth interviews and focus group discussions in Indonesia, and seminars and discussion with Korean experts.

[Figure 4-12] In-Depth Interviews with SMEs



Source: Researcher.

[Figure 4-13] Focus Group Discussion with Stakeholders



We organized focus group discussions and in-depth interviews with the management teams and coordinators of various deputies at BRIN, and some seminars and on-line meetings with policy experts in Korea, with the focus on the elements of the management regime at BRIN and the GRIs regarding their roles and contributions to the system of research and innovation in the two countries.

The Focus group discussions were held with the following organizations:

- Directorate for Strengthening Partnerships in the Research and Innovation Infrastructure (DSPRII)
- Directorate of Research and Innovation Funding (DRIF)
- Directorate of Technology Transfer and Audit System (DTTAS)
- Directorate of Utilization of Research and Innovation by Industry (DURII)
- Directorate of Utilization of Research and Innovation by Government, Society, and Micro, Small, and Medium-sized Enterprises (DURIGS &SME)
- Center for Technology Services

The team also conducted semi-structured in-depth interviews with two start-ups, PT Indomabs Biosaintika and PT Greenie.

3. Results and Analysis

3.1 Missions and Objectives of BRIN and the NST/GRIs

The genuine development of Korean S&T began when the military government of Park Chung Hee, which took power on May 16, 1961, announced the *First Five-Year Economic Development Plan* for 1962-1966 (EPB, 1962). Accordingly, the major objectives and strategies of South Korea's S&T policies in the 1960s were to build a technological infrastructure to support the government's strategy, including the creation of a government department responsible for S&T matters, the enactment of laws to support S&T development, the creation of new R&D performance capabilities, and the development of human resources. The government decided to establish the Korea Institute of Science and Technology (KIST), a new research institute with an excellent environment and high salaries, to conduct a wide range of industrial R&D, including technological feasibility studies, technological services for small and medium companies, and engineering studies on a pilot plant scale.

The main focus was on finding solutions for simple and practical problems arising from the application of technologies transferred from abroad. KIST led Korea's early stage of technological innovation and made a major contribution to solving many problems during the process of adapting and applying imported technologies, under a contractual R&D system with business (Kim, L.S., 1997: 139). In the 1970s, as labour-intensive industries were gradually replaced by heavy industry and chemicals, it was vital to establish an S&T infrastructure capable of modifying and internalizing foreign technology in order to meet the demand for the more sophisticated production technologies required by the machinery, steel, chemical engineering, shipbuilding, and electronics industries (OECD, 1996). To meet the ever-increasing demand for qualified scientists and engineers, Korea's S&T policy focused on expanding technical and engineering education in the related fields. Furthermore, a number of GRIs were established in the 1970s in such fields as machinery, shipbuilding, chemicals, marine science, electronics, and electricity to satisfy diverse industrial needs and to improve the institutional mechanism for adapting and internalizing

imported technologies.

The main objectives of the STI policy in the 1990s were to i) build the industry-led technology innovation system and strengthen industrial competitiveness; ii) promote the strategic drive of national R&D programs both in individual departments and across all relevant departments; iii) nurture all R&D performers and promote collaboration between them; iv) strengthen the base for science and technology innovation; v) increase government expenditure on R&D; vi) prioritize efficient and effective government investment; vii) pay greater attention to basic research; and viii) promote a scientific culture focused on globalisation and localisation.

During this period, the main objectives of Korea's GRIs were switched to the pursuit of efficiency and effectiveness, the creation of a competitive environment, and increased emphasis on performance and R&D quality, accountability and autonomous responsibility, the utilisation of R&D across all departments, and collaboration with other R&D performers such as universities and industry. Then, in January 1999, the Research Council system was introduced to support these changes in the objectives of the GRIs.

With the evolution of the Research Councils, the main objectives of the GRIs under the National Research Council of Science and Technology (NST) since 2014 have been to develop new technologies, support companies and industries, contribute to regional innovation, participate in global cooperation, provide give solutions to societal problems, and support the national agenda and government policy in the field of science and technology.

As the first variable, the missions and objectives show that Korea's research council, the NST, has been leading and driving the GRIs towards securing research capabilities, fostering GRIs trusted by citizens, and establishing a researcher-friendly support system. As of the end of 2022, the NST's vision is to lead the new future of the nation along with the top GRIs, with the aim of fostering GRIs that are trusted by the general public, securing collaborative research capabilities, and establishing a research-friendly supportive system. The main mission and objectives of the NST are to support national research projects and policies and to lead future-oriented national STI, create a sustainable convergence innovation ecosystem, and strengthen social responsibility.

Meanwhile, BRIN's objectives are to support the vision and mission of the President of the Republic of Indonesia by developing Indonesia into a sovereign and independent country,

and on the value of mutual cooperation (“gotong royong” in Indonesian). BRIN’s main vision is to become a reliable, professional and innovative National Innovation Agency that supports the vision and mission of the President and Vice-President of the Republic of Indonesia. The main mission and goals of BRIN are to 1) provide technical and administrative support as well as quick, accurate and responsive analyses to the President and Vice-President in conducting integrated research and innovation, 2) improve the quality of human resources and the research and innovation infrastructure, and 3) organize effective and efficient supervision, general administration, information and institutional relations services.

[Table 4-3] Comparison of the Objectives of BRIN and NST

	BRIN	NST
Vision	To become a reliable, professional, and innovative National Innovation Agency that supports the vision and mission of the President and Vice-President of the Republic of Indonesia. To develop an Indonesia that is sovereign and independent and embraces its personality based on the value of “gotong royong” (mutual cooperation).	To lead the new future of the nation along with the top GRIs. To foster GRIs that are trusted by the public, secure collaborative research capabilities, and establish a research-friendly support system.
Mission Goals	To provide technical and administrative support as well as quick, accurate, and responsive analyses to the President and Vice-President in conducting integrated research and innovation. To improve the quality of human resources and infrastructure for research and innovation. To organize effective and efficient services for supervision, general administration, information, and institutional relations.	To support national research projects and policies and lead future-oriented national STI, create a sustainable convergence innovation ecosystem, and strengthen social responsibility.
Role	1. Assist the President in carrying out government tasks in the fields of research, development, assessment,	1. Plan research and policy for the development of GRIs.

	BRIN	NST
	and application, as well as invention and innovation; and in implementing the national energy policy and the integrated national space policy. 2. Monitor, control, and evaluate the execution of the duties and functions of the Regional Research and Innovation Agency (BRIDA) in accordance with the provisions of the laws and regulations.	2. Adjust and align the functions of the GRIs (including establishment, integration and dissolution of institutes). 3. Evaluate research performance and management (including the management of corporations in which the GRIs have invested). 4. Support cooperative research between GRIs. 5. Assist in improving research performance and disseminating the results. 6. Propose policies to innovate national science and technology and strengthen competitiveness. 7. Support the internal audits of GRIs.

Source: The Head of the National Research and Innovation Agency's Decision No. 1/HK/2022 on the Vision, Mission, Strategic Objectives, and Key Performance Indicators of BRIN, 2022-2024 (2022) and the NST (2022).

Presented below are the lessons learned from the comparative analysis:

- a. South Korea's GRIs have experienced several discernible periods of change in their missions and objectives. In the 1980s, they focused on making greater contributions to national R&D programs and achieving better results. During the 1990s, they switched non-incrementally to more transparent research projects and more fundamental research actors, introducing the Project Base System and the Research Councils system. During the 2000s-2010s, the GRIs changed incrementally in line with the evolution of the PBS and RC systems.
- b. The NST's current vision is to lead the new future of the nation together with the global top level GRIs and to support, foster and systematically direct the GRIs towards contributing more to the national R&D policy and the development of the knowledge industry. BRIN aims to be a reliable, professional and innovative National Innovation Agency that supports the vision and mission of the President and Vice-President of the Republic of Indonesia, with the support of BRIN's

Research Organizations/Research Centers (OR/PR = Indonesian version of Korea's GRIs)

- c. As BRIN's vision tends to be supportive of the current government, the priority areas of research and innovation are likely to change with a change of government. The vision of the NST is to serve as a supporter of the GRIs, so that the priority areas of research and innovation follow the priorities of each GRI.
- d. The concept of collaboration among the GRIs and the support system provided by the NST is similar to the collaboration between the Research Organization/Research Center at BRIN, which is also supported by the Deputy for the Utilization of Research and Innovation, Deputy for the Facilitation of Research and Innovation, and the Deputy for Research and Innovation Infrastructure.

3.2 Strategies

BRIN's main strategies are as follows: to increase research excellence and scientific innovation; to produce policy-based evidence that aligns with the direction sustainable development; to enhance collaboration in the development and use of science knowledge based on development priorities; to expand the application of knowledge and technology to support a high-quality environment and resilience against disasters and climate change issues; to strengthen productivity and resource competitiveness in BRIN research and innovation; and to make BRIN's governance more effective, efficient, and accountable.

The crucial strategies of the GRIs to achieve NST's vision and missions are as follows: greater focus on the development of fundamental or original technologies; greater contribution to solving societal problems; more effective technology transfer and commercialization; more cooperative research and innovation with other GRIs, universities and SMEs; closer relations with regions; autonomous and accountable management; and a more stable research environment, as shown in [Table 4-4].

[Table 4-4] Comparison of the Vision and Mission of BRIN and NST

	BRIN	NST
Strategies	1. Increase research excellence and scientific innovation and produce policy-based evidence that aligns with the direction of sustainable development. 2. Enhance collaboration in the development and use of science knowledge based on development priorities. 3. Expand the application of knowledge and technology to support a high-quality environment, resilience against disasters, and climate change issues. 4. Strengthen the productivity and resource competitiveness of BRIN's research and innovation. 5. Make BRIN's governance more effective, efficient and accountable.	1. Focus more on the development of fundamental or original technologies. 2. Contribute to solving societal problems. 3. Pursue more effective technology transfer and commercialization. 4. Promote more cooperative research and innovation with other GRIs, universities and SMEs. 5. Forge closer relationships with regions. 6. Promote autonomous and accountable management. 7. Create a more stable research environment.

Source: Head of the National Research and Innovation Agency's Decision No. 1/HK/2022 on the Vision, Mission, Strategic Objectives, and Key Performance Indicators of BRIN, 2022-2024 (2022) and NST (2022).

The strategies of the GRIs have changed along with the stages of industrial development, political preferences, new management ideas and changing positions in the national innovation system. In the first stage, the main strategy of KIST and the GRIs was to provide strong support to industry and government, mainly focusing on R&D as requested by industry, and STI policy development for the government. Throughout the 1980s and 1990s, however, the GRIs' strategies shifted toward national R&D programs that had been expanded by central government departments. Since the end of 1990s, in line with the introduction of the PBS and RC systems, their strategies have evolved toward excellent research performance and outcomes, transfer of the results (technology) of research, and commercialization in an environment characterized by strong competition between universities and industry. Currently, the NST and the GRIs are also emphasising convergent collaboration and networking not only between GRIs but also between universities and

industries.

Like BRIN, the NST sets its strategies every three years, integrating each GRI's own strategies based on its characteristic fields and research ecosystem. However, the assignment time of each president of a GRI tends to vary, and it is still quite difficult to align the strategies of the NST and the GRIs.

BRIN's research organizations and research centers also have their own target strategies based on their research characteristics. BRIN needs to take into consideration not only the shifts in the strategies of the GRIs in response to changes in the national innovation system and other changes in the environment, but also the political pledges and agendas of the regime.

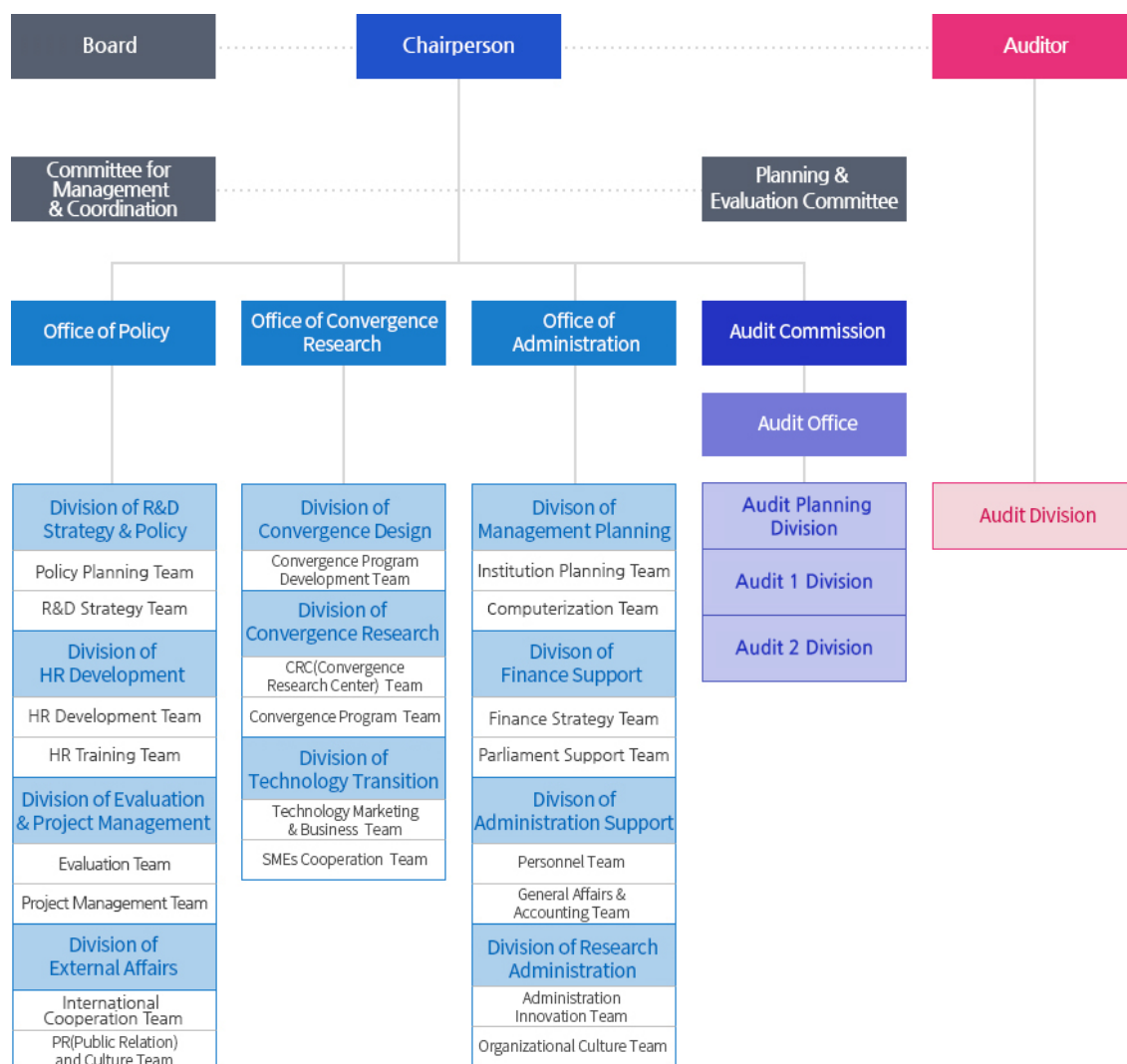
3.3 Governance

The Korean government created the Economic Planning Board (EPB) in July 1961, and then established the Technology Development Bureau, a government unit responsible for S&T policy within the EPB, in 1962. The first and second five-year plans for the promotion of S&T were formulated in line with two Five-Year Economic Development Plans for the periods 1962-1966 and 1966-1970. In addition, the *S&T Promotion Law* was enacted in 1967, and the Ministry of Science and Technology (MOST) was established as a lower level government department in 1967, in order to put science and technology higher on the agenda. The establishment of the MOST and its functions within the policy mechanism of the central government meant that S&T would become one of the national tools from the outset of national development, unlike in other less developed countries (MOST, 1997: 43).

After the creation of KIST in 1966, the enactment of the *Law for the Promotion of Specific Research Institutes* in December 1973 was designed to drive the proliferation of GRIs. After the dramatic development and expansion of the GRIs and the national R&D program in the 1980s and 1990s, the National Science and Technology Council (NSTC), chaired by the president, was launched in January 1999 to prioritize and coordinate national R&D programs and policies. The Minister for Science and Technology became the Secretariat of the NSTC, and the KISTEP was created under the MOST to evaluate the national R&D

programs and the MOST's role in allocating and managing the R&D programs. After aligning themselves with these changes in governance and policy governance, the GRIs were regrouped into three new Research Councils in order to separate them from the bureaucratic culture which inevitably stemmed from their supervision by specific departments and thereby use them beyond departmental boundaries and missions. The role of the Boards of Directors of the GRIs was transferred to the Boards of the relevant Research Councils, which have a wider range of responsibilities in all matters related to the GRIs. In the 1990s, the governance structure shifted towards a more coordinated approach with dynamic debates and the greater involvement of actors in the overall policy-making and implementation process. Then, in 2014, the Research Councils were merged into the NST and the related Framework Act was revised. Such changes gave every GRI an independent legal identity and institutional distinctiveness. None of the GRIs has an independent Board of Directors or any full-time auditors. Only the NST's Board of Directors may appoint a president to a three-year term at a GRI after an open competition. Each president occupies a full-time position and is responsible for managing the GRI in question. The NST now belongs to the Ministry of Science and ICT (MSIT), which runs a special team as a channel of communication between the NST and the twenty-five GRIs. As for the GRIs, they operate more than one hundred regional branches to support regional innovation and twelve overseas branches for international collaboration. The NST has a chairman who is appointed for a three-year term by the president, a board of directors that manages the GRIs, a full-time audit committee, and 130 staff including 2 directors with 3 offices and 12 divisions, (NST, 2020). [Figure 4-14] below shows the governance structure of the NST.

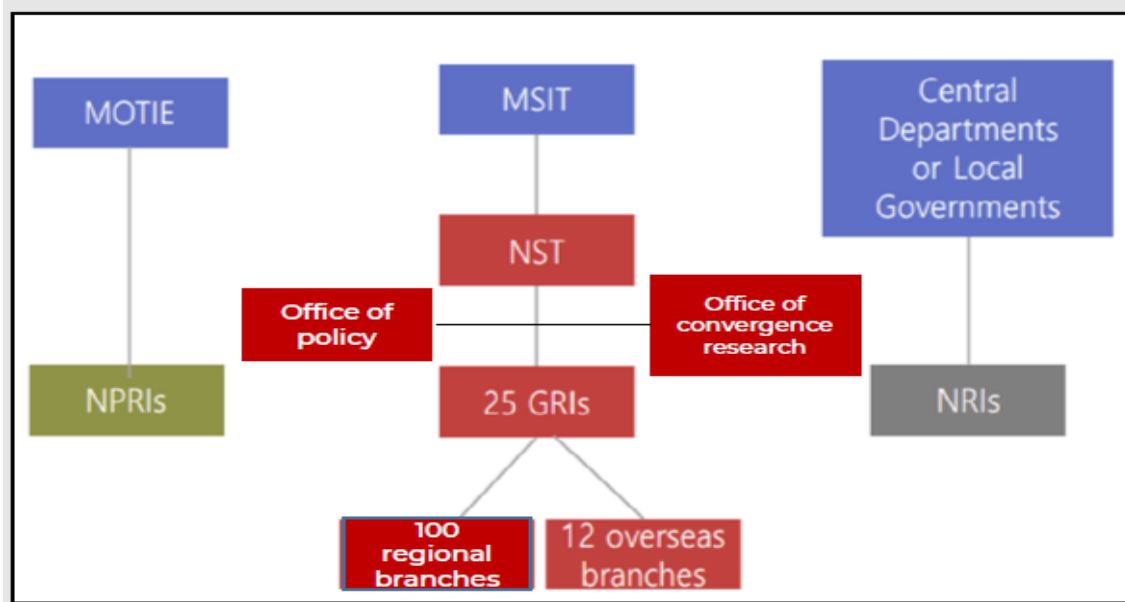
[Figure 4-14] The Governance Structure of South Korea's NST



Source: www.nst.re.kr

It is worth noting that there are fundamental differences in the institutional structure of the STI institutions of Indonesia and South Korea. The institutional structure in Korea is divided between the Ministry of Science and ICT (MSIT), which is responsible for STI policy and coordination of the R&D budget, and the NST, which serves as a management agency, along with the twenty-five GRIs, of national R&D and regional innovation.

[Figure 4-15] The Institutional Structure of STI in Korea



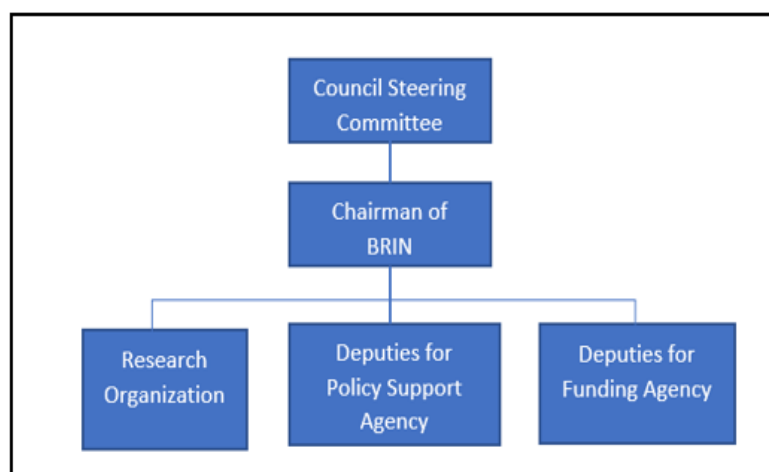
Source: NST (2022).

Meanwhile, Indonesia's BRIN is the sole organization responsible for implementing national research and innovation. BRIN has three main functional structures, i.e. deputies for policy support and deputies for the funding agency and the executing agency: The deputies for policy support include the Deputy for Development Policy (DKP), the Deputy for Research and Innovation Policy (DKRI), and the Deputy for Regional Research and Innovation (DRID). These three deputy's role is to support and develop policies based on evidence. While the DKP focuses on the national development policy, the DRID focuses on the regional development policy, and the DKRI focuses on data, operational regulations, monitoring, and evaluations to support policy-based evidence.

The research organizations (OR) act as executing agencies, which means that they conduct research and development and innovation activities. Currently, there are twelve such research organizations at BRIN, each of which is responsible for different areas of research activity. Each research organization consists of research centers with more specific research themes. The funding agency deputies consist of the Deputy for Infrastructure Research and Innovation (DIRI), the Deputy for Research and Innovation Facilitation (DFRI), the Deputy for Research and Innovation Utilization (DPRI), and the Deputy for Human Resources (DSDMI).

These deputies contribute to strengthening the national research and innovation ecosystem by providing infrastructure, funding schemes, and utilization schemes, and to enhancing the competencies of human resources in research and innovation. They also collaborate with external partners and provide matching funds in order to meet the funding requirements of the research organizations.

[Figure 4-16] The Governance of BRIN



Source: Author's elaboration

These comparative analyses have resulted in the following “lessons learned”:

- a. South Korea has established a research council system in which the NST acts as only one RC in order to support, foster, and systematically manage the GRIs with regard to the national R&D policy and the development of the knowledge industry. The NST is a private R&D entity that focuses on the GRIs with autonomy, whereas BRIN is a part of the government and an integrated entity dedicated to policy, funding, and R&D. As such, it is essential to figure out how to make BRIN's R&D organizations as effective and flexible as a private entity.
- b. Collaboration between GRIs is strongly encouraged, supported by the NST office of convergence, which manages the convergence technology development program between GRIs. It will be necessary to figure out how to make BRIN's R&D organizations more collaborative in the future with a view to developing Indonesia's technology-based industry.

- c. In sum, BRIN needs to enhance both autonomy in R&D and collaboration between its internal research institutes and deputies in order to promote convergent innovation in research and industry in the future.

3.4 Funding

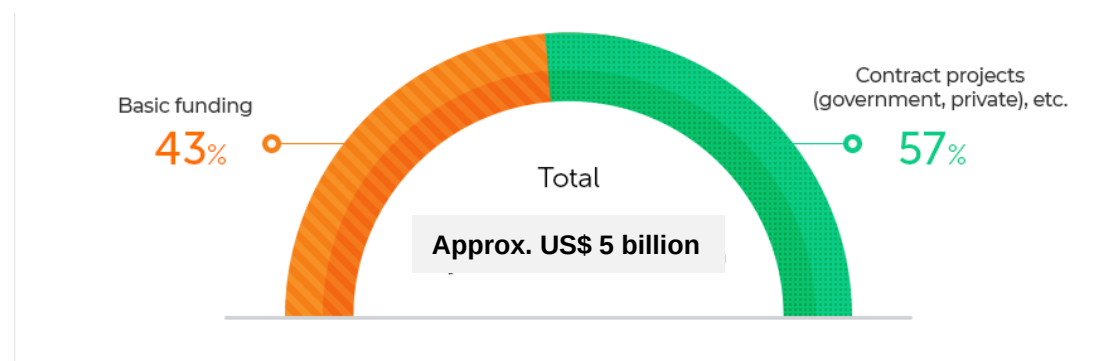
At the end of the 1970s, South Korea's total R&D investment represented about 0.8% of GNP, with government expenditure accounting for almost 70 % of the nation's total R&D investment. N/PRI and GRIs accounted for 86.4% of the total domestic R&D investment in 1969 and 56.4% in 1979, while industry's share was 10.2% in 1969 and 34.1% in 1979. Of the 86.4% for which N/PRI and GRIs were collectively responsible, GRIs carried out 51.6%. In 1973, KIST conducted about 170 projects, i.e. three times as many as in the early stage, and the portion of R&D funds originating from industry was 61%. Consequently, KIST received rave reviews as a Korean success story (MOST, 1997: 31).

The GRIs managed organizations by using grant-in-aid funding from the government and conducted research under project contracts with private companies until the mid-1990s. This kind of dual funding system for the GRIs began to change with the introduction of the Project Based System (PBS) in January 1996. At first, this system aimed to abolish the grant-in-aid funding and to introduce a new system of funding based on the full-cost pricing of all R&D projects, including the salaries of staff participating in R&D projects and the operating cost of the GRIs. The government decided to keep less than 30% of the institutional funding for the least stable operations of the GRIs. Since the establishment of the Research Councils in 1999, the funding system based on PBS has continue to evolve. In one sense the introduction of the PBS resulted in a highly unstable funding system and too much open competitive contracting. To remedy the problem of unstable GRIs that lacked institutional funding, the Finance Ministry started allocating core institutional R&D projects to the GRIs under a project based system that was free from any competition.

Nowadays, South Korea's GRIs benefit from two types of funding: institutional funding and project-based funding. In 2021, the GRIs' funding consisted of institutional funding (43%) from the central government and project-based funding (57%) from government departments and industries, etc. The GRIs must submit their annual budget plans for the

execution of institutional funding and project funding to the NST by April, whereupon the NST's Board of Directors reviews the plans and delivers the results to the MSIT in May. The MSIT then submits the total R&D budget to the Ministry of Economy and Finance (MOEF), including the institutional funding of the GRIs, by the end of August. Pursuant to the decision of the National Assembly in December, the MOEF distributes the institutional funding on a quarterly basis the next year. The GRIs must also obtain other project funding based on an open competition from nearly twenty government departments and private sectors. The total annual budget of the GRIs amounted to approximately five billion US dollars in 2021, including over two billion dollars of institutional or basic funding.

[Figure 4-17] GRI Budget, 2021



Source: <https://nst.re.kr/eng>

In 2021, South Korea's gross R&D investment amounted to about US\$ 100 billion, with some US\$ 30 billion of gross GERD, of which GRIs accounted for almost 17%. As government departments have tried to increase the number of collaborative projects with industry, GRIs have been expanding their collaboration with the industrial sector in a bid to win more PBS projects.

In addition, the NST provides funding for GRIs (Almost US\$ 100 million) to activate research for convergence among the twenty-five GRIs in the form of "matching funding" incentives. Project base funding sources come from twenty other government departments in the form of competition funding. These government departments usually emphasize collaborative research with industry, especially SMEs, which is why nearly half of the GRIs' research projects involve collaborative research with industry. This is supported by well-established

policies to increase the private sector's involvement in the innovation environment. Meanwhile, the government's technology licensing policy is hospitable to SMEs, as the companies participating in such research projects usually give priority to technology licensing and commercialization for two to three years.

BRIN acts not only as a research institution but also as a funding agency, providing research funds to the research organizations of industries or universities that collaborate with BRIN's own research organizations. The main difference between the funding of BRIN and that of the NST and the GRIs is the percentage of institutional funding: BRIN's funds mostly come from the government, i.e. institutional funding, whereas the NST and the GRIs mostly receive funds from government departments and industries through competitive research projects.

[Table 4-5] Comparison of the Sources and Management of Funding at BRIN and the NST

	BRIN	NST
Funding	2019: national GERD was only 0.27% of GDP. Investments in innovation in Indonesia are still dominated by the government (80%) (UNESCO, 2019).	2021: total national R&D investment amounted to about US\$ 100 billion, i.e. 4.81% of GDP, with government expenditure accounting for nearly US\$ 30 billion.
Funding source	Mostly from the government.	Nearly 90% of funding came from the government in the form of institutional funding and project base funding with open competition.

Source: Data processed by researchers (2022).

BRIN acts not only as an R&D organization but also a funding agency, providing research funds to the research organizations of industry or universities that collaborate with the research institutes of BRIN.

Another difference between BRIN and the NST and the GRIs lies in the percentages of their funding sources. BRIN receives most of its funds from government institutional funding, whereas the NST and the GRIs obtains more than 60% of its funding from project contracts with government departments and companies enrolled on the PBS scheme.

Some of the lessons learned are as follows:

- a. Korea's GRIs have a strong capacity not only to compete against other GRIs for performance-oriented projects, but also to develop convergent technologies with other GRIs. As such, BRIN needs to enhance these kinds of funding so as to improve research performance through both open competition and collaboration between BRIN's research organizations.
- b. The NST and the GRIs activate collaboration with industry through joint research projects and disseminate the results of research in order to support industry. In this respect, BRIN needs to introduce new funding policies in order to increase private sector involvement in research investment.

3.5 Organization Management

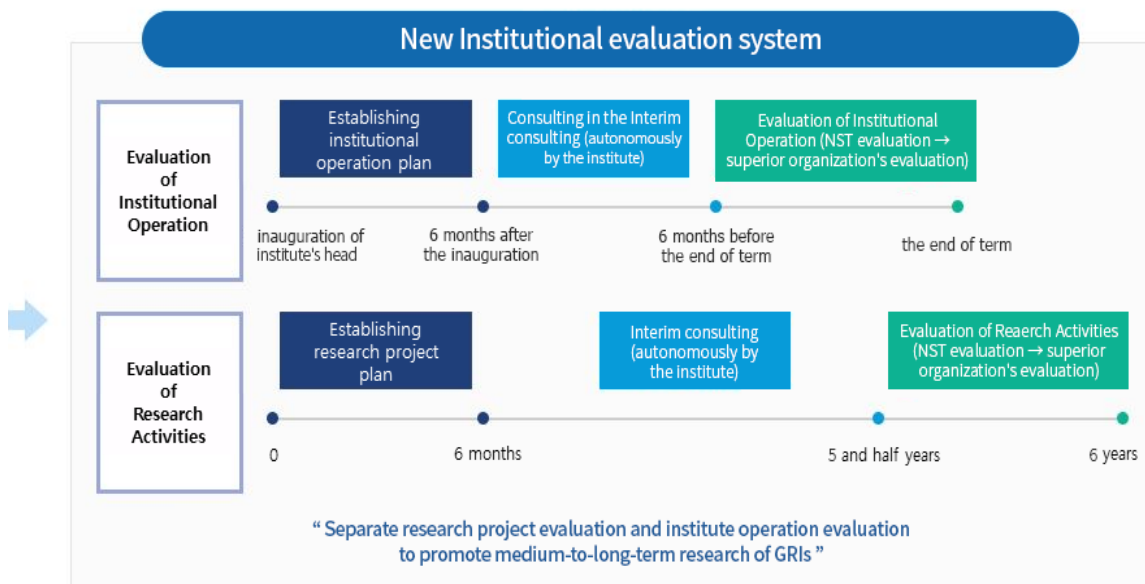
South Korea's N/PRIs were a part of the departments immersed within the bureaucratic culture and hence lacked autonomy. GRIs, including the KIST, were designed to guarantee not only the principle of autonomy, but also to ensure financial support for stable R&D from the government. They were also self-financing in relation to the overheads of internal research divisions and labs, which were their basic management units. Individual labs were compelled to earn sufficient funds from research contracts to cover their overheads for human resources and facilities. Despite the principle of stability and autonomy, the GRIs faced two major problems at the start of the 1980s: i) less autonomy under the government departments and ii) less stability due to a decline in the number of research contracts from industries which they themselves had grown.

In 1991, the NST introduced an annual evaluation of institutions, including an annual self-evaluation and a three-year term evaluation, the results of which were used to determine the performance-related allocation of budgets by the Finance Minister. This evaluation scheme has continued to evolve since the establishment of the Research Councils in 1999, based on the Act on the Establishment, Management and Promotion of Government-funded Research Institutes.

At present, the GRIs should be evaluated by the NST evaluation scheme within the term of

the president of each GRI. Each GRI president must submit a three-year management plan to the NST within 3 months of his or her inauguration. The first evaluation of a GRI's organizational management performance should be conducted eighteen months later by external experts appointed by the NST, while the final comprehensive evaluation of research performance, including organizational management, is conducted at the end of the president's term. The results of the final evaluation are reflected in the annual pay and reappointment of the president.

[Figure 4-18] The New Institutional System of GRIs Adopted by the NST



Source: www.nst.re.kr.

Another important aspect of the organizational management of the GRIs is to activate various kinds of commercialization and innovation via technology transfer, direct start-ups by researchers, and technology investment for a spin-off company with other external management experts. The GRIs usually operate their own technology licensing office, while the NST supports the TLOs with external experts and companies. Intellectual property management is playing an increasingly significant role in ensuring better commercialization from the outset of R&D.

Human resource management is also an important issue of organizational management. As

of 2021, the total number of regular researchers and other personnel at the twenty-five GRIs was about 23,000, of whom regular researchers and staff accounted for nearly 80%, while the rest were contract researchers, post-doctoral researchers, assistant staff, and foreign scientists. The number of regular researchers and staff is regulated by the MOEF.

The National Research and Innovation Agency (BRIN) is the government institution that is directly responsible to the President for conducting research and development, assessment, and implementation of inventions and innovations, as well as for implementing the country's nuclear power policy and integrated space operations. BRIN was formed by the President of the Republic of Indonesia in April 2021 under Presidential Regulation No. 74, 2019, which was later updated by Presidential Regulation No. 78, 2021, concerning the National Research and Innovation Agency.

BRIN comprises seven deputies and twelve Research Organizations (OR). The seven deputies responsible for supporting organizational performance include the following: (1) Deputy for Development Policy, (2) Deputy for Research and Innovation Policy, (3) Deputy for Human Resources of Science and Technology, (4) Deputy for Research and Innovation Infrastructure, (5) Deputy for the Facilitation of Research and Innovation, (6) Deputy for the Utilization of Research and Innovation, and (7) Deputy for Regional Research and Innovation.

The intermediary activities designed to facilitate relations between researchers (as inventors) and users are mostly carried out by the following three deputies: (1) Deputy for Utilization of Research and Innovation, (2) Deputy for Facilitating Research and Innovation, and (3) Deputy for Research and Innovation Infrastructure. In conducting intermediary activities, these three deputies cooperate in encouraging the commercialization of the results of research conducted by researchers at BRIN. There are two main commercialization schemes at BRIN: (1) commercialization through industry and (2) commercialization through SMEs, start-Ups, and communities.

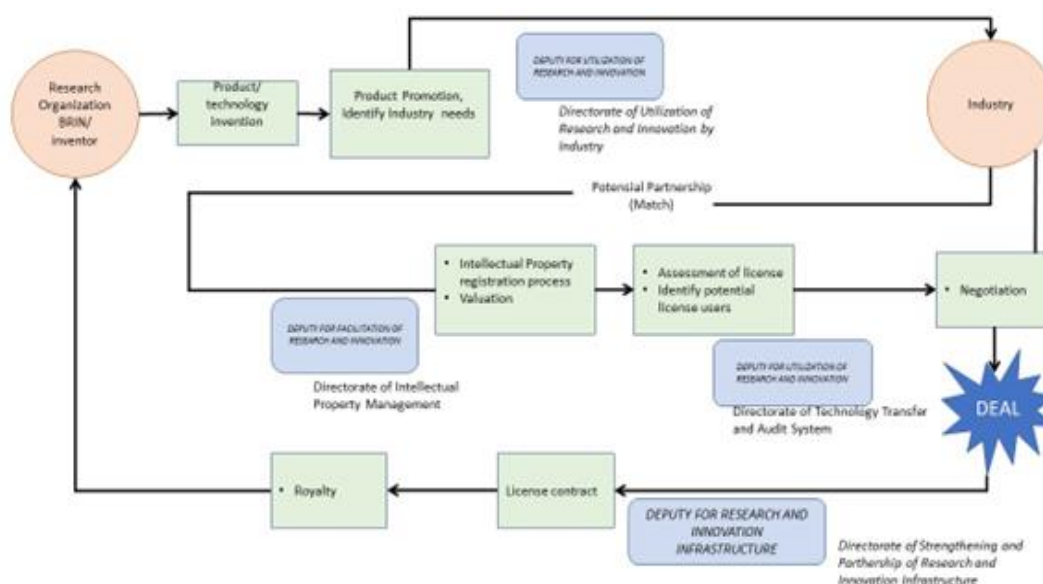
3.5.1 Commercialization through industry

The purpose of commercialization through industry is to link BRIN's research results to industry. The business process for the commercialization of Scheme 1 begins when BRIN researchers have developed a product or products that are ready to be implemented. Such products are then identified, mapped according to industry needs, and promoted to the industry by the Directorate of Utilization of Research and Innovation in under the Deputy of Utilization of Research and Innovation. If there is a possibility that any companies are willing to use the product or mass produce it, the process continues with the preparations for obtaining an intellectual property.

In an effort to obtain the intellectual property rights of the product, the Directorate of Intellectual Property Management of under the Deputy for the Facilitation of Research and Innovation accommodates the product. This Directorate also assists in calculating product valuations used by industry. Furthermore, the Directorate of Technology Transfer and Audit System in under the Deputy for Utilization of Research and Innovation identifies potential license users. If the results of the assessment reveal that the industry seems feasible and has the potential to develop a product, the negotiation process with that industry proceeds.

Once the negotiations result in an agreement, the process continues with the signing of a license contract. This process is handled by the Directorate of research and Innovation Infrastructure in the Deputy for Research and Innovation Infrastructure. After concluding the contract, the industry is obliged to provide royalties to the inventors.

[Figure 4-19] Commercialization Scheme through Industry



Source: Data processed by researchers (2022).

3.5.2 Commercialization through SMEs

The purpose of commercialization through SMEs is to facilitate SMEs' commercialization of the results of research conducted by BRIN's research organizations through a series of incubation processes. Incubation assistance provides help in the form of technical guidance and funding. In addition, this scheme is also available for use by SMEs seeking to develop their products.

Through the Directorate of Research and Innovation Funding in under the Deputy for Facilitation of Research and Innovation, BRIN provides open call proposals for incubation funding for product users such as start-ups, SMEs, and communities. Users can use the results generated by BRIN's researchers or their own products. When a start-up commits to using the research results of BRIN's inventors, collaborative contracts are required. After a proposal by a start-up is selected, the next incubation process begins. This process is facilitated by the Deputies for the Utilization of Research and Innovation, who provide technical guidance for increasing the skills of human resource, such as making business proposals, legalizing companies' product promotions, and Deputy for research and innovation infrastructure that facilitates the use of infrastructure during the incubation

[Figure 4-20] Commercialization Scheme through SMEs



BRIN operates its human resource management based on permanent employees, whereas GRIs have a mixed human resource management scheme with contractual and permanent jobs.

The differences in the roles of the TLOs of the GRIs and BRIN and human resource management are presented in the following table.

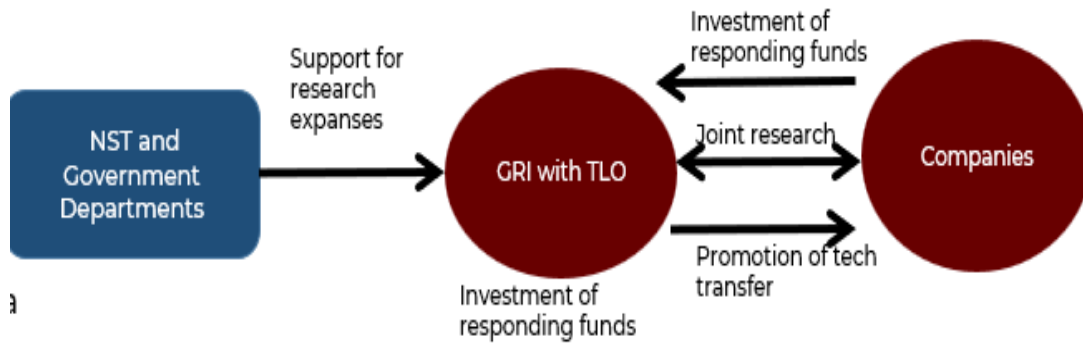
[Table 4-6] Comparison of the Organization Management of BRIN and NST

	BRIN	NST
	Centralized TLO	Decentralized TLOs
Nature of organization	TLO services are carried out by three deputies to serve twelve research organizations.	Each GRI operates its own TLO with autonomous commercialization.
Role	To assist the acquisition of IP, infrastructure research, product promotion, and tenant incubation.	To help the GRIs strengthen their IP management capabilities, promote more effective technology transfer, and activate commercialization.
Employees	Permanent employees	Mixed: Contractual and permanent employees. The NST never intervenes in a GRI's employment process.

Source: Data processed by researchers (2022).

In terms of technology transfer promotion to existing companies, the NST and government departments provide research funds to the GRIs based on the principle of competition. The GRIs conduct research with their own assets. Some projects can be performed as joint research with companies' responding funds. Companies' invested matching funds have priority in using the results of research through a technology transfer contract. Other research results without companies' matching funds are displayed and marketed to all the related companies via on-line means, off-line business meetings, and so on.

[Figure 4-21] Technology Transfer Pattern (Industry) of the NST and GRIs

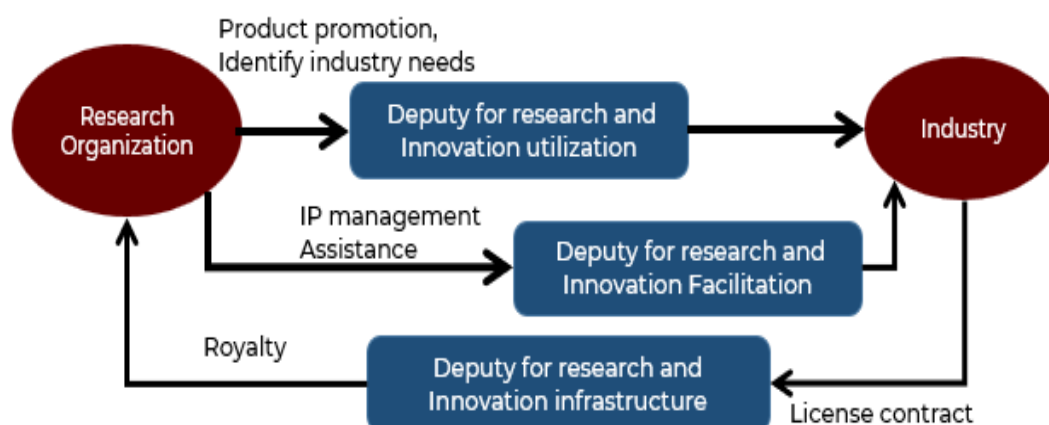


Source: Data processed by researchers (2022).

The purpose of commercialization through industry is to link the results of research produced by BRIN's research organizations with industry in order to have them mass produced by a company. The process of commercialization begins when BRIN researchers have developed a technology that is ready to be implemented. Such a product is then identified, mapped according to industry needs, and promoted to the industry by the Deputy for Research and Innovation utilization. If there is a possibility that companies are willing to use the product or mass produce it in the promotion process, the process will continue with the preparations for acquiring intellectual property (rights, which are carried out by the Deputy for Research and Innovation Facilitation.

If the industry is declared feasible and has product development potential, the process of negotiations with that industry proceeds. Once an agreement is reached, the process continues with the signing of a license contract. This process is handled by the Deputy for Research and Innovation Infrastructure. Upon concluding a contract, the industry is obliged to provide royalties to the inventors.

[Figure 4-22] The Technology Transfer Pattern (Industry) of BRIN



Source: Data processed by researchers (2022).

The TLOs of the GRIs assist researchers with establishing their own start-ups, providing them with various benefits and incentives. Researchers who found start-ups can take a leave of absence with salary for three years, and for a further two years without salary. The other scheme for activating commercialization involves GRI technology investment, which usually consists in acquiring the shares of spin-off companies. Some GRIs have jointly established and operated holding companies like venture capital in order to activate technology transfer and other types of commercialization.

Some lessons learned from the above analyses are as follows:

- a. It is necessary to conduct more demand-pull and market-need based research. Encouraging BRIN's research organizations and industries to conduct research together will have a significant effect on catalyzing technology transfer and commercialization.
- b. Another option for BRIN is to establish holding companies to accelerate commercialization. Korea's GRIs have established their own investment holding companies, Korean Science and Technology Holdings (KST) and ETRI Goldings, to support the development of start-ups.

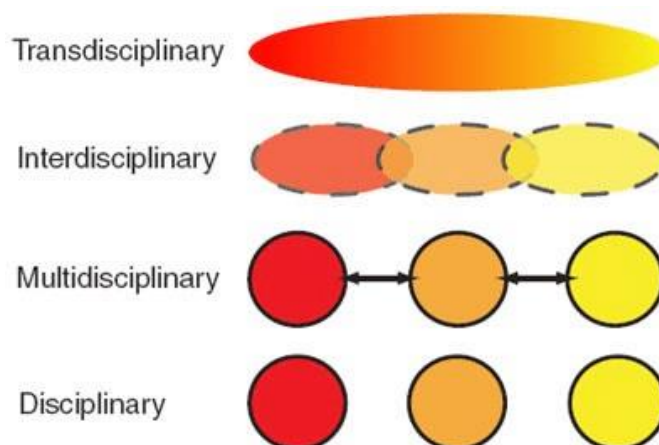
4. Policy Recommendations

4.1 Strengthening Research Convergence

More collaborative programs are needed to give BRIN more opportunities to develop convergent technologies and create new demand industries based on those technologies. Organizational and logistical challenges are essential for activating convergence research, supporting various degrees of integration between different disciplines, and adopting new convergence strategies through grants, workshops, and cooperative agreements.

One of the main convergent research goals is to solve issues and challenges facing society, such as climate change, disasters, energy and transportation issues, etc. Convergent research has two characteristic features: 1) challenging research driven by problems arising from scientific questions or acute societal demand; and (2) the integration of research across disciplines, i.e. interweaving different disciplines in order to find solutions to research challenges based on diverse knowledge, theories, methods, and data. In short, convergent research is similar to transdisciplinary research. Transdisciplinary research accompanies interdisciplinary and multidisciplinary research (International Center for Transdisciplinary Research in Darian-Smith and McCartney, 2016). The elements of transdisciplinary work are problem-oriented and based on the practical application of knowledge.

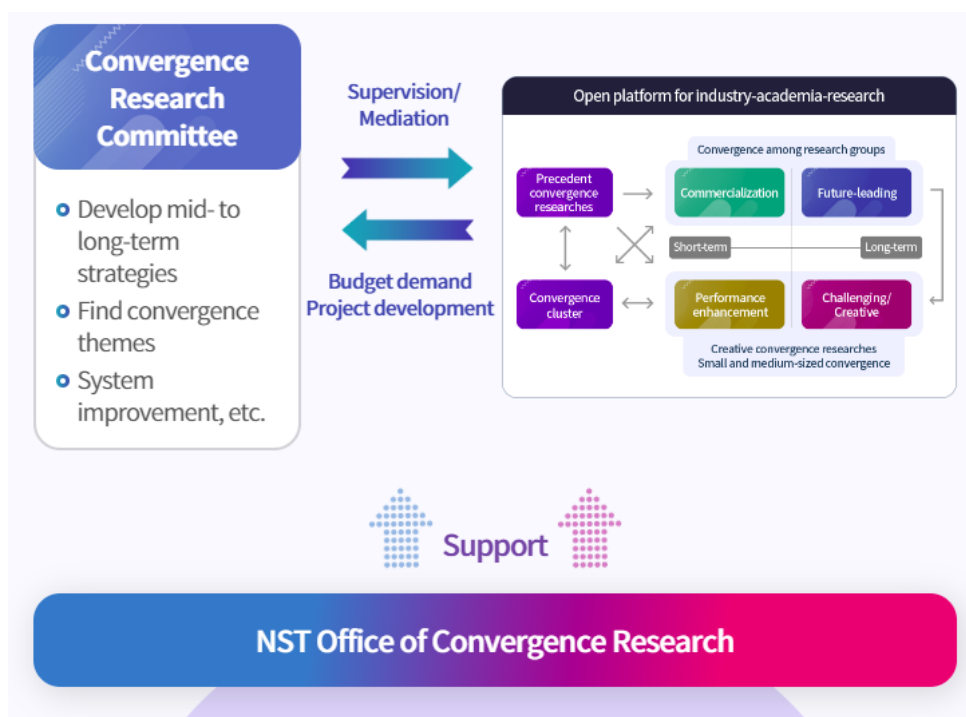
[Figure 4-23] The Concept of Convergent Research



Source: Nature in Darian-smith and McCartney (2016).

In South Korea, the main background to the foundation of the NST is the need to activate the convergence of research among GRIs. A new funding program of around 100 million dollars was introduced by the NST to support this convergent research. Currently, a paradigm shift is taking place in the research conducted by GRIs, from technology-oriented research to problem or mission-oriented research. Therefore, the focus of research is also shifting to solving specific social problems such as climate change, pandemics, smart cities, etc. As such, the convergent research strategy carried out by the GRIs is also expanding to involve collaboration between industry and universities to solve social problems and meet the demands of society and industry, and not only those issues set by the government at the national level but also those issues that are equally important at the regional innovation system level, in order to accelerate the innovation rate of local companies and universities. This is also supported by the existence of around 100 regional branches of GRIs that act as proponents of the regional innovation system.

[Figure 4-24] Implementation of Convergent Research in the NST



Source: <https://www.nst.re.kr> (2022).

The similarity between the objectives of the NST/GRIs and BRIN here are salient in which NST & GRIs act as a convergence platform in the innovation ecosystem. The formation of BRIN is also motivated by the need to consolidate research activities and resources to better respond to economic and societal issues through science and technology. The head of BRIN has issued statements on multiple occasions regarding the role of BRIN as a collaboration hub, an enabler, and an *open platform* in the innovation ecosystem (Handoko, 2022).

The implementation of convergent research at the NST is operated by the Office of Convergence Research (OCR) and the Convergence Research Committee (CRC) along with the GRIs. The roles of the Convergence Research Committee are to develop (solutions to mid-to-long problems and societal needs, identify convergence themes, and improve the convergence research system. Meanwhile, the convergent research ecosystem has been expanded to industry and universities, and the GRIs hope to establish a new platform for convergent research.

In the case of BRIN, convergence research can be implemented in several steps, as follows: (1) Determine the convergence research system through a scheme that can be carried out based on BRIN's strategic plan and/or national priority research in a top-down approach (It is also necessary to determine the source of funding); (2) Determine the framework and role of each research organization and deputy under the concept of convergent research at BRIN. The convergent research strategy could be implemented at BRIN via research funding programs, both from the national budget (APBN) and from external funding. This policy recommendation will specifically explore the arrangement of research funding programs from the national budget (APBN).

Currently, in the early stage of BRIN's formation, the tools and mechanisms used to support mission-oriented research seem to be scattered, and lack guiding actors; or, if they exist, their agency in integrating and connecting one program with the others is rather weak. Basic research funding programs for research organizations (OR) and research centers (PR) can be obtained through a scheme known as 'Program House.' The 'Program House' research theme is determined by each OR and the head of BRIN to support the solution of global and/or national issues. The research proposals submitted are expected to be able to solve these problems. However, this scheme does not strongly encourage collaboration and is still

not connected with other funding mechanisms at BRIN, thereby jeopardizing research continuity in the long run. Furthermore, this is exacerbated by the lack of coordination among the units at BRIN, as shown by data obtained through interviews with industry partners and several members of the deputies at BRIN.

In a mission-oriented research term, Mazzucato (2018) implies the importance of sustained funding and the ability to encourage interaction and learning in order to experiment with explore solutions. Thus, the strategies proposed are focused on providing an arena for interactions between the ORs at BRIN and other research institutes/academia, companies, or other parties such as government institutions, through various integrated tools and mechanisms including funding policies and support programs and structures.

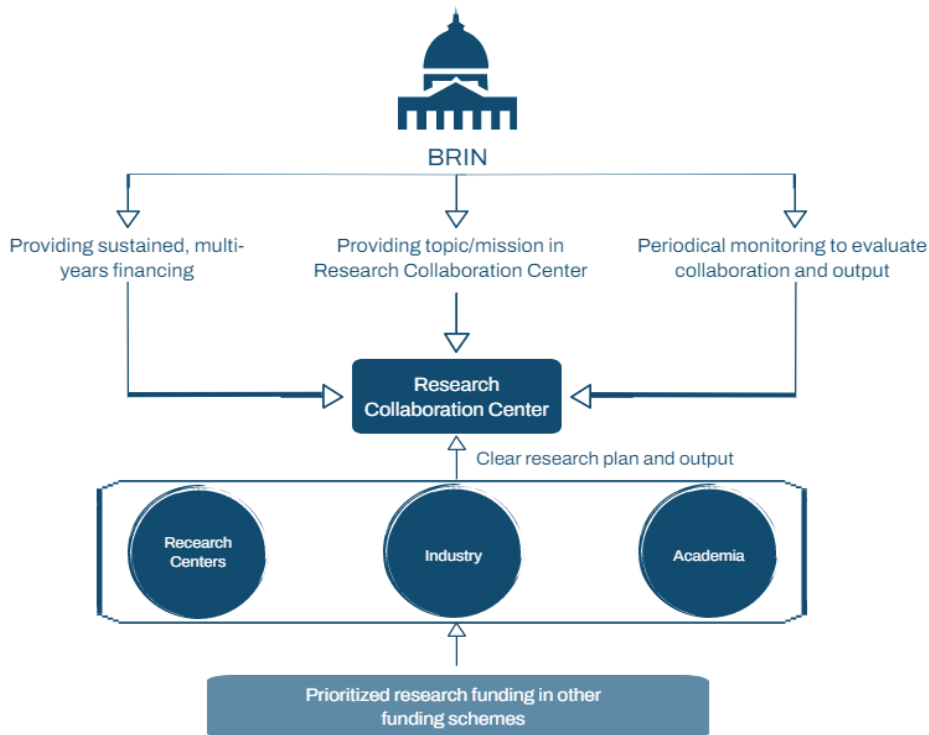
The following three strategies for research convergence in mission-oriented research are proposed based on the findings of the analyses of Korea's policies and BRIN's interviews and data.

4.1.1 Encouraging Convergence through Research Funding Policies

- a. Reinforcement of the existing funding scheme for research convergence in BRIN through the finding program of the Research Collaboration Center (Pusat Kolaborasi Riset).

This funding scheme specifically facilitates collaboration between research centers at BRIN with the research organizations of industry or universities with specific research topics. The topic/mission of each collaboration needs to be determined according specific top-down guidelines, contrary to the current mechanism in which the topic/mission is determined by the researchers in a bottom-up manner. This mechanism also needs to be strengthened with a clear research plan that supports the mission presented in a proposal for collaboration, periodical monitoring and evaluation of collaborations and their targeted outputs, support for financing collaboration, and the prioritization of research proposals in other research funding schemes. This mechanism will need strong coordination across all units that facilitate research funding at BRIN.

[Figure 4-25] The Reinforcement Mechanism of the Existing Funding Scheme for Research Convergence at BRIN

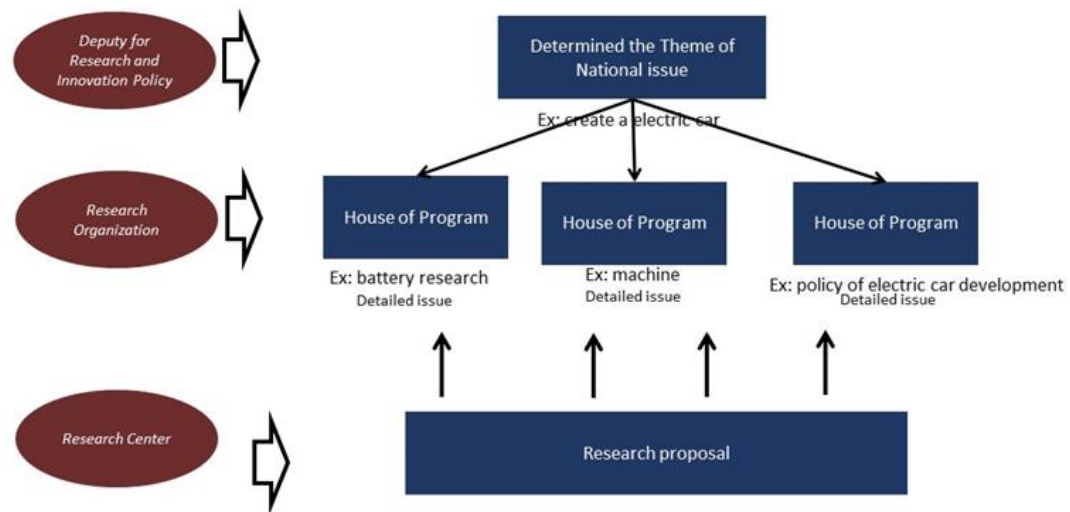


Source: Data processed by researchers (2022).

- b. Setting of a minimum percentage for funding from the 'Program House' for research that contributes to solving pre-determined specific national issues, with collaboration with industry/academia as a requisite.
- c. To encourage research convergence at BRIN, it is necessary to determine which national themes and issues are to be resolved in the short and long term as the guideline for research activities at BRIN's research centers. These pre-determined issues then need to be translated to the operational level of the OR, and the details of problem solving are divided according to the expertise of each OR. The funding scheme for this OR research competition is called the 'Program House' (Rumah Program), in which each research center (PR) submits proposals for research aimed at solving the issues. Setting minimum percentage of funding in

'House Program' for convergent research provides assurance for research continuity and helps make sure the mission is achieved.

[Figure 4-26] Mechanism of Program House (Rumah Program) in BRIN



Source: Data processed by researchers (2022).

4.1.2 Supporting Convergence through Programs

a. Facilitating meetings and discourse

Convergence research requires interdisciplinary and knowledge integration of the technical aspects and the applied standards (Sick & Broring, 2022). BRIN needs to help the actors in convergent research to meet and discuss specific issues that need solving. This program can be conducted mainly by the intermediary units at BRIN, such as the Deputy for the Utilization of Research and Innovation.

b. Conducting periodical foresight research and discourse

Systemic mission-oriented policies must be based on a clear and sound diagnosis and prognosis (foresight) (Mazzucato, 2018). In order to achieve a sound diagnosis, it is necessary to conduct periodic research on foresight with regard to emerging and critical issues, supported by discourse or scientific forums involving the related stakeholders. This will help BRIN to determine the themes and national issues and choose the right policies for implementing mission-oriented convergent research.

4.1.3 Supporting Convergence through a Structural Brace

a. Clarification and explicitation of the unit's role inside BRIN

In an attempt to eliminate any hindrances to coordination, which is highly necessary for convergent research, BRIN also requires a clear division of the tasks involved in order to realize the convergence of mission-oriented research, i.e.:

- The Deputy for Research and Innovation Policy – the Directorate of Research, Technology and Innovation Policy Formulation, in collaboration with the Deputy for Development Policy and other related parties such as industry and other government institutions, are the designers of the master plan for national priority programs that aim to advance science and technology and need to be completed within five years. In addition, it is necessary to determine which social and technical issues will be the annual research themes to be implemented in BRIN's research and funding policies.
- The Deputy for Research and Innovation Policy – the Directorate of Research, Technology and Innovation Policy Evaluation act as the evaluators in the convergence research program.
- The Deputy for Research and Innovation Facilitation plays a role in funding the research convergence scheme.
- The research center (PR) is the implementer of convergent research.

b. Empowerment of interface entities or intermediary units

Intermediary institutions that help to link the actors in convergent research are important, especially with regard to their role in creating a boundary space for learning (Stewart & Hyysalo, 2008; Martiskainen & Kivimaa, 2018). Hekkert, et al., (2020) also assert the importance of such entities in coordinating a mission-oriented innovation system. After analyzing the relevant literature and considering BRIN's characteristics as a government institution, we synthesized the various requisite intermediary functions of BRIN, as shown in [Table 4-7] below.

[Table 4-7] BRIN's Intermediary Functions

No.	Intermediary Function	Description
1	Scanning, processing, and dissemination of information	Acquisition, processing, and dissemination of knowledge and information regarding technological capacities (human resources, infrastructures, and scientific products), including the management of knowledge protection, such as scientific collections or patents.
2	Mobilization of resources	Enabling access to innovation resources (funding, infrastructure, and scientists).
3	Bridging actors and creating networks	Creation of an innovation network involving diverse stakeholders from industries, universities, and regulatory bodies.
4	Technology gatekeeper	Enabling access and channelling technology to industries and communities.
5	Prototyping/piloting, testing, and technical consulting	Facilitating industries' prototyping and testing of technology, or technical consulting.

Source: Data processed by researchers (2022).

Substantial efforts will need to be made to realize this recommendation. Furthermore, it is imperative that a clear division of labour be made among the intermediary units at BRIN (such as the Deputy for Research and Innovation Infrastructure, the Deputy for Research and Innovation Facilitation, the Deputy for the Utilization of Research and Innovation, and the Center for Technology Services) in carrying out their roles as intermediary units, in order to facilitate collaboration between research centers (PR) and their partners in industry or academia.

It is also necessary to map the intermediary units' human resource capabilities and the planning of capacity building commensurate with intermediary roles (e.g. articulating industrial needs, negotiation capability, possession of knowledge regarding the transfer of technology, and property rights regulations and practices, etc.) according to a long-term vision.

4.2 Encouraging BRIN-Industry Research Collaboration

Placing an industry as a research partner needs to be a prerequisite of the BRIN research financing scheme, as this enables the results of research conducted by BRIN's research institutes to be directly absorbed by industries. This program is carried out as follows:

4.2.1 Creating a co-investment program with the private sector for research development

This could be implemented by creating a new research grant scheme that invites industry or private sector contributions (in cash or in other forms) and rewards the contributors with co-investment by the government (usually a multiple of the industry funding). Furthermore, BRIN could support and monitor the effects of Indonesia's new super-tax deduction policy in attracting contributions from both the non-governmental and private sectors.

The incentives for industry to conduct research and development are now a standard policy solution to boost national investment in research. Most countries, whether they are mature or developing knowledge economies, have now adopted tax incentives (as deductions or credits) for industry expenditure on research and development (Appelt et al, 2019; OECD, 2017).

Some economies (generally those with highly developed industrial sectors and/or very high levels of foreign investment) have been successful in securing significant research and development spending by industry.

That, however, does not guarantee that all research actors are collaborating and producing research of the highest quality possible. Strategies for attracting more industry expenditure on research and development by co-investing government funds, as mentioned in *Making Indonesia's Research and Development Better* (2020), are becoming increasingly important. Making such co-investment conditional will help to forge partnerships between industry, university, and/or government or non-governmental research actors. In fact, such 'linkage', 'consortium' or 'collaboration' schemes are quite often linked to national research missions.

4.2.2 Improving the Capacity of BRIN's Human Resources in Licensing Technology Services and Incubation Services

It is necessary to ensure that the government creates a specific program particularly for BRIN to enable it to enhance its human resource capacities in order to increase the quality and competitiveness of its technology licensing and incubation services.

The downstream (*hilirisasi*) process of utilizing research results is still a significant issue in Indonesia. When it comes to commercialization, there are certain requirements which must be met by a given product innovation. For instance, the results must be registered, and the research must comply with certain standards. Often, the results of a research program are ready, but when it is about to be registered, the results have to be proven to be correct three times. Furthermore, when the results are about to enter an industry, the latter might say that they are far from ready because they still need to be validated, and thus should be supported by several documents and data testing.

BRIN could act as an intermediary between research actors and industry by creating guidelines on the standards and processes to follow, and by assisting with the conclusion of mutually acceptable research and downstream agreements. This would ensure results of better quality and minimize the possibility of the results being rejected by industry and the private sector. BRIN could also use its convening power through the human resource support in licensing technology and incubation services to help identify the needs of a given industry or company and help it to connect and communicate with the research actors.

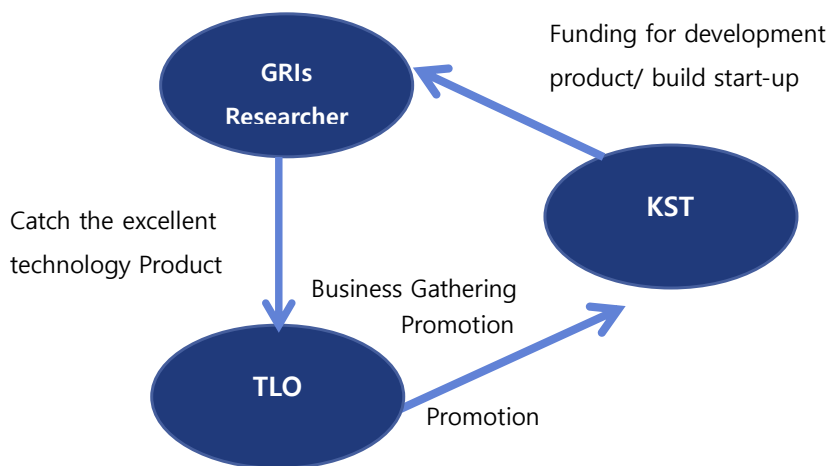
4.3 Accelerating the Growth of Startups as Commercialization Path Development

South Korea's GRIs can opt for one of the following three tracks of technology commercialization: (1) transfer a developed technology to existing companies; 2) engage in direct technology investment from inside and outside, usually for spin-off companies; and (3) create a new startup through their researchers or groups. To support their development, the Korean government established the Korea Science and Technology Holdings Company.

In order to catalyze, the Korea Science and Technology Holdings Company (KST) was established in 2013 by seventeen GRIs as a means to support the three kinds of commercialization pathways mentioned above. The KST's mission is to provide funding for innovation-based start-ups and to create jobs in order to increase national economic development through R&D productivity (<https://stip.oecd.org>).

Operationally speaking, the KST and the TLOs of the GRIs are interconnected and interdependent. The TLOs find technologies to transfer or invest in, while the KST reviews the possibility of investment. Promotion is carried out through business gatherings supported by researchers, TLOs, and the KST. If there is a product match, the KST and TLOs can develop the commercialization pathway after conducting an economic feasibility analysis of all kinds of start-ups. Then, start-up financing is carried out by the KST or other venture capital.

[Figure 4-27] Technology Commercialization through the KST



Source: Data processed by researchers (2022).

Even though the KST and the GRI TLOs are operationally interdependent, the KST is managed separately from the operation of the GRIs. The KST is led by a three-year term CEO who is not an employee of the GRIs. The government usually does not intervene in the KST. The KST also pays its employees independently, and conducts its own HR capacity building programs independently.

KST funding is carried out via a combination of government and private funding. As a funding initiator, seventeen GRIs invested in KST's funds. The KST has invited many other VCs and private industries to invest in it; and, based on data from the kstholdings.co.kr site, there are now some 112 KST subsidiaries.

One of the lessons learned with regard to boosting the technology commercialization of GRIs in South Korea is the formation of a holding company. Regulations related to holding companies have yet to be specifically stipulated in the laws and regulations of Indonesia. In general, the form of holding company based on the practice in Indonesia can be classified into the following two types:

- a. Investment Holding Company: This is a type of company whose business activities are limited to the ownership of a part or all of the shares of its subsidiaries, but which does not participate in the operational activities of the subsidiaries.
- b. Management Holding Company (Operating Holding Company): This is a type of company which, besides owning a part or all of the shares in its subsidiaries, participates in managing their business operations and in making their business decisions.

Currently, BRIN does not have a holding company. As such, the best way to support the commercialization of technology is through a TLO comprising three deputies, i.e. (1) the Deputy for Research and Innovation Facilitation, (2) the Deputy for Research and Innovation Use, and (3) the Deputy for Research and Innovation Infrastructure.

Of the two aforementioned types of holding company, BRIN tends to refer to the second type. BRIN's holding company aims to help commercialize research products produced by BRIN's researchers. Commercialized products need to meet certain technology readiness level (TRL) requirements. So that a product can be made ready for production, referring to the KST, BRIN's holding company also provides funding assistance to start-ups. The difference here is that if KST finances start-ups founded by GRI researchers, then BRIN's holding company will finance start-ups that utilize the results from research conducted by BRIN's researchers because they have not been allowed to establish a start-up

independently.

To achieve this goal, the implementation is carried out according to the following steps:

4.3.1 Formulating the BRIN Holding Company's Objective Framework

BRIN's holding company could acquire investments from the following two activities:

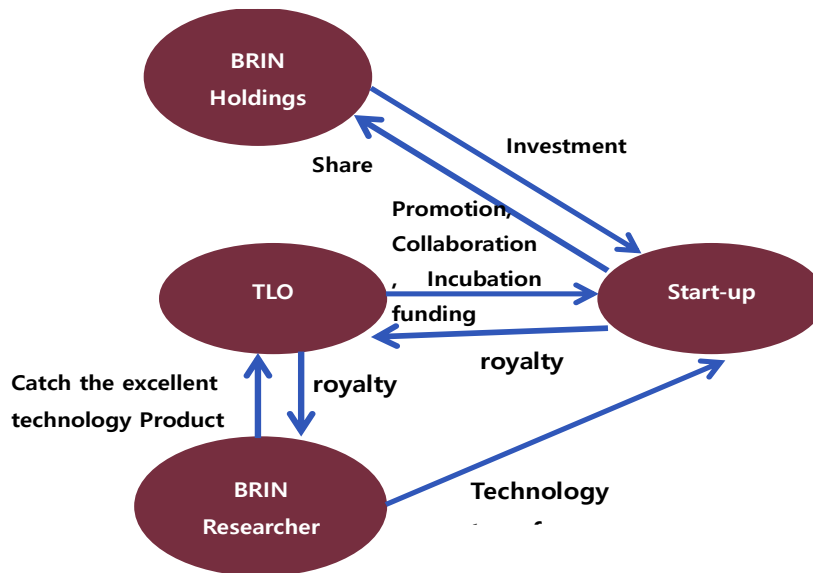
- a. Establishment of a subsidiary with research center competencies (science-based company)

Learning from the KST, this activity focuses on supporting the commercialization of results produced by BRIN's researchers. There are two schemes that could be implemented within the framework of this science-based company: investment schemes through the development of new start-ups, or joint ventures with existing industries.

- Establishing a new start-up

BRIN Holdings could support the development of start-ups by investing in selected start-ups under the incubation funding program conducted by the TLO. First, the TLO promotes excellent technology products to start-ups. Then, start-ups that are willing to produce the research must submit a proposal to the TLO, which then selects the most favorable proposal, which in turn receives incubation training. Upon graduation, if the business is considered to have profitable prospects, BRIN Holding will invest in it.

[Figure 4-28] New Start-Up Development Scheme through BRIN Holdings



Source: Data processed by researchers (2022).

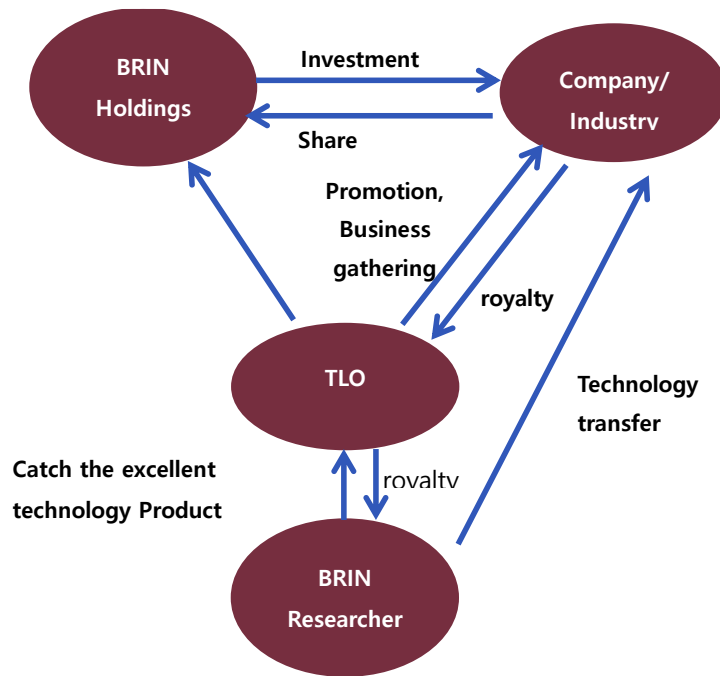
The development process of start-ups in Indonesia is generally quite rapid. Every year many new start-up founders (owners) appear. Currently there are at least, if not more than, 1,500 local start-ups in Indonesia. This condition will be advantageous for BRIN researchers who rely on start-ups as a means of commercialization.

▪ Joint venture

The joint venture scheme between BRIN holdings and the existing industry will consist in investing in and collaborating on the production of research products by BRIN researchers. In the early stages, the TLO will hold a business gathering to which BRIN holdings and industry will be invited, and promote excellent technology products produced by BRIN's researchers.

During the business gathering, if an industry is interested in a BRIN product and appears willing to produce it, a contract will be signed, and BRIN Holdings will participate in investing a certain amount of funds in the development of that product. The results of the share will be divided according to the agreement made. In addition, research royalties will be paid by the industry through the TLO.

[Figure 4-29] Joint Venture Scheme via BRIN Holdings (1)



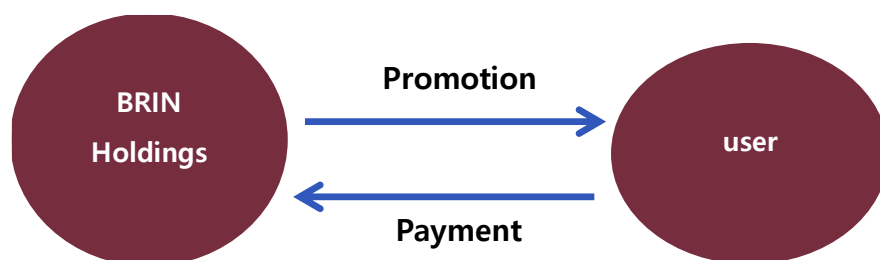
Source: Data processed by researchers (2022).

b. Establishing a service-based subsidiary company (service-based company)

A holdings business involving companies other than science-based companies aims to provide services to the community. This method consists in leasing a number of assets owned by BRIN. BRIN's ownership of a number of buildings, areas of science and technology, and botanical gardens can be used as a source of income.

This service to the community does not always involve payment but rather the introduction of technology and knowledge to the community. This is made possible by opening a number of areas and botanical gardens as spaces for learning, especially in the fields of biology and natural science.

[Figure 4-30] Joint Venture Scheme through BRIN Holdings (2)



Source: Data processed by researchers (2022).

4.3.2 Funding the BRIN Holdings Company

Initially, BRIN Holdings Company could be funded based on the case of the KST in Korea, which includes the incoming state budget for GRIs and additional income generated by technology transfer royalties in particular. The research center could set aside additional funds to build BRIN Holding. Meanwhile, during the next stage, BRIN Holding can invite the private sector/industry to invest.

Currently, the financing of research infrastructures based on collaboration between the government and the private sector is not included in Presidential Regulation No. 38, 2015, *Government Cooperation with Business Entities in Providing Infrastructure*. The importance of such collaboration in providing research infrastructure as part of an effort to support the improvement of national innovation needs to be considered for inclusion as one of the points in the Public Private Partnership policy.

4.3.3 Management System of BRIN Holdings Company

Referring to the KST, BRIN Holding's management system is also planned to be managed by professionals. Its employees will be different from those of BRIN or the Research Center. The plan is for BRIN Holdings to be led by a CEO, with the following three departments under it:

■ **Department of Management Science-based Companies**

- Investment support for start-ups and existing companies.
- Joint projects with other parties.

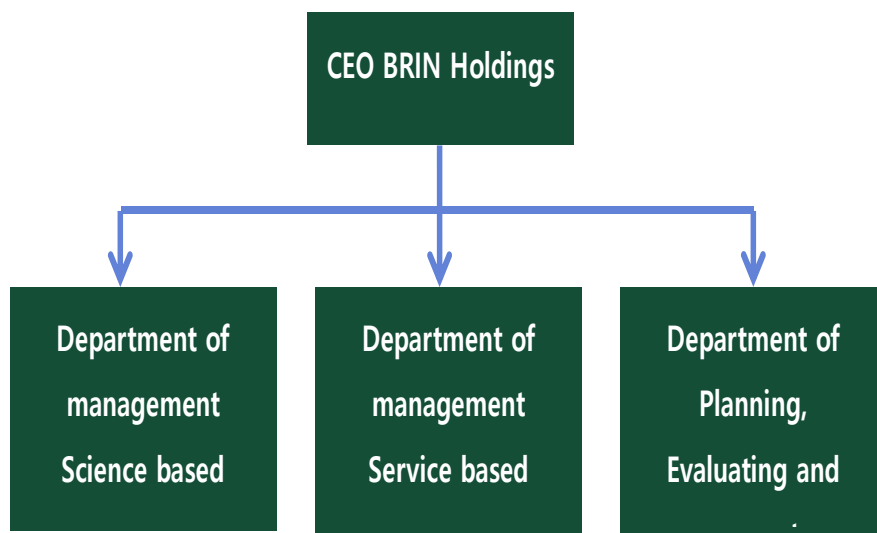
■ **Department of Management Science-based Companies**

- Management of BRIN's Assets.
- Maintenance of research assets.
- Joint projects with other parties.

■ **Department of Planning, Evaluation and Support Management**

- Design of policy regulations for technology commercialization.
- Determination of strategic projects.
- Evaluation of technology commercialization activities.
- Support for financial management.
- Accounting.

[Figure 4.31] Management Structure of BRIN Holdings



Source: Data processed by researchers (2022).

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CHAPTER 5

Conclusion



Chapter 5. Conclusion

1. Summary

1.1 Designing the STI Policy Think Tank for Green Growth Policy Formulation: Lessons Learned from South Korean Practices

South Korea's government has developed a comprehensive green growth strategy as one of the industrializing economies. The country has emerged as a world leader in environmentally-friendly technologies, and its STI policy think tank has had a clear impact on South Korea's green and economic policymaking. Korea could bridge the gap between STI and the development of green economic growth by establishing the Green Technology Center. It would be beneficial to learn more about the connections between STI policy think tanks and ministries and agencies that deal with environmental and economic issues in South Korea.

In the context of Indonesia, the formation of the National Research and Innovation Agency of Indonesia (BRIN) in the middle of 2021 was expected to contribute to national GEG (green economy growth) by optimizing STI aspects. As a new public institution, BRIN is responsible not only for R&D activities but also for developing and implementing STI policies. Furthermore, BRIN played a role in supporting national GEG development through its active participation in the G20 Meeting held at the end of 2022. It is in line with BRIN's function to contribute to GEG development in Indonesia based on STI. As a result, BRIN must learn from the GTC's practices in South Korea in terms of resources in order to formulate GEG policies based on STI at the national level.

In this regard, this research aims to contribute to the formulation of STI policy think tanks for Indonesia's national GEG policy. To do this, we frame the main part and the discussion in order to reveal the role of STI policy think tanks in both the GTC and BRIN. We have

analyzed the practices of the GTC in South Korea in order to identify the three key elements, namely, resources, strategy, and governance, of the institution on STI policy think tank, rather than the external environment surrounding the think tank. In particular, this study focuses on the Directorate for the Formulation of Research, Technology, and Innovation Policy (Directorate of PKRTI) under the Deputy of Research and Innovation Policy (Deputy of KRI). The Directorate of PKRTI (hereinafter, “PKRTI-BRIN”) is a unique agency in that it has dual functions that must be delivered simultaneously, as both an STI policy formulator and an STI policy and management researcher. Other deputies and ORs under BRIN’s structure do not perform a similar role to this directorate.

As history and country status, the GTC goes ahead of the PKRTI-BRIN. The GTC is more senior and is supported by an advanced economy ecosystem. Meanwhile, the government created the PKRTI-BRIN at the end of 2021 within the developing economy ecosystem. The GTC has global and national networks which are widely dispersed around the world, encompassing universities, firms, government agencies, and communities. Meanwhile, the PKRTI-BRIN has one global organization and multiple national organizations, but they are not specifically identified. Strikingly, the GTC has a specific focus on research fields and expertise related to green technology, green policy, and climate change management. All research areas are related to STI policy and cooperation aimed at sustaining green economy growth (GEG) in Korea and overseas. The motive for establishing the GTC is very clear and specific: a response to global climate change policy. It is the GTC’s vision and mission, as a leading policy think tank, to propose information and policy recommendations on green technology to the government of South Korea and whoever needs such information whether in South Korea or overseas. In turn, the GTC has an impact on the specific research and cooperation divisions on green technology and climate change policy. Unlike the PKRTI-BRIN, it was established due to the weak coordination in formulating STI policy, making it difficult to establish the NIS. The scope of PKRTI-BRIN’s STI research is broad, and is not confined strictly to green technology. Its research topics include STI policy formulation, STI talent management, R&D institution dynamics, aeronautics policy, renewable energy, main research plans, and the innovation ecosystem among others.

1.2 Developing STI Human Capital in Indonesia

Indonesia is promoting the development of science and technology human resources based on Law No. 11, 2019, on the Science and Technology Framework Act in Indonesia. BRIN is interested in increasing both the quantity and the quality of human resources in the science and technology fields. By benchmarking Korea's experience, BRIN aims to establish a policy for the development of Indonesia's STI human capital.

To this end, this study presents specific policy measures and the implications by synthesizing Indonesia's current status, major issues, and Korean cases. The current status of and major issues surrounding STI human capital in Indonesia were identified as follows. First, the government's R&D budget investment is small, and the budget allocation system is complicated, so researchers cannot secure a sufficient budget for research. The problem of obtaining research funds is a factor that hinders researchers' commitment to research and the creation of research results. Second, the supply of human resources in science and technology needs to be increased. In 2018, the number of researchers per million was just 215, which is low compared to other ASEAN countries. Third, the number of Ph.D. holders is small. Among the employees of government agencies, only 24% of researchers, 8% of engineers, and 12% of instructors hold a doctorate. In order to come up with an improvement plan based on the current situation of Indonesia's STI human capital, BRIN conducted in-depth interviews with officials from various institutions, as the Ministry of Culture and Education, BRIN, and the LPDP, in relation to the need to nurture scientific and technological human resources in Indonesia.

The case study of Korea is composed of the following contents. First, it presents the current status and characteristics of STI human capital in Korea. Second, it introduces the contents of the Special Act on Support for Scientists and Engineers for Strengthening National Science and Technology Competitiveness, which became the basis for Korea's scientific and technological human resource development policy after the 2000s. Next, it discusses the primary support measures for human resource development in Special Act. Finally, it presents three Korean cases for promoting the research capabilities of graduate schools. This study examined the following: 1) KAIST, a research-centered university established by the government to foster high-quality human resources; 2) UST, jointly established by

government-funded research institutes to foster master's and doctoral students; and 3) BK21 projects promoted by the Ministry of Education to strengthen the research competitiveness of public and private universities.

Based on the current situation in Indonesia and case studies of Korea, this study suggested three policy proposals for the development of S&T human resources in Indonesia. The first proposal consists in expanding the program for funding and training human resources; the second in improving the method of operation of the Kedaireka Initiative (Indonesian Sovereignty in Design) to strengthen industry-university cooperation in the STI fields; and the last in reorganizing the classification system of public research institutions such as BRIN, a civil servant organization. The study also presents measures for strengthening the autonomy of researchers.

1.3 Improving the Management Regime of BRIN for Mission-oriented Research & Innovation

Indonesia has suffered from the problem of weak innovation capacity, especially in terms of institutional indicators. As such, in 2021, the Indonesian government restructured its STI organizations by integrating the National Research and Innovation Agency (BRIN), including the BPPT, LIPI, LAPAN, BATAN and many other R&D institutes, in a drive to reinforce the national innovation capability. In its early stages, BRIN sought to establish an efficient and effective management system. In benchmarking Korea's NST and GRIs cases, this chapter presents a comparative analysis based on the management regime framework, which consists of five elements: objectives, strategies for achieving those objectives, governance, funding schemes, and organizational management.

BRIN's main vision is to become a reliable, professional, and innovative National Innovation Agency that can support the vision and mission of the President and Vice-President of the Republic of Indonesia, while its strategy for achieving this vision is to increase research excellence and technology commercialization. BRIN has three main functional structures, or deputies, which serve as policy support, funding, and execution agencies, respectively. BRIN acts not only as a research institution but also as a funding agency, providing research funds

for research organizations in industry or universities that collaborate with BRIN's research organizations. BRIN's funds come mostly from the government, such as institutional funding. BRIN also pursues more innovative organizational management schemes, especially with effective commercialization capability.

Korea's National Research Council of Science and Technology (NST) and government-funded research institutes (GRIs) have been analyzed using the same management framework. There are some similarities and differences between the GRIs and BRIN. For instance, their vision and mission are similar, namely, to become excellent research hubs and effective innovation platforms. Some policy recommendations derived from the comparative analysis include: 1) strengthening convergent research, 2) enhancing collaboration between BRIN and industry, and 3) catalyzing more start-ups with the development of various commercialization pathways.

2. Policy Recommendations

2.1 Designing an STI Policy Think Tank for Green Growth Policy Formulation: Lessons Learned from South Korean Practices

The study has explored the practices of the prominent GTC in South Korea, as outlined in the section on the scope of research. There are three key elements, namely, the resources, strategy, and governance of the institution on the GTC which could be adopted by the PKRTI-BRIN in Indonesia. However, adapting key elements of the GTC to the PKRTI-BRIN is not a simple task; such elements cannot be fully and directly adapted from the GTC to the PKRTI-BRIN. Indeed, adjustments or incubation may be required before in-cooperating elements of the political and government system of South Korea, a developed country, into Indonesia, a developing country; in other words, the situation in these two countries is vastly different. Korean cases can influence how a public think tank works and sustains itself. Besides, history, value, and culture are unveiled factors that influence the work style of think tanks in both countries.

Both the GTC in South Korea and the PKRTI-BRIN in Indonesia are public R&D institutions that provide STI policies based on the results of research and development activities. Although the GTC can provide practical lessons for the DPKRTI-BRIN, it is no secret that establishing a world-leading think tank requires ample resources, specific and long-term policies or strategies, and appropriate institutional governance. We will elaborate on those elements as follows.

First, regarding the Indonesian government's strategies, the main player in its GEG strategy is Bappenas (RPJPN, RPJMN, GG program, GEI, CE, G20 working group), while the Ministry of Industry plays a role in the CE strategy along with Bappenas and various other public agencies. Meanwhile, the Ministry of Environment and Forestry, the Ministry of Finance, Bappenas, and local government arrange the green growth program. As for the G20 Working group, the DPKRTI-BRIN is involved, along with other public agencies including Bappenas. Therefore, it is necessary to reinforce strategic planning for the development of

green technology initiated by the DPKRTI-BRIN, by involving those ministries and other public agencies. In addition, the DPKRTI-BRIN plays close attention to such issues as global green technology and climate change, connecting national strategies with global strategic plans.

Second, with regard to resources, there is no specific resource for moving GEG policy in the DPKRTI-BRIN. Existing resources should be moved to specific resources to support green technology development. STI research and its funding should be directed to contribute specifically to green technology, green policy, GE, GG, and other SDGs issues. Furthermore, the relevant networks should be widened by involving global climate change and green technology organizations, countries affiliated with green economy goals, SDG research centers, and multinational green industries. At the level of national networks, the involvement of public agencies (ministries and local government agencies), industries, R&D research centers, universities, and NGOs is more focused on two aspects: STI that contributes to the development of green technology and STI that contributes to the development of the GE.

Third, institutional governance is one of the important aspects. BRIN's vision and mission concern national research, technology and innovation development, while green technology and climate change are small parts of its duties. In the DPKRTI-BRIN, the establishment of a "national green technology policy program" (NGTPP) is necessarily required. At BRIN, such a program is equal to an organizational structure which functions to operate specific tasks at the top management level (deputy at BRIN). Like the GTC, the NGTPP's tasks are to conduct policy and management research for green technology development, provide policy consultation and advocacy on green technology issues, cooperate with national partners dealing with green technology and climate change issues, and produce public information concerning STI contributing to GEG. In the early stages, the NGTPP will be asked to respond to and meet green technology demands in Indonesia.

The government fully funds the DPKRTI-BRIN, which is regulated by governmental laws and managed by bureaucrats. It means that all activities within the DPKRTI-BRIN are adjusted to the rules of the game as governed by the government. The emergence of the NGTPP in the DPKRTI-BRIN will require diverse governance under the BRIN's system. BRIN should shed on

semi-autonomous authority for the NGTPP in managing resources, in involving outside experts of BRIN, in determining priority research areas, in communicating with stakeholders, and in collaborating with domestic and overseas partners. The NGTPP should focus on creating the Indonesian style of public organization governance.

In the context of the DPKRTI-BRIN, a think tank is perceived as an entity that “bridges or transmits the STI knowledge produced by academicians to the STI policy produced by stakeholders”. Agenda setting and policy formulation are core areas of a think tank’s activities. It is a conventional thing because STI policy think tanks should play a role in the implementation and evaluation areas. The emergence of the NGTPP is an opportunity to deliver and enhance the STI think tanks at BRIN. In addition, the NGTPP is expected to become a leading think tank in Indonesia because of its new working area. It plays a role in the following four areas at once: the role of an expert in the agenda setting phase; the role of advocacy in the policy formulation phase; the role of a consultant in the policy implementation phase; and the role of supervisor-monitor in the policy evaluation phase.

Proposing a new organizational structure for the NGTPP under the DPKRTI-BRIN is related to new institution, new policy, and new governance. The creation of the NGTPP as a new program will not wholly change the hierarchical structure of the DKRI and BRIN. Program(s) can be added or removed at the DPKRTI-BRIN level, because the program is not part of the echelon structure. Therefore, it is feasible to create the NGTPP within the hierarchical structure of the DPKRTI-BRIN. The NGTPP will be led by a program coordinator who is accountable to the director of the PKRTI-BRIN. Moreover, it will be connected to other programs within the internal and external DKRI environments (i.e. other deputies and research organizations at BRIN).

2.2 Developing STI Human Capital in Indonesia

2.2.1 Expansion of Human Resource Capacity-Building Programs

The Indonesian government is operating various programs designed to improve the capacity and quality of human resources, especially in the field of science and technology. A representative program in this regard is the master's and doctoral human resources training

project. Current projects support only a small number of universities, and only a limited number of research areas are eligible to receive financial support, so in some cases it may be difficult to raise funds. To improve the quality of research and innovation, it is necessary to adjust the priorities of research projects carried out by the Indonesian government and organically combine various human resource support projects. In addition to scholarships, it is necessary to expand the programs for S&T human resource development and strengthen the capabilities of private companies.

First, the master's and doctoral manpower training programs should be reorganized. For example, the Indonesian government could consider combining the BRIN Research Scholarship with the Ministry of Finance LPDP Scholarship. The goal would be to integrate BRIN's degree application program with the LPDP program in order to unify sources of funding and provide scholarships to researchers, lecturers, and engineers in government agencies, thereby making it possible to increase the proportion of doctoral degree holders in the field of S&T in Indonesia.

Second, it is necessary to diversify the programs not only to support the acquisition of degrees but also to strengthen the capabilities of the researchers at BRIN. It is necessary to improve the competency of BRIN's employees using various methods, such as workshops, international training, and conferences, and to expand the relevant educational programs. For example, a non-degree research program (Non-Degree by Research) could be designed to strengthen BRIN's human network and research capabilities.

Finally, there is a need to strengthen cooperation between BRIN/universities and private companies. In the future, policy priority should be given to enhancing private companies' human resources and research capabilities. Strengthening cooperation with the private sector, which has been neglected so far, will not only improve the nation's research capacity and companies' ability to execute research projects, but will also contribute to improving efficiency in the allocation of the government's R&D budget and to strengthening the innovation capacity of the private sector.

2.2.2 Strengthening Industry-University Cooperation: Kedaireka Initiative (Indonesian Sovereignty in Design)

Indonesia operates the Kedaireka initiative (Indonesian Sovereignty in Design) as a platform that connects universities and industries, and that enables industries and universities to obtain necessary information, collaborate, and receive financial support. Kedaireka provides more than IDR 1 trillion to higher education and vocational programs and seeks to strengthen cooperation on STI with universities and industry. Here, the overall aim is to strengthen the role of S&T in promoting the national development strategies and solving social problems. Financial support is focused on five areas: green economy, blue economy, digital economy, tourism, and health infrastructures.

South Korea's experience raises two implications concerning the implementation of the Kedaireka initiative. First, it is a case of establishing a customized talent training system between universities and industries with industry-academic cooperation. In Korea, when a company needs to retain its employees, it requests a university to design a program, and the university then opens a program to strengthen the capabilities of the participants. In addition, retraining programs for incumbent workers, such as mentoring, can be operated in industry-academia linkage.

Second, South Korea's experience emphasizes cooperation with SMEs in research funding. The Korean government often requires universities to present a model for cooperation with private companies as a condition of receiving government support. Fostering local industries is also an essential issue in Indonesia. The Kedaireka initiative serves as a hub to induce cooperation between universities and the private sector. In operating the Kedaireka program, the government could support the researchers of public research institutes as well as university members and induce them to transfer their developed technologies to local SMEs, industries, and local governments. As a result, the Kedaireka initiative could link the delivery of education, research, and community services.

2.2.3 Improvement of the Research Personnel Regulations to Disseminate Research Results

In Indonesia, lecturers and researchers at national universities and BRIN researchers are subject to the same regulations as civil servants. The regulations for public officials have limitations in dealing with such issues as start-up ownership and human resource flexibility. In particular, when a researcher from a public institute starts a business, there are restrictions on ownership and budget execution. The Ministry of Finance manages the research budget of the start-up, and the start-up is owned by the institution. Therefore, the current system, which restricts autonomy due to such civil servant regulations, urgently needs to be improved.

This study presents two main alternatives. The first is to establish an intermediary agency for start-up and technology transfer. The members of the Technology Transfer Office (TTO) can alleviate potential rigidity in the operation of government agencies by staffing them mainly with private personnel. The establishment of the Technology Transfer Office is feasible according to government regulation no. 72 of 2012.

The second alternative is to reorganize the classification system of public research institutions such as BRIN, a civil servant organization. Government-funded research institutes in Korea are classified and operated as quasi-governmental institutions, rather than as government ones. Because they are not an official government organization, it has autonomy and independence in executing its budget and using its human resources. However, in the case of Indonesia, the concept of a third sector public sector (i.e. Korean quasi-governmental institutions) is unfamiliar, so it may be challenging to convert the status of a researcher into a civilian by introducing Korea's R&D system as it is. Alternatively, it may be possible to consider a plan to operate such institutes by partially easing the current regulations, including incentive payments for research achievements and recognition of research achievements. However, deregulation must also be accompanied by researchers' responsibility. In this case, it is necessary to establish a system for evaluating whether researchers carry out their duties faithfully within the institution. However, it is possible to minimize the potential side effects of deregulation through continuous monitoring.

2.3 Improving the Management Regime of BRIN for Mission-oriented Research & Innovation

2.3.1 Strengthening the Research Convergence

More collaborative programs are needed to provide BRIN with greater opportunities to develop convergent technologies and create new demand industries based on those technologies. At BRIN, the implementation of convergence research could be carried out as a series of steps, as follows: (1) determine the convergence research system through a scheme that can be carried out based on BRIN's strategic plan and national priority research in a top-down approach; and (2) determine the framework and role of each research organization and deputy according to the concept of convergent research at BRIN. This policy recommendation will specifically explore the arrangement of research funding programs from the national budget (APBN).

1. Encouraging convergence through research funding policies

- a. Reinforce the existing funding scheme for research convergence at BRIN via the funding program of the Research Collaboration Center (Pusat Kolaborasi Riset).
- b. Set a minimum percentage of funding from the 'Program House' for research that contributes to pre-determined specific national issues, with collaboration with industry/academia set as a requisite.

2. Supporting convergence through programs

- a. Facilitate meetings and discourse.
- b. Conduct periodic foresight research and discourse.

3. Supporting convergence with a structural brace

- a. Clarify and explain the unit's role inside BRIN.
- b. Empower the interface entities or intermediary units.

2.3.2 Encourage BRIN-Industry Research Collaboration

Placing an industry as a research partner is a prerequisite of BRIN's research financing scheme. Industry can directly absorb the results of research conducted by BRIN's research institutes. This program consists in:

1. creating a co-investment program with the private sector for research development; and
2. improving the capacity of BRIN's human resources in licensing technology services and incubation services.

2.3.3 Accelerating the Growth of Start-ups as a Path to Commercialization

One of the lessons learned in boosting the commercialization of GRI technologies in Korea can be applied to the formation of a holding company at BRIN. Regulations related to holding companies have yet to be specifically regulated in the laws and regulations in Indonesia. In general, there are two types of holding company based on practice in Indonesia:

- a. Investment Holding Company: This is a type of company whose business activities are limited to the ownership of a part or all of the shares of its subsidiaries but which does not participate in the operational activities of those subsidiaries.
- b. Management Holding Company (Operating Holding Company): This is a type of company which, apart from owning a part or all of the shares of its subsidiaries, also participates in managing the business operations of, and making business decisions for, its subsidiaries.

Of these two types of holding company, BRIN tends to refer to the second type. BRIN Holdings Company aims to help commercialize the research products of researchers at BRIN. This goal is achieved by carrying out the following steps:

1. Formulate the BRIN Holdings Company's objective framework; for BRIN Holdings

Company, it is possible to have investments from two activities, i.e.:

- a. Establishment of a research center competency-based subsidiary (science-based company).
- b. Establishment of a service-based subsidiary company (service-based company).

2. Funding of BRIN Holdings Company

3. Management System of BRIN Holdings Company

2.4 Monitoring Policy Recommendations and Policy Success

“Policy transfer” refers to the process by which policies implemented by other governments or international organizations are transferred to and adopted by a specific government. To ensure the success of a policy transfer, it is necessary to modify and improve policies that are more appropriate for one's own country rather than simply transferring other countries' policies as they are. In other words, if the time and circumstances are different, the situation or context of the policy problem will also be different. As a result, the nature of the policy problem changes, so we should consider this in the design of policy alternatives.

The K-innovation project has been working to improve Indonesia's SIT governance for three years, from 2020 to 2022. South Korea's experience based on the Framework Act on Science and Technology or the reform of government research institutes can be applied to the development of Indonesia's STI system. During the project, Indonesia implemented follow-up measures to the Law of 2019-11 and established BRIN. Moreover, Indonesia is currently facing the task of improving the system for evaluating public R&D and fostering more abundant S&T human resources. A successful response to these challenges will upgrade the economy of Indonesia, a country with great growth potential.

It is important to ensure that the recommendations presented in this report are incorporated into Indonesia's policy practice and successfully implemented. Policy advice related to governance has a number of limitations that make it difficult to quantify and evaluate output performance. However, if we intentionally set our project goals and

performance indicators, we can create more and better outcomes in developing STI governance in Indonesia.

To this end, in the following project planning, it is necessary for the two partners, Korea and Indonesia, to set clear performance indicators and consider how to monitor performance. For effective monitoring, we need to gather information on the operational status of the proposed policy recommendations and the difficulties faced in practice. Monitoring activities will also help us to consult on and develop policies characterized by wider acceptability and greater effectiveness.

As regards monitoring, the following measures should be considered: First, it will be necessary to establish a monitoring system after project completion. Second, it may be possible to increase the acceptability and effectiveness of project suggestions through interim monitoring in the course of implementing the project. Third, the effectiveness of policy transfer should be increased by conducting a comparative analysis of the goals and environmental changes in the monitoring process. Fourth, one should consider establishing a regular monitoring and feedback system for the implementation process and making certain organizational changes in the early project planning stage. More active mutual communication, interaction, information exchange, and data collection will also be required to introduce and activate a new monitoring system.

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ABBREVIATIONS

A	Academia
ABGC	Academia, Business, Government, and Community
ADB	Asian Development Bank
ASN	Aparatur Sipil Negara (State Civil Apparatus)
B	Business
BAPPENAS	Badan Perencanaan Pembangunan Nasional (Ministry of National Development Planning)
BATAN	Badan Tenaga Nuklir Nasional (National Nuclear Energy Agency)
BERD	Business Contribution to Expenditure on Research & Development
BGC	Business, Government, Community Actors
BGH	Bangunan Gedung Hijau (Green Building)
BK21	Brain Korea 21
BKN	Badan Kepegawaian Nasional (National Civil Service Agency)
BLU	Badan Layanan Umum (General Service Bodies)
BPPT	Badan Pengkajian dan Penerapan Teknologi (Agency for the Assessment and Application of Technology)
BRIDA	Badan Riset dan Inovasi Daerah (Regional Research and Innovation Agency)
BRIN	Badan Riset dan Inovasi Nasional (National Research and Innovation Agency of Indonesia)
C	Community
CE	Circular Economy
CIPE	Center for International Private Enterprise
CIPG	Center for Innovation and Policy Governance
COVID19	Coronavirus Disease 2019
CTCN	Climate Technology Center and Network
DEKRTI	Deputi Evaluasi Kebijakan Riset, Teknologi, dan Inovasi (Directorate of Research, Technology, and Innovation Policy Evaluation)
DKP	Deputi Kebijakan Pembangunan (Deputy of Development Policy)
DKRI	Deputi Kebijakan Riset dan Inovasi (Deputy of Research and Innovation Policy)
DPIRTI	Direktorat Pengukuran dan Indikator Riset, Teknologi, dan Inovasi (Directorate of Research, Technology, and Innovation Measurement and Indicators)

DPKRTI	Direktorat Perumusan Kebijakan Riset, Teknologi, dan Inovasi (Directorate of Research, Technology, and Innovation Policy Formulation)
DPR	Dewan Perwakilan Rakyat (People's Representative Council)
DRIDA	Deputi Riset dan Inovasi Daerah (Deputy of Local Research and Innovation)
DUDI	Dunia Usaha dan Dunia Industri (Business and Industrial World)
DWG	Development Working Group
EKPKA	Program of Space and Aeronautics Policy Evaluation
EKPKT	Program of Nuclear Policy Evaluation
EKPRTI	Program of STI Policy and Management Evaluation
G	Government
G20	Group of Twenty
GDP	Gross Domestic Product
GE	Green Economy
GEG	Green Economy Growth
GEI	Green Economy Index
GERD	Gross Expenditure on Research and Development
GG	Green Growth
GGF	Green Growth Framework
GGGI	Global Green Growth Institute
GND	Green New Deal
GPB	Green Planning and Budgeting Strategy
GRI	Government-sponsored Research Institutes
GTC	Green Technology Center
GTP	Green Technology Policy
GTPI	Green Technology Partnership Initiative
HCI	Human Capital Index
HEI	Higher Education Institutions
HRST	Human Resources in Science & Technology
ICT	Information & Communication Technology
IGG	Inclusive Green Growth
IP	Intellectual Property
IRL	Innovation Readiness Level
IS	Innovation System

KAIS	Korea Advanced Institute of Science
KAIST	Korea Advanced Institute of Science and Technology
KemenPANRB	Kementerian Pendayagunaan Aparatur Negara dan Reformasi Birokrasi (Ministry of State Apparatus Empowerment and Bureaucratic Reform)
KIST	Korea Institute of Science and Technology
KIT	Korea Institute of Technology
LAPAN	Lembaga Antariksa dan Penerbangan Nasional (National Institute of Aeronautics and Space)
LCDI	Low Carbon Development Indonesia
LIPI	Lembaga Ilmu Pengetahuan Indonesia (Indonesian Institute of Sciences)
LPDP	Insentif Riset Sistem Nasional (Educational Fund Institute of Management Insinas)
MBKM	Merdeka Belajar Kampus Merdeka (Independent Learning-Independent Campuses)
MSIP	Ministry of Science, ICT and Future Planning
NDC	Nationally Determined Contribution
NDE	National Designated Entities
NGO	Non-governmental Organization
NGTPP	National Green Technology Policy Program
NIEs	Newly Industrializing Economies
OECD	Organization for Economic Cooperation and Development
OIS	Open Innovation System
OR	Research Organization
ORC	Organizational Readiness for Change
PDB	Product Domestic Bruto
Pemkab	Pemerintah Kabupaten (District Government)
Permen	Peraturan Menteri (Ministerial Regulation)
PIPRI	Program for STI Measurement and Indicator Analysis
PKKPTI	Program for Technology Development and Innovation Policy Studies
PKMRI	Program for STI Policy and Management Research
PKPERI	Program for Development of the Research and Innovation Ecosystem
PKPKPK	Program for Study and Formulation of Space and Aeronautics Management Policy
PKPKPN	Program for Study and Formulation of Nuclear Management Policy

PKPME	Program for the Development of STI Evaluation Methods
PNBP	Penerimaan Negara Bukan Pajak (Non-Tax State Revenue)
PPDIRI	Program for the Collection and Management of Research and Innovation Data
PRIPJRI	Program for Formulation of the STI Master Plan and Roadmap for Research and Innovation
PTN	Perguruan Tinggi Negeri (Public Universities)
PTS	Perguruan Tinggi Swasta (Private Colleges)
R&D	Research and Development
RIRN	Rencana Induk Riset Nasional (National Research Master Plan)
RPJMN	Rencana Pembangunan Jangka Menengah Nasional (Indonesia's Mid-to-Long-Term Development Plan)
RPJPN	Rencana Pembangunan Jangka Panjang Nasional (Indonesia's National Long-Term Development Plan)
S&T	Science and Technology
SDGs	Sustainable Development Goals
SMEs	Small and Medium-Sized Enterprises
STI	Science, Technology, and Innovation
SWOT	Strengths, Weaknesses, Opportunities, Threats
TEC	Technology Executive Committee
TEEB	The Economics of Ecosystems and Biodiversity
TKATRI	Programs of STI Foresight and Trend Analysis
TRL	Technology Readiness Level
TTO	Technology Transfer Office
UN	United Nations
UNDESA	United Nations Department of Economic and Social Affairs
UNDP	United Nations Development Program
UNEP	United Nations Environment Program
UNFCCC	United Nations Framework Convention on Climate Change
UST	University of Science and Technology
WALHI	Wahana Lingkungan Hidup Indonesia (an Indonesian Environmental Organization)
WFA	Work From Anywhere
WRI-NCE	World Resources Institute – The New Climate Economy